Passive Attitude Control and Stabilization for Microsatellites

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by

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Abstract

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Based on the breakthroughs of Microspace missions, New Space ventures capitalize on radical innovation to usher in a new era where spacecraft form part of the infrastructure for global service provision in Telecommunications and Earth Observation. Confronted with a congested space domain, spacecraft developers are advised to comply with design-by-disposal practices. This thesis expands on observations of CanX-7 on-orbit behavior, a 3U satellite developed by SFL and equipped with an innovative drag sail mechanism. A passive stabilization solution is suggested and tested to further improve upon the lessons learned from the mission's success. Furthermore, this thesis investigates the design, testing, and validation of a passive control solution for attitude death-mode avoidance leading to power outage. The solution involves the sizing and testing of a permanent magnet to be placed on board of satellites whose mission requirements do not allow for solar panel mounting on all faces of a cubic bus geometry.

To my parents and my sister

Acknowledgments

“ *I am indebted to my father for living, but to my teacher for living well*”
Alexander the Great

With these words, Alexander the Great paid eternal respect to his teacher Aristotle for equipping him with the mental armor required to conquer the whole then known world.

This quote resonates my deep gratitude for Dr. Zee, who, similarly to a general in the field, has been guiding his troops in victory for more than 23 years. On my first day as a student in his lab, I vividly recollect him mentioning that “here, we are working together as a team, we are and operate like the Navy SEALs”. Drawing the above resemblance was only natural, inspiring, and memory-lasting for me, having previously served in the army and coming from a family military tradition.

Strongly similar to a paternal figure that nurtures a child’s inquisitive nature while teaching and expecting respect to house rules, Dr. Zee has mentored me throughout my journey at the Space Flight Laboratory. Through some of my harder times, he could see through the sincere effort, recognize hard work, and give advice that has still left a strong impact on me. It is only natural that a team led by Dr. Zee is bound to innovate and sustainably succeed with such a level of motivation as if every day is the first day at work.

It took little time for me to realize that Dr. Zee’s mentality has truly flowed down to his mission managers and staff. Forged through his teachings of the Microspace philosophy, the art and logic of building small, innovative and cost effective spacecraft, all the mission managers that I had the privilege to work for gave me a unique perspective through which my skills improved exponentially. I pay strong respect to my first mission manager, Brad Cotten, for welcoming me to the SFL family and trusting me with what would eventually form a large part of my thesis. I am deeply thankful for my second manager, Laura Bradbury, for trusting me with equally interesting work on the art of how to comply to the Microspace doctrine and “always avoid death modes”. I am equally grateful for the critically important tasks that my third manager, Karan Sarda, gave me, pertaining to the fast-paced acceptance testing of attitude-control hardware, hinting towards what would eventually become the new norm for hardware testing in the New Space era.

It would be an omission to not pay my gratitude to some notable lab experts. I deeply thank Neils Roth for always being available to explain difficult concepts in plain language, no matter how much of his busy time this would take. I am grateful to Chris Coggon, for always being the optimistic, resourceful, and full of advice facilities manager that every laboratory would wish they had. I am grateful for Mihail Barbu, whose intuitive understanding of electronics and avuncular style bred my deeper interest for electronic design. I am grateful to Robert Magner, for always helping and guiding me with my immediate tasks, as well as

recognizing my contributions, in a warm and friendly manner. I will be forever grateful to my colleague students, having met them in such a thriving environment and tapping from both their emotional and cognitive intelligence, through what I experienced as a flawless collaboration.

Thank you again, Dr. Zee, for making the Space Flight Laboratory the warm family that I have proudly joined.

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Nomenclature

Acronyms

ACS	Attitude Control Subsystem
ADC	Attitude Determination and Control
ADS	Attitude Determination Subsystem
AOCS	Attitude, Orbit and Control Subsystem
CDF	Cumulative Distribution Function
COM	Center of Mass
ECEF	Earth-Centered-Earth-Fixed
ECI	Earth-Centered-Inertial
FK5	Fifth Fundamental Katalog
GEO	Geostationary
GHGSat-D	Greenhouse Gas Satellite Demonstrator
GNB	Generic Nanosatellite Bus
IADC	Inter-Agency Space Debris Coordination Committee
IAU	International Astronomical Union
IERS	International Earth Rotation Services
IGRF	International Geomagnetic Reference Field
J2000	12:00 January 1, 2000 (Terrestrial Time)
LEO	Low Earth Orbit
LNA	Low Noise Amplifier
LTDN	Local Time of Descending Node
LVLH	Local Vertical
NEMO	NExt-generation Monitoring and Observation
NORAD	North American Aerospace Defense Command
NTS	Nanosatellite Tracking of Ships
OASYS	Onboard Attitude System Software
RAAN	Right Ascension of the Ascending Node
SFL	Space Flight Laboratory
SGP4	Simplified General Perturbations
TLE	Two Line Element
UTIAS	University of Toronto Institute for Aerospace Studies

Math

v	Scalar quantity
$\underset{\rightarrow}{\mathbf{v}}$	Vector quantity
$\mathbf{V}$	Matrix quantity
$\mathbf{0}_{\mathrm{nxn}}$	n by n matrix of zeros
$\mathbf{1}_{\mathrm{nxn}}$	n by n identity matrix
$\mathscr{F}_{\mathrm{a}}$	Frame a
$\underset{\rightarrow}{\mathscr{F}}_{a}$	Vectrix, a matrix whose columns are the basis vectors of frame a
$\mathbf{v}_{\mathrm{a}}$	Column vector containing the components of $\underset{\rightarrow}{\mathbf{v}}$ defined in frame a
$\vert\vert \cdot \vert\vert$	The two-norm of a vector

$$\mathbf{v} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}, \quad \mathbf{v}^{\times} = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix}$$

Chapter 1

Introduction

1.1 The NewSpace Revolution

The "NewSpace" revolution has emerged as a new business model in the satellite industry over the last several years. Resembling yet another space race, it is predominately driven by large scale commercial initiatives rather than political agendas or governmental finance, seeking to provide broadband connectivity and Earth Observation (EO) services that scale. Since 2018, companies such as SpaceX, Amazon, and OneWeb have been filing applications for the assignment, allocation and allotment of frequency spectrum to the International Telecommunications Union (ITU) for satellite networks as large as 30,000 satellites, hinting towards their ambitious goals [1]. For reference, the US NORAD system currently tracks more than 23,000 man-made objects in space, larger than 10 cm [2]. As of August 2020, SpaceX has launched 654 Starlink satellites within less than 13 months, all of which were manufactured in-house [3]. Whether those opportunities remain financially viable in the long term is an unanswered question, particularly amidst the uncertainty of a global pandemic, and the future ahead. Past experience has illustrated both commercial successes (Iridium) as well as failures (Teledesic). The bankruptcy of OneWeb and the efforts for re-acquisition and continuation of development demonstrate the fragility of such large-scale initiatives in the absence of global market stability, while emphasizing the perseverance and commitment to maintain momentum.

1.2 Small Satellite Market Prospects

A key characteristic of the NewSpace revolution lies in the fact that it inherits the heavy legacy of the Microspace paradigm and transforms the commercial offering to be service based, rather than being limited to satellite manufacturing [4]. In the presence of such a broader business scope and the need for a competitive and viable price point, those ventures have brought small satellite manufacturing in-house, a step towards deeper vertical integration. Satellites are considered as the infrastructure needed to provide a service, which in turn requires support by a broader number of services such as sales and marketing, satellite operation, data analytics etc. As the Microspace philosophy pushed and broke the microsatellite utility barrier, making space more accessible with smaller, smarter, and more competitive platforms, the NewSpace paradigm capitalizes on those gains to advance in a new era.

The driving role of NewSpace ventures is highlighted by market intelligence reports that provide insight on the prospects of the small satellite industry over the years to come. It is of interest to examine those prospects as they evolve through time since last decade and through the next one.

Figure 1.1: Summary of Euroconsult's 2020 report titled: "Prospects for the small satellite market" [4]

Euroconsult's latest report [4] is illustrated in Figure 1.1. It reports that the top three applications for the microsatellite industry will remain in the top three positions, but with shifted ranks. There is a projected clear preference for telecommunication satellites that offer broadband connectivity. However, absolute revenue growth of all categories is notable, regardless of their rank. During the 2010-2019 decade, the top three microsatellite applications were Technology (39%) followed by Earth Observation (32%) and Telecommunications (11%). In the upcoming decade (2020-2029), the Telecom segment is projected to lead the market(56%), followed by Earth Observation (15%) and Technology applications (12%). Commercialization opportunities for space are projected to become increasingly abundant.

It is of interest to note that the average mass per satellite is expected to triple in the next decade, while the manufacturing and launch value is expected to quadruple. This hints towards the maturity of the Microspace technology and the ever-improving miniaturization of components and payloads. Microspace ventures contributed significantly to this effort during the previous decade, with radical innovation and major breakthroughs, both in terms of hardware, as well as software. Euroconsult reports that the commercial Microspace sector will grow its market share from 54% to 76%, the institutional sector will shrink from 23% to 7%, while the demand for military applications will remain proportionally the same throughout the two decades, at 6%.

1.3 The NewSpace challenges

The projected growth of the small satellite market does not necessarily translate to equitable access to those opportunities by new and old players. In fact, vertical integration, which leads to improved economies of scale, can push the satellite product offering to such low pricing as to raise the barrier to entry even higher. The plethora of applications for space utilization does leave room for specialization and diversification of the market, yet, the original Microspace ventures are faced with challenges to which they need to adapt. Amidst such an environment, there are two key elements that will sustain growth in the Microspace industry; radical innovation, as well as timing [5]. The following paragraphs expand on the importance of those elements and form the high-level motivational basis for the original contributions of this thesis.

Disrupting the market with a new technology and incrementally building upon it based on lessons learned is a proven recipe for success and the cornerstone of the smallsat developer business model, particularly for the Space Flight Laboratory [5]. Well known successful cases include the Alouette-Award-winning CanX-4 and CanX-5 satellites for achieving unprecedented formation flying capabilities [6]; the NTS (CanX-6) satellite for being the first of its kind nanosatellite equipped with a space-based ship tracking AIS payload, built by COM DEV; the CanX-7 mission and its radically innovative drag sail design, proving its deorbiting performance up to this day. This series of innovations led to more advanced concepts that have now turned into successful missions; Hawkeye Pathfinder Mission, the Hawkeye Constellation Mission, NorSat-3, NEMO HD, and many more. For example, with regards to the Hawkeye Pathfinder Mission, the Pathfinder satellite is scheduled to go on display when the National Air and Space Museum's renovation is completed, slated for 2025 [7].

The timing of delivering innovative technologies forms the other half of the success equation for sustainable Microspace growth. Within this scope, is it critical to consider the two satellite developer strategies that have emerged; disruptive technology development, and operational capability [8]. Waiting and theorizing on the best possible design for a spacecraft with the highest performing capability is a time consuming process that can delay innovation, effectively entering the market possibly too late. Nimble and agile decision making is required to support radical innovation. The combination of innovation and agility without overemphasizing performance can lead to attracting public interest and customers early. The subsequent early profits will spiral only towards faster quality iterations of a proven design. Therefore, while the NewSpace service providers will be investing in a battle of competition based only on the basis of cost due to the advantages offered by economies of scale, the Microsat business model will still be the instrument that best fosters a high rate of iterative innovation. Leveraging this will enable Microspace organizations to adapt in a volatile environment.

1.4 Space Traffic Management in the NewSpace Era

At the cusp of this wave of disruption, a new reality has been slowly emerging, challenging the NewSpace revolution with an increasingly alarming problem; the space domain has become increasingly congested, convergent, contested, and competitive. One of the paradoxes of space is that it is "empty but crowded" at the same time [9]. There is a narrow family of orbits which attract most satellite applications due to their unique properties. For example, the geostationary belt (also known GEO belt) is used for applications where constant line-of-sight visibility is desirable to a target area or asset. Sun-synchronous orbits are best suited for Earth Observation applications due to the repeating nature of their passes over the equator, as well as their high latitudes which facilitate nearly whole Earth coverage. Even among those orbit families, certain orbital "slots" are preferred. For example, the density of satellites in GEO is clearly higher over the continental United States and Central Asia, while there are almost no satellites for orbital slots above the Pacific Ocean. Sun synchronous orbits with specific orbital nodes are preferred due to their optimized sunlight interval times and optimal timing for Earth observation applications. All this leads to a congested orbital environment, where the chances of a conjunction in space increase with the subsequent launch of every batch of mega constellation satellites. The relatively small amount of conjunctions and anti-satellite tests in space have already led to the generation of tens of thousands of debris, which over time spread out due to the Kessler effect and challenge spaceflight safety for existing and planned missions, for a broad range of orbits. Plans of constellations comprised of 30,000 satellites necessitate active measures for ensuring spaceflight safety, sooner rather than later. The world's leading space powers have long recognized the problem of congestion and space debris. The United Nations has signed the Outer Space Treaty, while the IADC has established guidelines for design-by-disposal methods to ensure that the amount of space objects remains manageable and under control [10].

The sustainable use of space formed part of the motivation behind the efforts of the Space

Flight Laboratory to develop CanX-7. A truly innovative design that has proven to work actively towards space debris mitigation, the CanX-7 satellite deployed its four drag sail modules after a 7 month mission using the AIS Space based receiver for space-based aircraft tracking. Among many other of its crown jewel missions, the CanX-7 satellite has proven that satellites equipped with this module can effectively contribute to the effort of reducing space debris, providing the innovative answer with which the majority of NewSpace ventures may inevitably be required to comply, not by recommendation alone, but possibly by a strict regulatory provision prior to launch.

The drag sail is an add-on module which all SFL satellites can accommodate. With a particular focus on the NewSpace satellite designs offered by SFL, the inclusion of a drag sail for a family of low Earth orbits can ensure proven compliance to space debris mitigation guidelines, with minimal operational risk, at the end of the mission's lifetime.

1.5 Thesis Scope

The scope of this thesis revolves around original contributions towards a better understanding of technologies that promote spaceflight safety with radical innovation. More specifically, the thesis expands on observations of the on-orbit behavior of CanX-7 satellite, a 3U satellite developed by SFL and equipped with an innovative drag sail mechanism. In continuation to those observations, a passive stabilization solution is suggested, tested, and validated for future drag sail equipped missions, to further assist in spaceflight safety and improve the design based upon lessons learned from the mission's success. It is shown that the combination of a drag sail module with a suggested passive stabilization technique ensures the reliability of the de-orbiting process of any drag sail equipped satellite, without additional risk or cost for other critical satellite subsystems.

In addition, this thesis investigates the design, testing and validation of a passive control solution for attitude death-mode avoidance; the solution includes the sizing and testing of a permanent magnet to be placed on-board the GHGSat-C1 and -C2 satellites delivered by SFL for the purpose of measuring and reporting greenhouse gas emissions. In line with the efforts of spaceflight safety and sustainable operations, the use of a passive attitude solution, combined with the highly innovative attitude control hardware and software is a perfect mix of innovation, performance, and reliability.

Chapter 2

CanX-7 Drag Sail Performance

This chapter provides necessary background information about the CanX-7 satellite, built by SFL. Furthermore, it includes an evaluation of its primary payload performance, a drag sail module used for safely de-orbiting the satellite back in Earth's atmosphere.

2.1 The CanX-7 Mission

The CanX-7 satellite is a 3U Cubesat developed by SFL and launched in September 26^{th} 2016 from the Satish Dhawan Space Centre in India. Its primary mission is to demonstrate the deployment, operation and performance of an innovative drag sail that is used to de-orbit the spacecraft from its nominal orbit at the end of its operational life. As a secondary payload, CanX-7 includes an ADS-B receiver, developed by the Royal Military College of Canada with software support from SFL. Its purpose is to provide space-based ADS-B telemetry for aircraft tracking. [11].

There is an abundance of literature and resources related to the CanX-7 satellite bus design, as well as payload performance. Preliminary analysis of the mission concept and trade-off studies during the early prototyping stages of the drag sail module can be found in [12]. Details about the mechanical design and the deployment of the drag sail can be found in [13]. Structural design analysis of the CanX-7 satellite are provided in [14]. For an overview of the thermal control design, as well as the specifics of the attitude control subsystem hardware and software, [15] is a thorough reference. Regarding the design, simulation, testing, and on-orbit performance of the aircraft traffic surveillance payload (ADS-B) aboard CanX-7, consult [16]. Additional simulations performed for the payload with respect to the receiver's power can be found in [17].

The CanX-7 satellite is currently operational, and all telemetry shows that it functions nominally. As shown in subsequent sections, the ADS-B payload has been successful in collecting AIS telemetry, while the drag sail has been highly effective in reducing the satellite's orbital altitude for more than four years, following its deployment. More details about the sail's performance are illustrated in the next sections.

2.2 The CanX-7 Satellite Platform

This section provides information about the design of the CanX-7 bus, more specifically the mechanical design and the attitude determination and control sensors and actuators.

2.2.1 Mechanical Design

The CanX-7 bus in its stowed configuration is illustrated in Figure 2.1 below. In this configuration, the satellite's deployable UHF dipole antennas are folded against the four faces. In addition, the satellite is equipped with a deployable boom, which houses an inspection camera for observing the drag sail deployment event, as well as monitoring the sails' health status at any point on-orbit. The boom also houses a magnetometer sensor. The placement of the magnetometer in that location (externally to the main satellite bus structure) allows for the collection of accurate measurements with minimal electromagnetic noise, which is normally produced within the structure due to payload operation, as well as cabling and solar panel wiring assembly (also known as solar string loops). This is critical in the design of the spacecraft, because the attitude determination suite is comprised by this magnetometer only, while the control law ($\dot{B}$, also known as B-dot) is using only those measurements to extrapolate the satellite's angular rates. In this stowed configuration, the magnetometer boom is folded against the panel of the $-Z$ axis.

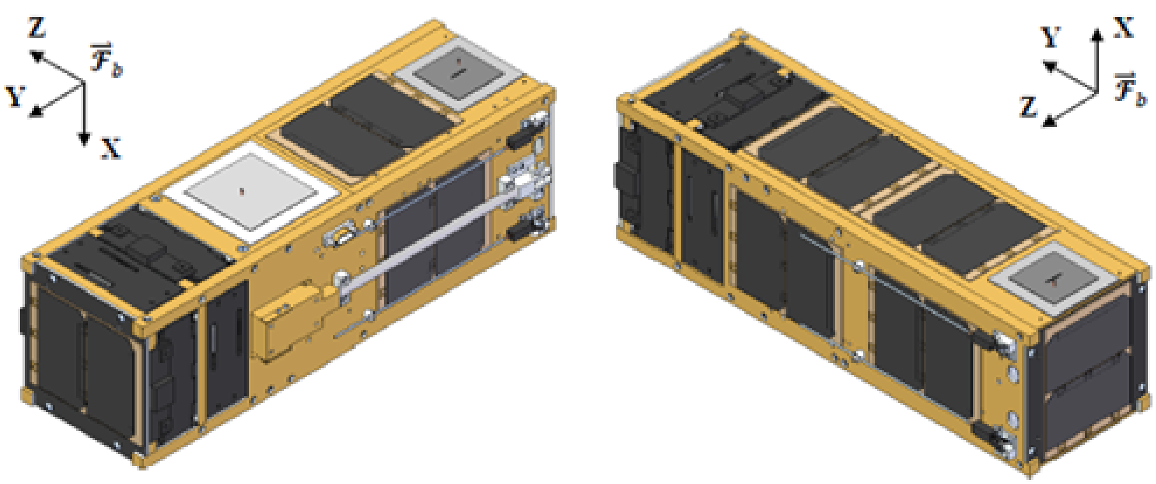

Figure 2.1: CanX-7 exterior view, stowed configuration [15]

The CanX-7 bus in its deployed configuration is shown in Figure 2.2. As discussed, the

UHF dipole antennas and the boom are the only deployable items of CanX-7.

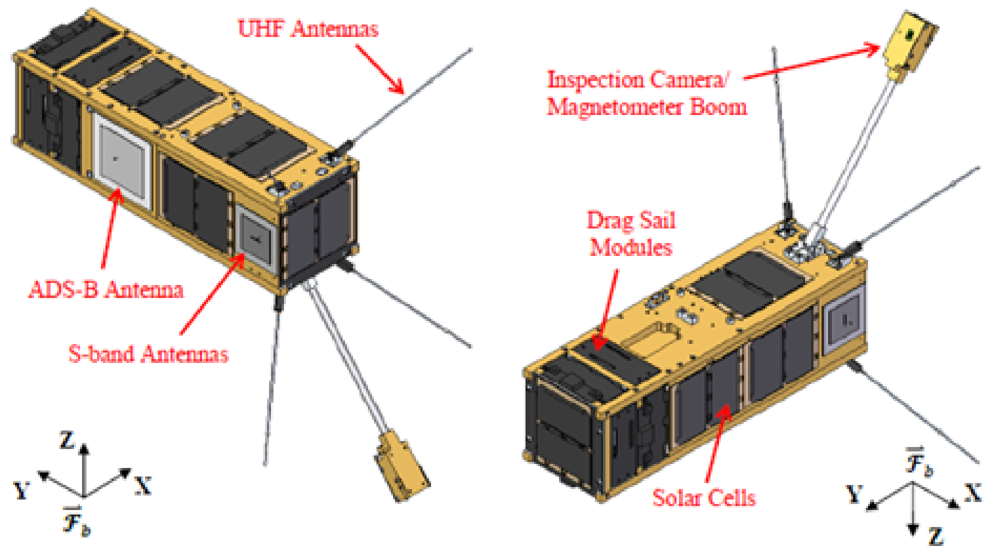

Figure 2.2: CanX-7 exterior view, deployed configuration [15]

The Y axis is the longitudinal axis across the longer dimension of the satellite, while the X and Z axes are the transverse axes of the satellite's coordinate system. This system is also known as the CanX-7 *body frame*, denoted as $\underrightarrow{\mathscr{F}}_b$. For a detailed description about the definition of a satellite's body frame, refer to Section 3.5. The drag sail modules are located at the top of the longitudinal $+Y$ axis of the spacecraft. An illustration of the drag sails in their deployed configuration is shown in Figure 2.3.

2.3 Attitude Determination and Control Hardware

The CanX-7 attitude determination suite comprises of a single three-axis magnetometer. As discussed, the magnetometer is installed on the deployable magnetometer boom, external to the spacecraft. This configuration minimizes spurious measurements of parasitic magnetism from the spacecraft's bus. Minimal measurement noise is critical for the improved performance of the control law. CanX-7 implements $\dot{B}$ control, which requires knowledge of the magnetic environment as well as the satellite's angular body rates in all axes. In order to extrapolate the body rate measurements from a time-series of magnetic measurements, the instrument needs to be as accurate as possible. With respect to actuators, CanX-7 uses three magnetorquers (vacuum core magnetic coils), mounted orthogonally with respect to the spacecraft mechanical structure, and aligned with the body frame $\underrightarrow{\mathscr{F}}_b$ as defined in Figures 2.2 and 2.1 above [15].

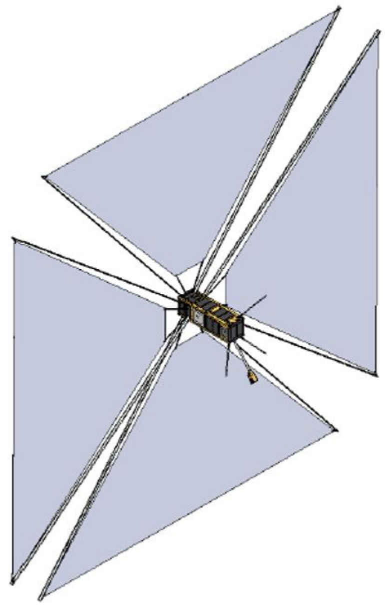

Figure 2.3: CanX-7 with its drag sails deployed [15]

2.4 On-Orbit Behavior of CanX-7

After the successful completion of its main mission to demonstrate the operation and performance of the ADS-B receiver, CanX-7 deployed its sail modules on May 3rd 2017. The deployment was observed through the satellite's telemetry, as well as from the ground. The deployment of the sails resulted in an increase of the spacecraft's projected surface area to the orbital velocity vector, translating to a decrease of the satellite's ballistic coefficient. Since then, the satellite's Semi-Major Axis (SMA) has been decreasing at an accelerated rate. This is shown below in Figure 2.4, which is derived and processed from historical Two Line Element information from Space-Track.org. In the same figure, the deployment time of the drag sail is marked by a dashed vertical line.

Figure 2.4 shows that, since the time of deployment, CanX-7 has reduced its semi-major axis by almost 100 kilometers over the course of a little more than four years. It is also clear that the altitude is reduced at a non-linear rate, reassembling more of an exponential decrease. This is expected due to the roughly exponential increase of atmospheric density as the orbital altitude is reduced. As more data becomes available from CanX-7, the orbital altitude decrease is expected to fit closer to an exponentially decreasing curve. Considering this pattern, it is expected that the de-orbiting CanX-7 will accelerate over time. Such findings solidify the success of the mission with every passing day and prove that SFL's drag sail technology is a solution towards space debris mitigation with design-by-disposal practices for a family of orbits within the LEO regime. It also proves that the satellite is expected to de-orbit well before the recommended 25-year IADC guideline.

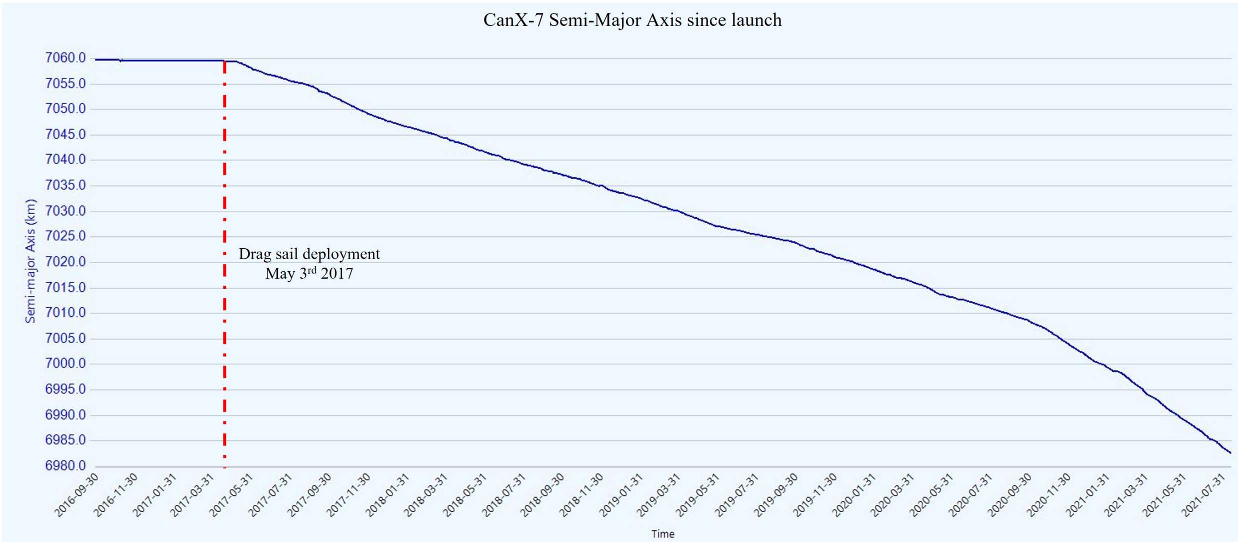

Figure 2.4: CanX-7 semi-major axis since launch

2.5 CanX-7 Tumbling

During the first months after the deployment of the drag sail modules, CanX-7 occasionally demonstrated spin-ups of its angular velocity. Those spin-ups would occur with a maximum frequency of once per month. In their most aggressive form, the increase in the magnitude of the angular velocity was up to 1 deg/sec per day. The magnetorquers on-board of CanX-7 are sufficiently capable of cancelling out disturbances of that magnitude. However, it is of interest to examine the reasons as to why those spin-ups may have occurred. In the future, the intention is to use the drag sails as an end-of-life deployable passive solution, without the need to exercise means of active attitude control.

Figure 2.5 below illustrates the increase of angular velocity of CanX-7 between July 13$^{\text{th}}$ and July 22$^{\text{nd}}$, 2017. Some noise in the data is present due to measurement noise of the magnetometer, as well as process noise in the computation of the angular rates based on magnetometer measurements. On July 19$^{\text{th}}$, the magnetorquers on CanX-7 were used to detumble the spacecraft, illustrated by a sudden decrease in body rates.

By examining the top plot in Figure 2.5, we note that the spin-up occurs along the Z axis, marked in yellow color. In the same plot, the X-axis body rate is marked in blue color, and the y-axis body rate is marked in red color. Recall that the Z axis is one of the transverse axes of CanX-7 body frame $\mathscr{F}_{\text{b}}$. This means that the satellite is not spinning about the longitudinal axis; this is an important observation from the telemetry. Within the same figure, the initiation of the spin-up as well as the detumbling command are labeled, following the corresponding increase and decrease of the satellite's body rates. In addition, we note that there are periods where no data is available for the body rates. Recall that the body rates are computed based solely on magnetometer measurements. The lack of data is attributed to attitude singularities related to the magnetometer measurements. More specifically, these singularities are formed during periods of time when the local magnetic field vector of the spacecraft forms a small angle with the angular velocity vector of the spacecraft. When that angle becomes sufficiently small, the algorithm that extrapolates body rates from magnetometer measurements would encounter singularities, thus being incapable to determine the body rates correctly. The bottom part of Figure 2.5 illustrates the magnitude of the angular velocity over time, marked in blue color.

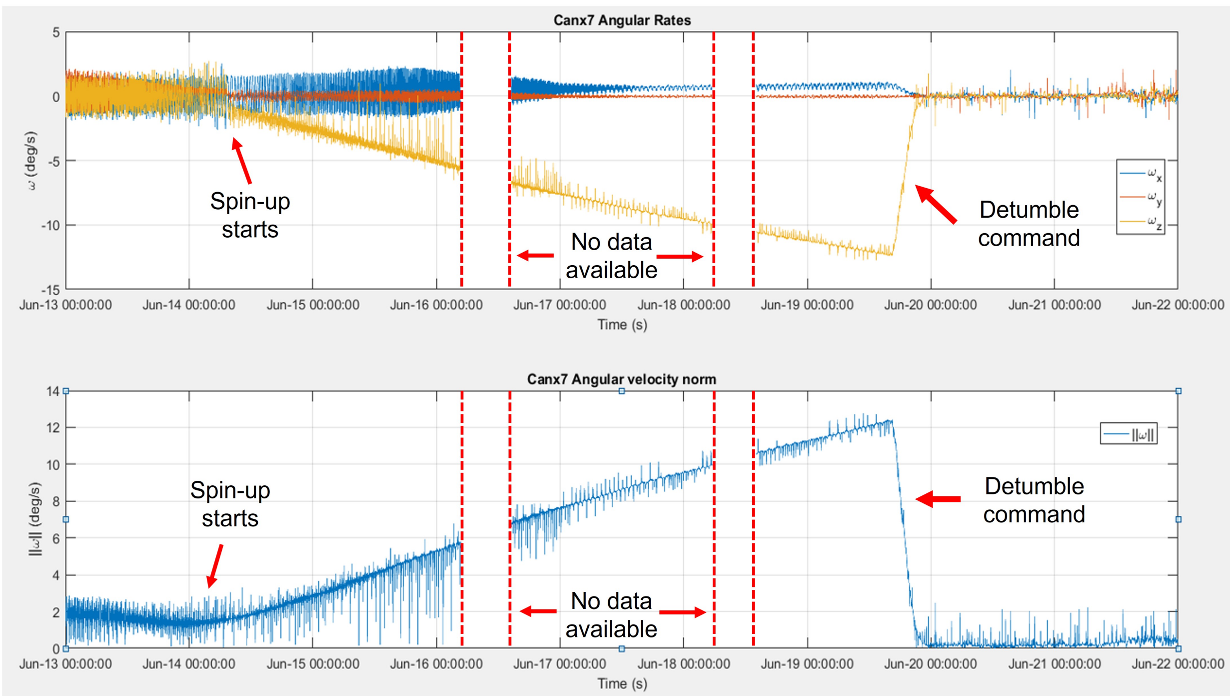

Figure 2.5: CanX-7 body rates from July 13$^{\text{th}}$ 2017 to July 22$^{\text{nd}}$ 2017

2.6 Permutation Spin-Up Analysis

The spin-up behavior of CanX-7, as observed in 2.5, poses a unique opportunity to examine its underlying reasons. In this section, we will demonstrate how the spin-up can be replicated through simulation and variation of key design parameters. Furthermore, we demonstrate that the variation of the same design parameters can lead to a potential spin-up on other popular SFL platforms, such as a NEMO class platform that is equipped with drag sails. The motivation is to identify a set of initial conditions that lead to a linear increase in angular velocity over time, similarly to the observed -on-orbit behavior. With this set of initial conditions, in Chapter 3 we will examine the performance of passive stabilization solutions that can provide rate-dependent damping of the satellite's rotation, through the installation of soft ferromagnetic rods in the satellite structure.

According to Figure 2.3, recall that CanX-7 satellite has a small mass and a large area with the drag sails deployed. Because of this small-mass to high-area ratio, examining the causes of a spin-up in this platform can define a worst-case scenario for satellites of larger mass (and moment of inertia), with equal or smaller number of drag sails installed. Even though there is no need for a mitigation plan, given that CanX-7 is able to easily cancel out the occasional spin-up. investigating low risk stabilization solutions to complement the drag sail modules would be an improvement for future missions. In this section, a permutation analysis for the possible causes of spin-up is investigated. The permutation analysis was performed for both CanX-7 as well as for NEMO class spacecraft. In the case of NEMO class spacecraft, NorSat-3 was chosen as the sample mission to analyze. NorSat-3 is an SFL spacecraft built for the Norwegian Space Agency and equipped with a Navigational Radar Detector to assist with maritime awareness operations [18] [19]. The spacecraft was successfully launched on April 29$^{\text{th}}$ 2021. The satellite was not equipped with drag sail modules, yet this analysis formed part of the mission design study, up to the point of the Critical Design Review.

As noted earlier, the observed spin-up of CanX-7 is relatively fast in nature, considering the low magnitude of environmental disturbances. Under the practical assumption that sudden changes in angular velocity should be the result of the highest environmental disturbance in magnitude, we will explore the various combinations with which the satellite orientation and geometry can trigger a spin-up event. At a high-level, the parameters of most interest for such a condition are the initial body rates of the spacecraft, the initial attitude of the bus with respect to the orbital velocity vector, as well as the relative orientation of each drag sail with respect to the spacecraft bus. An exhaustive brute force method that evaluates the increase of the body rate norm by randomly selecting values for the above parameters within a certain parameter search space is computationally inefficient. Therefore, a number of corner cases are investigated on the extremes of the parameter search space, based on logic and intuition, while being mindful about computational efficiency. The variety of parameter sets are known as *permutations* , since they are programmatically indexed in such a way that their order matters within the input function of the simulation. The definition of those parameter sets follows.

In general, the number of possible k ways to select r things, each having n possibilities, is defined by:

$$k = n^r \tag{2.1}$$

This is applicable to parameter sets that can allow for repetition of values for every selection. In this analysis, the following three parameters were allowed to vary:

- *Initial Body Rates* $\boldsymbol{\omega}_\mathrm{b}$ - We define $\boldsymbol{\omega}_\mathrm{b}$ as the column matrix representing the satellite's angular velocity vector with components $\boldsymbol{\omega}_\mathrm{b} = \begin{bmatrix}\omega_1 & \omega_2 & \omega_3\end{bmatrix}^T$, expressed in the satellite's body frame $\mathscr{F}_\mathrm{b}$. ω_1 corresponds to the satellite's angular velocity about the $+X$ axis of $\mathscr{F}_\mathrm{b}$, ω_2 corresponds to the satellite's angular velocity about the $+Y$ axis, and ω_3 corresponds to the satellite's angular velocity about the $+Z$ axis. Each component of $\boldsymbol{\omega}_\mathrm{b}$ ($r_\omega = 3$) was allowed a value of -1, 0 or 1 *deg/sec* ($n_\omega = 3$). The total number of permutations corresponding to variations in body rates k_ω is therefore:

$$k_\omega = {r_\omega}^{n_\omega} = 3^3 = 27 \tag{2.2}$$

  The value of 1 *deg/sec* per axis as the extremum of the search space was heuristically chosen, considering that it is a small value yet well within the rate control capabilities of any actively controlled SFL spacecraft.

- *Initial Satellite Attitude* - We define $\boldsymbol{\theta}_\mathrm{o}$ as the column matrix representing the initial attitude of the spacecraft, with components $\boldsymbol{\theta}_\mathrm{o} = \begin{bmatrix}\theta_1 & \theta_2 & \theta_3\end{bmatrix}^T$, expressed in the satellite's Local Vertical Local Horizontal (LVLH) frame $\mathscr{F}_\mathrm{o}$. For details about the exact realization of the LVLH frame, please consult Section 3.5 in Chapter 3. First, consider the satellite body frame $\mathscr{F}_\mathrm{b}$ initially aligned with the satellite's LVLH frame $\mathscr{F}_\mathrm{o}$. Each component of $\boldsymbol{\theta}_\mathrm{o}$ corresponds to a principal rotation of the satellite's body axis away from the corresponding LVLH axis with which it was aligned, following a rotation sequence of 1-2-3 Euler angles. For example, θ_1 represents a principal rotation $\mathbf{C}_1(\theta_1)$ of the satellite's $+X$ axis by θ_1 degrees away from the LVLH frame $+X$ axis. The number of allowable attitudes is six, so the number of initial attitude permutations k_θ is:

$$k_\theta = 6 \tag{2.3}$$

  The initial attitudes are:

  - $\begin{bmatrix}0 & 0 & 0\end{bmatrix}$ - This is the default attitude. The satellite's body frame $\mathscr{F}_\mathrm{b}$ is aligned with the LVLH frame $\mathscr{F}_\mathrm{o}$
  - $\begin{bmatrix}90 & 0 & 0\end{bmatrix}$ - In this attitude, the satellite's body frame is rotated about its $+X$ axis by 90 degrees. Positive direction of rotation follows the right-hand rule.
  - $\begin{bmatrix}180 & 0 & 0\end{bmatrix}$ - The satellite's body frame is rotated about its $+X$ axis by 180 degrees.
  - $\begin{bmatrix}270 & 0 & 0\end{bmatrix}$ - The satellite's body frame is rotated about its $+X$ axis by 270 degrees.
  - $\begin{bmatrix}0 & 0 & 90\end{bmatrix}$ - The satellite's body frame is rotated about its $+Z$ axis by 90 degrees.

- $\begin{bmatrix} 0 & 0 & 270 \end{bmatrix}$ - The satellite's body frame is rotated about its $+X$ axis by 270 degrees.

The permutation space is limited to these attitudes because they capture a complete rotation about each body axis in 90 degree increments, with the exception of the satellite Y axis. It is reminded that the Y axis is the longitudinal axis for CanX-7 (see Figure 2.2), therefore any amount of principal rotation about the Y axis has no effect on the aerodynamic force, since the projected sail area would only change in orientation, but with the same magnitude in the end. Also, the principal rotation of the Z axis about 180 degrees is omitted because it is effectively the same as the default attitude, as far the projected effective drag area is concerned.

- *Drag Sail Twist* - Each of the drag sail modules (In the case of NEMO class spacecraft $r = 2$, and in the case of CanX-7 $r = 4$) was allowed to rotate about the spacecraft axis on which it was installed by a value of 1 or -1 degrees ($n = 2$). The number of permutations k_t corresponding to sail twist, for a total of two sails installed (in the case of NorSat-3) is therefore:

$$k_t = {r_t}^{n_t} = 2^2 = 4 \tag{2.4}$$

  The NEMO class design that is considered for this study called for the installation of two drag sail modules instead of four, as in the case of CanX-7. In order to visualize the effect of twist on the overall spacecraft geometry, the CanX-7 and NorSat-3 simplified structures in their deployed configuration are shown in the figures below.

Figure 2.6 shows a simplified model of the CanX-7 structure with the sails deployed, without any sail twist applied, and serves as a visual reference. The small red dots represent the center of pressure for each of the satellite surfaces. The center of pressure is the point of application of an aerodynamic force on that surface. In this model, all drag sails as well as all satellite faces are simulated to contribute to the aerodynamic disturbance torques.

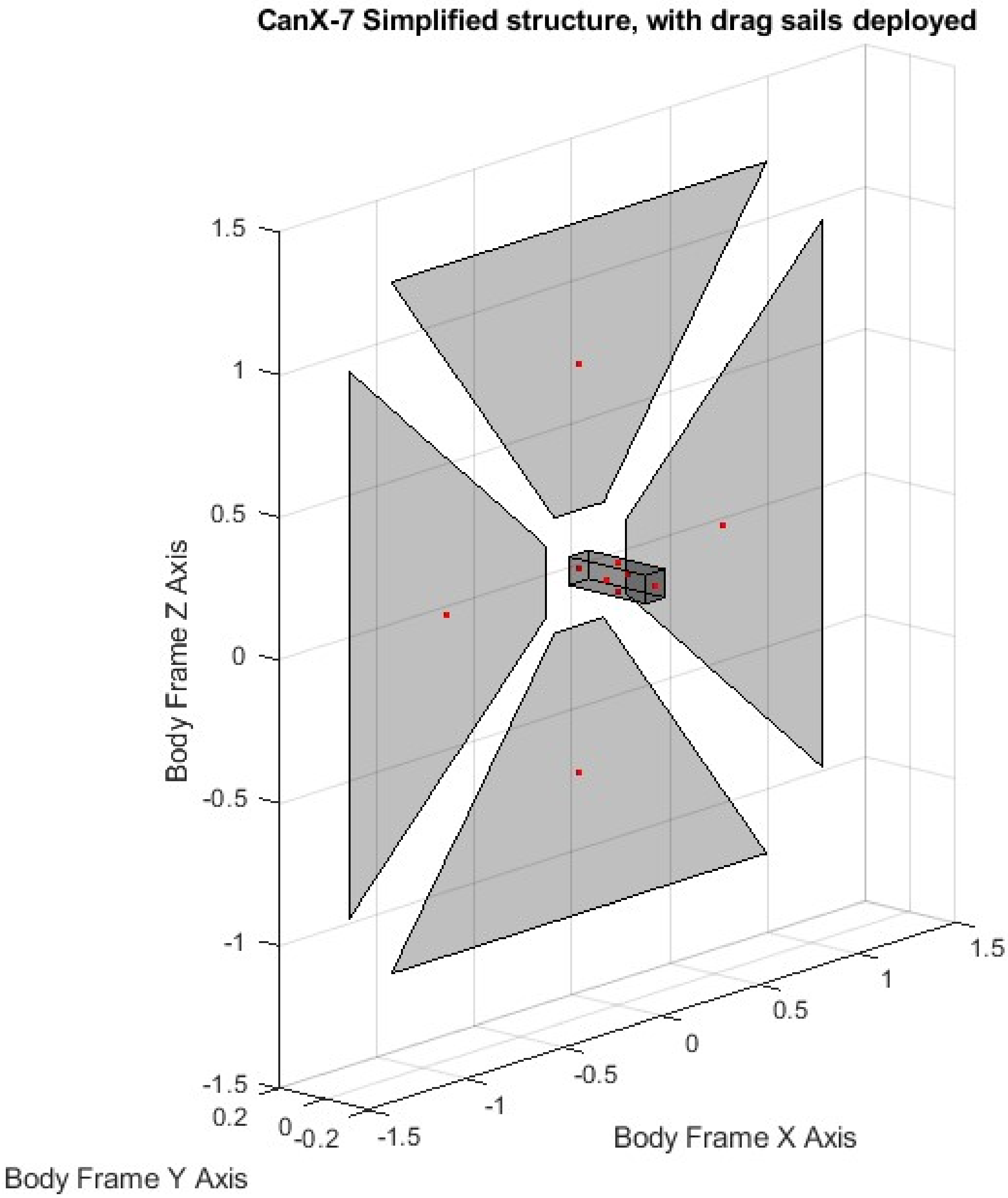

Figure 2.6: Simplified structure of CanX-7 with the sails deployed

Figure 2.7 below shows a simplified model of the NorSat-3 structure with the sails deployed and slightly twisted. In this figure, the red dots represent the points of pressure of each surface, and the blue dot in the middle of the structure represents the location of the satellite's Center of Mass. Note that the special Navigation Radar Detector (NRD) antenna of NorSat-3 is also considered and simulated as an aerodynamic surface. A side view of the same structure is illustrated in Figure 2.8. The purpose of this visualization is to show the direction of twist. One of the sails is twisted clockwise with respect to the body $+X$ axis, while the other sail is twisted counterclockwise with respect to the body $+X$ axis, by an equal amount (5 degrees in this case, for illustration purposes), thus reassembling a "windmill" configuration.

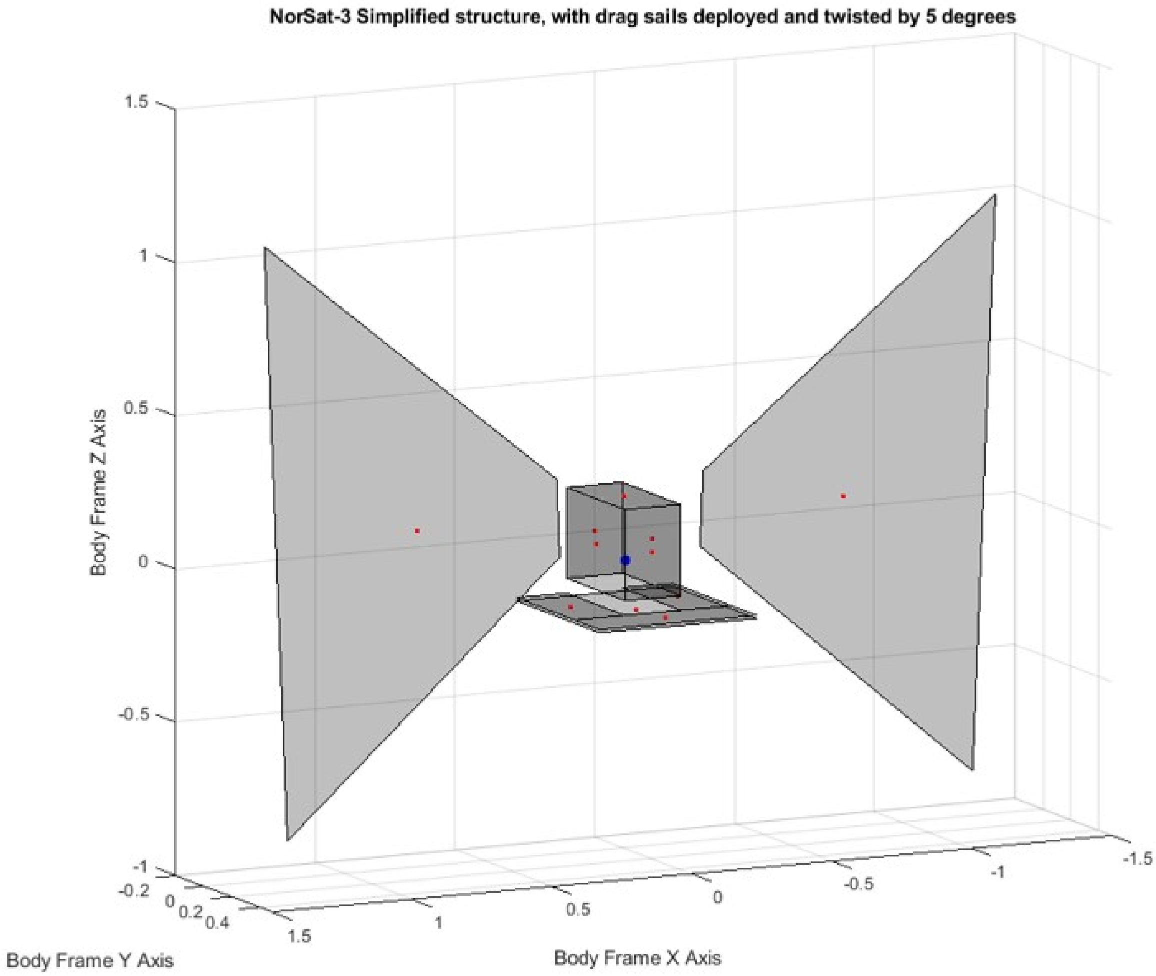

Figure 2.7: Simplified structure of NorSat-3 with the sails deployed and twisted

The visualizations in figures 2.7 and 2.8 illustrate the twist sail with an exaggerated rotation, for clarity. This variation can be envisioned as a "twist" of the sail modules, considering the somewhat flexible nature of their geometry. The twist of the sails is also a way to aggregate and represent the effect of differential billowing of the sails that can lead to rotation through solar pressure, or any other unmodeled effect. This twist offset is assumed to be constant and rigidly fixed throughout the simulations. Assuming any flexibility related to the drag sails would require treatment of flexible dynamics and integration of those equations of motion in the simulation, which is outside the scope of this thesis.

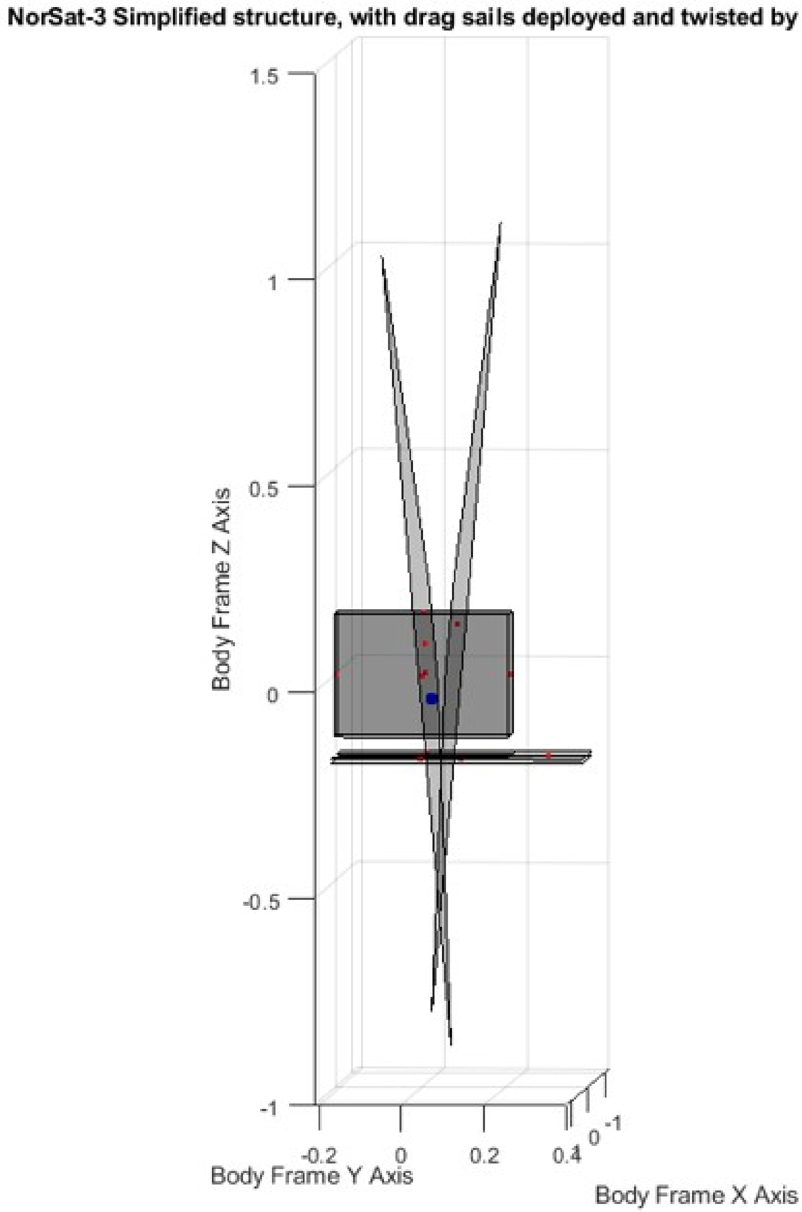

Figure 2.8: Side view of NorSat-3 simplified structure with the sails deployed and twisted

The total amount of permutations is computed as the product of all the above permutations from each of the three parameters that are allowed to vary. Therefore, the total number of simulations s is:

$$s = k_{\omega} \cdot k_{\theta} \cdot k_{t} = 27 \cdot 6 \cdot 4 = 648 \ simulations \tag{2.5}$$

The permutation method ensures that all corner cases are captured, without the need for a more aggressive brute-force randomization approach. In order to evaluate which of those cases yields the highest spin-up, the histogram of the angular norm difference at the beginning and end of the simulation is plotted, shown in Figure 2.10. We define the *angular velocity norm difference* as the difference between the magnitude of the satellite's angular velocity at the start of the simulation and the magnitude of the satellite's angular velocity at the end of the simulation. Each of the simulations will generate a single value of the angular velocity norm difference, making it easy to handle as a statistical figure of merit and therefore draw a histogram.

With 648 simulations as our sample and considering the CanX-7 bus first, we identify one case where the increase in the angular velocity norm is noticeably higher than the rest. For CanX-7 ,this case comprises of the following parameters:

- Initial body rates $\boldsymbol{\omega}_{\mathrm{b}}$ expressed in Body Frame $\mathscr{F}_{\mathrm{b}}$: $\boldsymbol{\omega}_{\mathrm{b}} = [0\ ,\ 1,\ 0]\ deg/sec$
- Initial attitude in 1-2-3 Euler Angles with respect to the LVLH frame $\mathscr{F}_{\mathrm{o}}$: [90,0,0] degrees.
- Drag sail twist: -1 degree for the +X sail and +1 degree for the –X sail, similarly to the orientation shown in Figure 2.8 for NorSat-3.

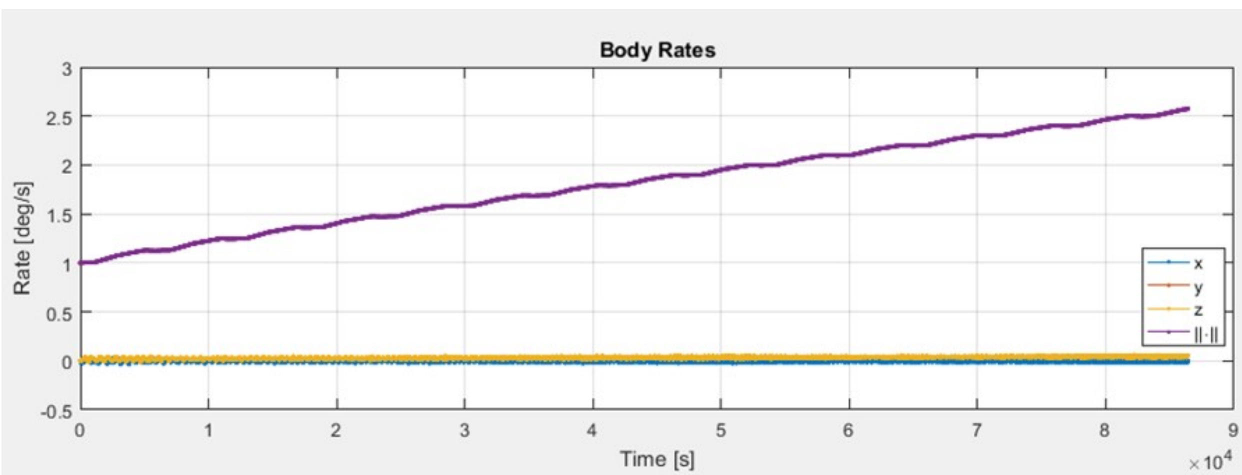

Figure 2.9: CanX-7 angular rates of the worst-case simulation parameter set (one day simulation)

Figure 2.9 illustrates the CanX-7 body rates during the simulation. It is noted that the increase of the angular velocity is linear and approximately 1 deg/sec per day. In addition, it is observed that the spin occurs and is maintained almost exclusively about the z-axis, which is the same axis that was observed to spin in the telemetry of CanX-7 in Figure 2.5. Recall that this axis *is not* the longitudinal axis of CanX-7, but it is one of the two transverse axis of the rectangular bus structure, which is an axis perpendicular to the sails. This matches the telemetry and the simulation for the CanX-7 spin up.

With 648 simulations as our sample and considering the NorSat-3 bus structure this time, we identify one case where the increase in the angular velocity norm is noticeably higher than the rest. This case is identified by first aggregating the angular velocity norm difference of all 648 simulations, and plotting a histogram, shown in Figure 2.10. The case that generated the largest spin-up resides in the right most bin of the histogram and is comprised of the following parameters:

- Initial body rates $\boldsymbol{\omega}_\mathrm{b}$ expressed in Body Frame $\mathscr{F}_\mathrm{b}$: $\boldsymbol{\omega}_\mathrm{b}$ = [1 , -1, -1] *deg/sec*
- Initial attitude in 1-2-3 Euler Angles with respect to the LVLH frame $\mathscr{F}_\mathrm{o}$: [0,0,90] degrees
- Drag sail twist:: -2 degree for the +X sail and +2 degree for the –X sail.

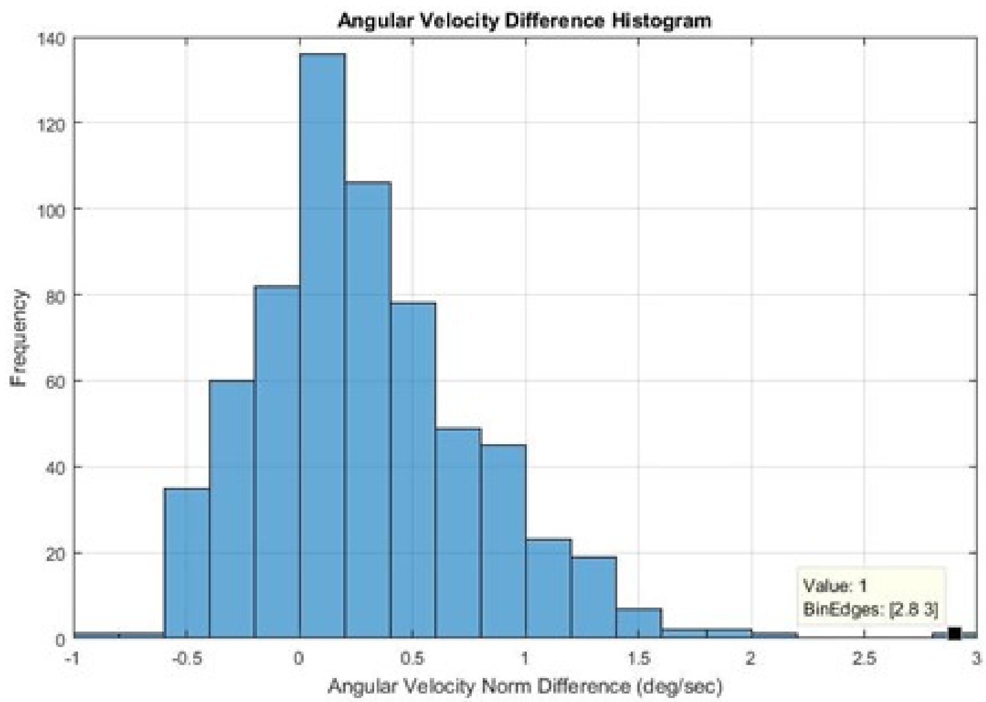

Figure 2.10: Angular Velocity Difference Histogram for the permutations simulated for NorSat-3 spin-up

The body rates of NorSat-3 for a simulated period of one day are shown in Figure 2.11 below. After repeating the simulation for a longer duration but with the same initial conditions, the spin up behavior continued unbounded for many days, resulting in high angular rates with no signs of exiting that condition. The spin-up is of the order of approximately 2 *deg/sec* per day, which is about double of the equivalent on-orbit behavior of CanX-7. The latter set of parameters is the baseline for assessing the performance of a passive stabilization solution in Chapter 3. Although somewhat aggressive, the above spin-up configuration of 2 *deg/sec* per day will provide some safety margin for any uncertainties or omitted varying parameters, when sizing the passive solution.

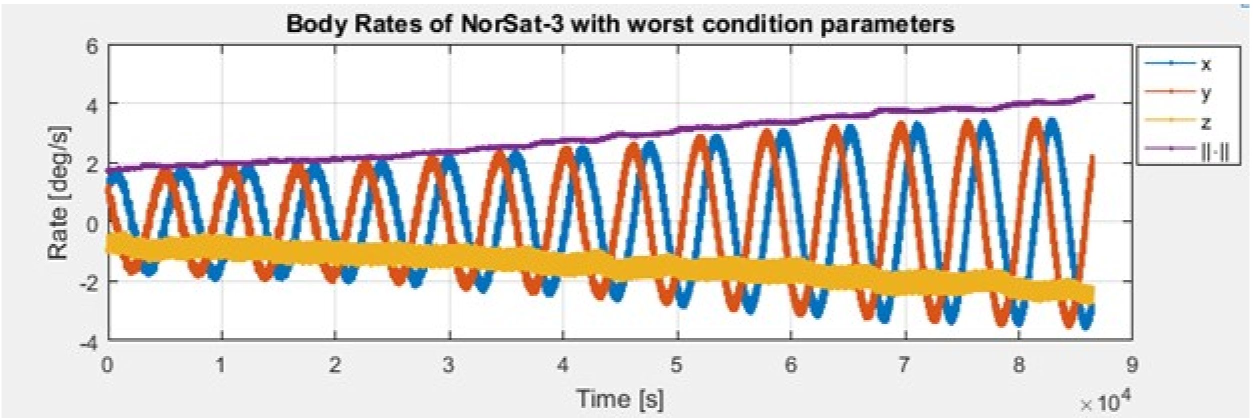

Figure 2.11: NorSat-3 body rates of the worst-case simulation parameter set (one day simulation)

At this point, an important remark needs to be made. As the satellite will be progressively reaching lower altitudes over time, the aerodynamic drag will become stronger, and it is expected that it will act as a "shuttlecock" such that the sails will allow it to stabilize aerodynamically. If at some point the satellite is exposed to the aforementioned unfavourable conditions, then the spin-up might potentially get stronger at lower altitudes. Therefore, any attempt to stabilize this motion needs to explore solutions that provide dissipative torque, rather than other forms of disturbance torque that might "fight" against this aerodynamic stabilization behavior. For example, within the realm of passive magnetic solutions, the addition of a permanent magnet on the satellite would not be an advisable strategy, since the magnetic disturbance torque would force the spacecraft to diverge from an aerodynamically stable orientation and end up to an orientation that wants to align the dipole moment of the magnet with the Earth's magnetic field vector. A more adequate course of action would be the installation of soft ferromagnetic rods that do not carry a fixed dipole moment but rather work based on the principle of magnetic hysteresis. In this case, the rods would provide a dissipative rate dependent torque that *resists changes in rotational motion*, eventually leading to passive stabilization, if sized properly. The beneficial effects of magnetic hysteresis and how well it synergizes with drag sail equipped satellites is discussed in the next chapter.

Chapter 3

Passive Stabilization with Hysteresis Rods

This section expands on the author's original contributions on sizing, testing and validating a passive stabilization solution that can help in mitigating a tumbling condition due to the use of drag sail modules. The solution involves the installation of thin cylindrical rods made out of soft ferromagnetic materials that exhibit magnetically hysteretic behavior. This addition introduces a rate-dependent, magnetically induced dissipative torque to detumble a satellite and continuously provide stabilizing damping.

3.1 Fundamentals of Magnetic Hysteresis

When a ferromagnetic material is exposed in a varying magnetic field, the magnetic domains within the material undergo a series of changes. The magnetic domains are volumetric regions within the material that possess a common, distinct magnetic dipole in both magnitude and direction [20]. Magnetic domains are analogous to the grains found within a solid material's micro-structure. Any magnetic domain region, theoretically, can be as small as the Bohr magneton, the fundamental magnetic quantity. The changing external field can enlarge/shrink the domains by displacement of their boundaries, rotate the domains to align with the direction of the applied field (i.e. easy magnetic axis), or eliminate part of a domain wall, thus forcing the domain to join an adjacent one. Some of these processes are reversible and some others irreversible, but all of those mechanisms have something in common; they lead to energy losses in the form of heat. It is important to note that the ferromagnetic material does not react instantaneously to the externally applied varying magnetic field; it rather follows the change with a time delay, generating a magnetic flux though its volume that is different from the externally applied field. The time delay with which the material reacts to the changing magnetic field gives rise to the definition of *magnetic hysteresis*. Under these conditions, the material will exert a torque that will try to *resist the change* in the magnetic flux due to the changing magnetic field. The magnitude of the generated torque depends on the frequency and amplitude of the externally applied field, as well as the dimensions and other properties of the ferromagnetic material.

More specifically, a ferromagnetic material exhibiting hysteretic behavior is characterized by a sigmoid (i.e. arctangent) relationship between the external magnetic field strength and the induced magnetic flux density through the material's volume. This relationship is empirically summarized in the form of the characteristic *hysteresis major loop*, which is the region enclosed in the field-flux plot (H-B plot) bounded by this sigmoid relationship. An example of a major loop is illustrated in Figure 3.1 below. The horizontal axis of the H-B plot captures the value of the externally applied magnetic field strength, while the vertical axis captures the induced flux density within the soft ferromagnetic medium. It is noted that the term "soft" relates to the material's ability to *change* its dipole moment, *with a delay*, based on the direction and magnitude of the externally applied magnetic field. In contrast to permanent magnets (ferromagnetically "hard") whose dipole moment does not change expect when exposed to extreme temperatures or extreme magnetic fields, a soft hysteresis material will get magnetized and demagnetized based on the magnetic field that is exposed to. In the absence of a magnetic field, a soft ferromagnetic material will eventually demagnetize, which is not true for hard ferromagnetic materials that retain their dipole moment (i.e. they are more *coercive*).

The material's hysteretic behavior is bounded to reside within the limits of the characteristic major loop. The two outer sigmoid curves that form the major loop are known as the *right* (ascending) and *left* (descending) boundaries. Knowledge of the closed-form expression of the major loop is crucial for phenomenological modelling. It is important to note that different parts of the sigmoid curve correspond to and are dominated by a different magnetization process. For more details about the different magnetization stages of the hysteresis loop, consult [21], and [22]. For highly accurate modelling of hysteresis, each magnetization process is modelled separately, and the total magnetization is computed as the sum of all contributions from each process. The latter is outside the scope of this thesis.

Flattley and Henretty [23] have empirically demonstrated that the major hysteresis loop of a material is well approximated by an arctangent function. This arctangent model of ferromagnetic hysteresis is used for the simulation and analysis in subsequent sections.

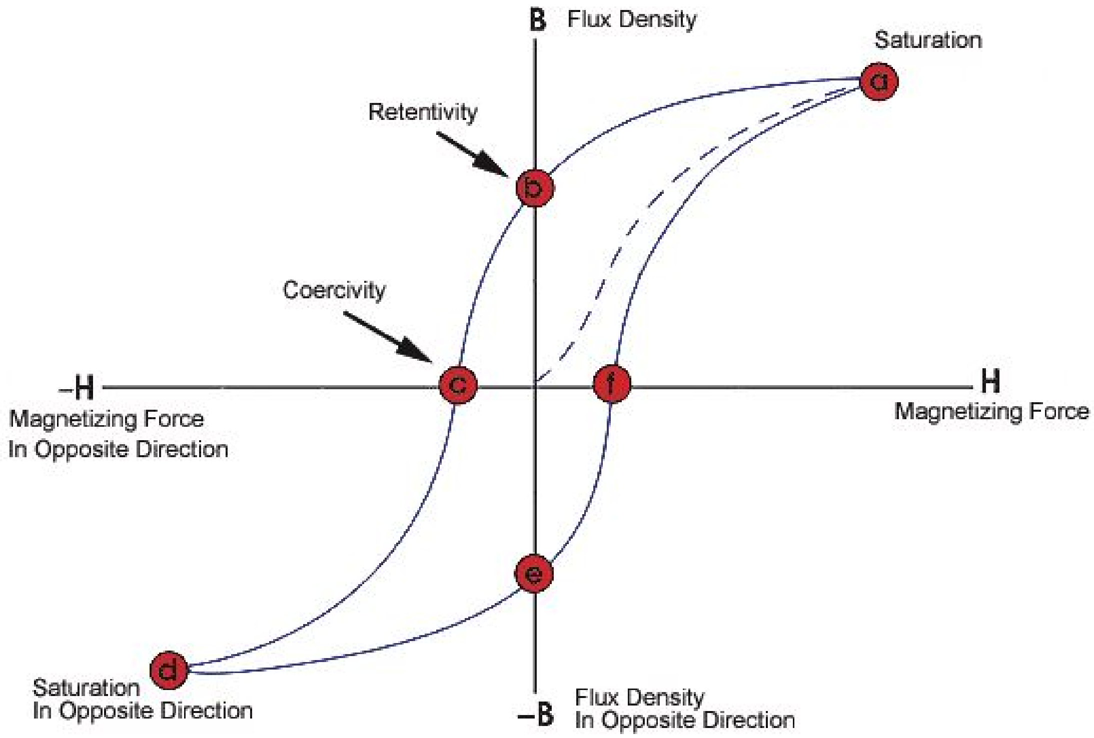

Figure 3.1: Example of a typical hysteresis major loop, characteristic of a soft ferromagnetic material [24]

3.2 Literature Review on Spacecraft Stabilization with Magnetic Hysteresis

The first satellites equipped with rods of soft ferromagnetic material to passively detumble was TRANSIT-1A and TRANSIT-1B [25]. In recent years, hysteresis rods have been used in the design of passive magnetic attitude control systems for CubeSats. A list of satellites that have used hysteresis rods as part of their attitude control system is provided in [21].

The empirical model of [23] is formulated by comparing and fitting simulated data with experimental data collected from a specific rod and test setup configuration. Kumar, Mezanej and Heck [25] later use the same model for the development of an attitude dynamics simulator. Santoni and Zelli [26] have experimentally determined the magnetic properties of hysteresis rods made of a specific nickel-iron alloy (*HyMu 80®*), presenting a methodology for modeling some of the hysteresis parameters of a rod based on its physical specifications, length and radius, rather than ideal material properties. Park, Seagraves and McClamroch[27] provide analytical expressions and the equations of motion for a satellite equipped with hysteresis rods as well as a permanent magnet, based on the work of [23] and [25] above.

With regards to testing the hysteresis rod efficiency, Santoni and Zelli [26], [28], Ivanov [29] and Gerhardt [30] all present different experimental setups. Of particular importance is the setup of the latter, making use of a Helmholtz cage to provide isolation, while also testing the effect of *mutual magnetization* when more than one rod is installed along a spacecraft axis.

With respect to the theoretical framework of hysteresis and domain theory, a qualitative approach is given by [31], while [21] and [22] also provide insight to the fundamental mechanisms that drive hysteretic behavior.

Finally, considering the post-launch performance verification of systems equipped with hysteresis rods, there are noticeable gaps reported between the expected and the actual passive stabilization results on-orbit. The Delfi-C3 cubesat required 60 days to achieve passive stabilization, while the simulated results were predicting detumbling in 1 orbit [32]. RAX-2 also witnesses a discrepancy between the expected and the actual stabilization time that was required for detumbling [33]. Much of these discrepancies seem to stem from poor characterization of the hysteresis rods, as well as magnetization of the rods due to the presence of a permanent magnet within the same satellite bus, mutual magnetization between multiple hysteresis rods, and the presence of other unmodelled sources of parasitic magnetism.

3.3 Simplification and Assumptions

The following study assumes that the hysteresis rods are mounted parallel to each of the spacecraft's orthogonal axes. This simplifies the rotation algebra required by eliminating one additional transformation from the spacecraft's body frame to the hysteresis rod's frame. This way, the latter frame does not need to be instantiated.

With regards to the use of multiple hysteresis rods per satellite body axis, the issue of mutual magnetization between the rods arises. Gerhardt and Palo [34] show experimentally that mutual magnetization between rods is insignificant when rods are perpendicular to each other. However, the phenomenon is significant when the rods are mounted parallel to each other and at a distance of less than 30%-40% of their length. This leads to decreased overall dipole efficiency (dipole moment per volume of hysteresis material). Considering that each rod needs to have the longest allowable possible length (roughly 150mm for NEMO class spacecraft) for maximum dipole efficiency, additional space limitations arise to fit the rods inside the spacecraft bus, as well as mass budget limitations. For this study,it is assumed that one rod per spacecraft body axis is used, for a total number of three rods mounted orthogonally within the spacecraft. It is shown that this simple design is sufficient to meet the detumbling requirement. It is noted that the requirement for this analysis is the cancellation and reduction of angular velocity of a satellite with the configuration provided at the end of Chapter 2. No quantitative requirement has been established for the rate of detumbling due to the addition of hysteresis rods. However, a reasonable quantitative requirement is the goal to cancel out the spin-up behavior of a satellite equipped with drag sails, as explained in Chapter 2. Specifically for the spin-up study of NorSat-3, this requirement would be to size the hysteresis rods such that the spin-up conditions of linearly increasing body rates of 1 deg/sec per day are effectively nullified to zero body rates. Within this chapter, it is shown that this goal is successully met.

When studying the phenomenon of hysteresis in ferromagnetic materials, the frequency dependent losses due to Eddy currents should also be taken into consideration. When the magnetizing field is changing quickly (i.e. the external magnetic field is not *quasi-static*), it will change the shape of the hysteresis loop due to the opposing effect of the induced field in the material. The magnetic domains, during the process of aligning their dipole with the externally applied magnetic field, induce Eddy currents which oppose the change of the magnetic field in the material. This leads to reduced effectiveness of the applied external excitation, leaving Eddy current losses to dominate, as the frequency of the magnetizing field increases. The problem was first examined theoretically by Shockley, Williams and Kitttel (Williams et al 1950; Cullity, 1972), while Bertotti (Bertotti, 1998) examines the problem with statistical principles in mind. An analytical calculation of the Eddy Current losses is found in [22]. The algorithm has been integrated in simulation packages such as the open-source *Modelica Standard Library* and its effect can be quantitatively evaluated. A simulation example in *Open Modelica* that illustrates the effect of the frequency on the shape of the hysteresis curve is shown in Figure 3.2 below. The figure illustrates the hysteresis response of an soft ferromagnetic inductor when exposed to a sinusoidal magnetic field, at three distinct frequencies of 0, 10 and 100 Hz. As the frequency of the sinusoidal field increases, Eddy current losses dominate, leading to a drastic change in the hysteretic response of the material. This behavior is very important because it provides us with the important insight that testing of soft ferromagnetic materials needs to be performed at very slowly alternating magnetic fields. The Earth's magnetic field, in combination with the relative orbital motion of a satellite around the Earth and the satellite's attitude leads to the satellite's hysteresis rods being exposed a very slowly changing magnetic field. Therefore, the test equipment that needs to be used to characterize the hysteresis rod properties needs to produce a very slowly changing magnetic field with a period that is of the same order of magnitude as the combined effects of frequency of rotation (i.e. and by extension angular velocity) of the satellite and its orbital motion.

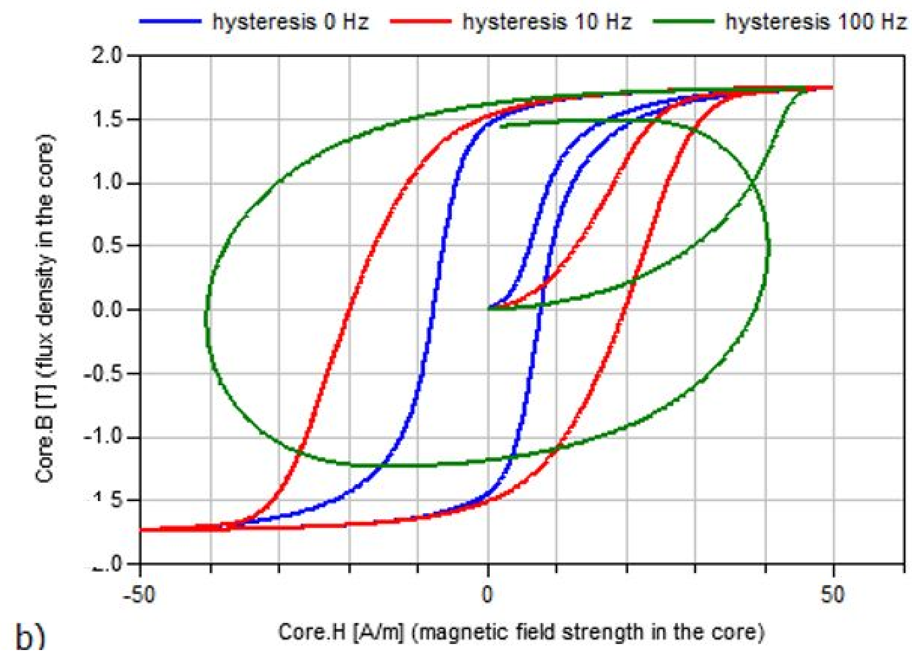

Figure 3.2: Simulated hysteresis curve of an inductor with a ferromagnetic core, for sinusoidal excitation fields of 0, 10 and 100 Hz [35]

With regards to NorSat-3, considering that the magnetic field does not change much in magnitude or direction locally in comparison to the angular velocity of the spacecraft, the frequency of the field can be loosely related to the spacecraft's body rate. For a single axis body rate of 10 deg/sec or 0.1745 rad/sec, the corresponding frequency of the field f is $f = \omega/2\pi = 0.023Hz$. The frequency is very low and therefore the modelling of Eddy current losses can be excluded from the simulation without any loss of fidelity. However, this points towards the need for reliable testing methodology to accurately determine the magnetization characteristics of the rods. Performing the test with high excitation frequencies would lead to the misleading result that the material is over-performing and that has the ability to dampen rotational motion faster than expected.

In subsequent sections, we review the equations of spacecraft attitude kinematics and dynamics. Then, we illustrate the equations of hysteresis damping torque. At first, we provide some necessary background on the notation used in the next sections, as well as the coordinate frames and relevant transformations from one frame to another.

3.4 Notation

Vectors are typically expressed in relation to a right-handed orthonormal set of vectors called a *frame*. In this thesis, *vectrix* notation adopted from Dr. Peter Hughes will be used to manipulate physical vectors in multiple reference frames [36]. A brief description of vectrices follows.

The *vectrix* $\underrightarrow{\mathscr{F}}_A$ corresponding to the frame $\mathscr{F}_\mathrm{A}$ is given by:

$$\underrightarrow{\mathscr{F}}_A \triangleq \begin{bmatrix} \underrightarrow{\hat{\mathbf{a}}}_1 & \underrightarrow{\hat{\mathbf{a}}}_2 & \underrightarrow{\hat{\mathbf{a}}}_3 \end{bmatrix}^T \tag{3.1}$$

A vectrix possesses simultaneously the characteristics of both a vector and a matrix. The *elements of the fundamental vectrix* $\underrightarrow{\mathscr{F}}_A$ are a set of *basis unit vectors*. Each of those orthonormal basis vectors is a column vector in three dimensional space. For instance, using the vectrix $\underrightarrow{\mathscr{F}}_A$, a three-dimensional vector $\underrightarrow{\mathbf{v}} \triangleq \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}^T$ can be expressed in terms of its components in frame $\mathscr{F}_\mathrm{A}$. For example, the vector $\underrightarrow{\mathbf{v}}$ is expressed in frame $\mathscr{F}_\mathrm{A}$ with the components $\boldsymbol{v}_\mathrm{A} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}^T$ as:

$$\underrightarrow{\mathbf{v}} = \underrightarrow{\mathscr{F}}_A^T \boldsymbol{v}_\mathrm{A} \tag{3.2}$$

Equation 3.2 holds for any frame of choice. Therefore, we can relate the components of $\boldsymbol{v}_\mathrm{A}$ to the components of $\underrightarrow{\mathbf{v}}$ in some other frame $\mathscr{F}_\mathrm{C}$, using the rotation matrix $\mathbf{C}_\mathrm{AC}$ where $\mathbf{C}_\mathrm{AC} = \underrightarrow{\mathscr{F}}_A \cdot \underrightarrow{\mathscr{F}}_C^T$ and $\boldsymbol{v}_\mathrm{A} = \mathbf{C}_\mathrm{AC}\boldsymbol{v}_\mathrm{C}$. Rotation matrices provide one of the most fundamental representations of attitude; we can define a frame $\mathscr{F}_\mathrm{B}$ fixed to a spacecraft body, and represent the attitude of the spacecraft in some arbitrary frame $\mathscr{F}_\mathrm{Q}$ using $\mathbf{C}_\mathrm{BQ}$. A rotation matrix can parameterized by a set of angles called *Euler Angles*, θ_i, which describe a series of principal rotations (rotations about the frame axes). More specifically, the rotation matrix $\mathbf{C}_1(\theta_1)$ describes the rotation of a frame about the first axis (x-axis) of the frame by an angle equal to θ_1. For the 3-2-1 *Yaw-Pitch-Roll* Euler rotation sequence, the rotation matrix $\mathbf{C}$ is given by:

$$\mathbf{C} = \mathbf{C}_1(\theta_1)\mathbf{C}_2(\theta_2)\mathbf{C}_3(\theta_3) \tag{3.3}$$

Another parameterization is the Euler axis angle, according to which the rotation of a body can be expressed as a rotation of ϕ about an axis along $\underrightarrow{\mathbf{a}}$, according to:

$$\mathbf{C} = \cos\phi\mathbf{1} + (1 - \cos\phi)\boldsymbol{a}\boldsymbol{a}^T - \sin\phi\boldsymbol{a}^\times \tag{3.4}$$

An alternative attitude parametrization is the use of *quaternions*, which are used extensively for attitude simulations due to their computational efficiency, and lack of singularities for rotation sequences. In this case, a rotation sequence can be described using the 4-parameter set $\mathbf{q} = \begin{bmatrix} \boldsymbol{\epsilon}^T & \eta \end{bmatrix}^T$, where:

$$\boldsymbol{\epsilon} = \boldsymbol{a}\sin(\frac{\phi}{2}), \quad \eta = \cos(\frac{\phi}{2}) \tag{3.5}$$

A thorough treatment on the properties of frames, vectors, vectrices, and rotations can be found in [36].

Unless otherwise explicitly noted, bold and lower-case letters will be used to denote column matrices, bold and capital letters will be used to denote two-dimensional matrices. Finally, scalar values will be denoted as italicized, non-bold letters, with occasional subscripts where applicable.

3.5 Coordinate Frames

In this section, we describe the coordinate frames that are relevant to our analysis. Some frames are related to orbital propagation, while others are related to the spacecraft's attitude. Throughout this thesis, the term *coordinate frame* will be used interchangeably with the term *coordinate system*. A *coordinate system* is the realization of a coordinate frame centered at a specific origin point in space, using a triplet of orthonormal right-handed basis vectors. Left-handed coordinate systems exist as well, but they do not form part of this discussion, since they are not commonly encountered. For example, some manufacturers of rate sensor or magnetometer components use a left-handed coordinate system to define the positive direction of the units' measurements positive direction on each axis.

The discussed coordinate systems are distinguished in two categories; Earth-based systems and satellite-based systems. Earth-based systems are realized with their origin attached at the center of the Earth, while satellite-based systems are realized with their origin attached at the satellite's Center of Mass (CoM).

3.5.1 Earth-Centered Inertial (ECI) Frame $\mathscr{F}_{\mathrm{i}}$

The *Earth-Centered Inertial* (ECI) system $\mathscr{F}_{\mathrm{i}}$ is also known as *geocentric-equatorial* system [37]. The origin of the system is the center of the Earth, and it is defined by the Earth's axis of rotation and its X-axis pointing towards the vernal equinox. The Z-axis points towards the North pole, and the Y-axis completes the right-hand rule. Even though $\mathscr{F}_{\mathrm{i}}$ is attached at the Earth's center, this system does not rotate with the Earth. A realization of the ECI frame can be quite intricate and laborious, because the fundamental X-Y plane and the vernal equinox move slowly over time. Therefore, any realization must specify the direction of the vernal equinox at a specific *epoch* (date), leading up to the realization of a quasi-inertial or *pseudo-inertial* frame, in reality. For example, the *J2000* frame is a popular quasi-inertial frame instantiated with the direction of the vernal equinox captured precisely at 12:00 January 1st 2000. Its motion is modeled using the IAU-1976 model (for modelling the precession of the Earth's axis of rotation) and the IAU-1980 model (for modelling the nutation of the Earth's axis of rotation). For more information about the effects of precession, nutation, and polar motion, the reader can consult [37]. The above perturbations also play a role in the coordinate frame transformation from ECI to any other frame (e.g. ECEF), and vice versa. In addition, different applications might allow for a number of simplifications in terms of whether some of these effects are taken into account, according to the required level of detail.

3.5.2 Earth-Centered Earth-Fixed Frame $\mathscr{F}_{\mathbf{e}}$

The *Earth-Centered-Earth-Fixed* (ECEF) frame $\mathscr{F}_{\mathrm{e}}$ is also known as the *International Terrestrial Reference Frame* (ITRF). It is defined with its origin at the center of the Earth. Its X axis is aligned with the Prime Meridian, while its Z axis is aligned with the North Pole. Finally, the Y axis is instantiated by completing the right-hand rule. The ECEF frame rotates with respect to the ECI frame at an approximately constant angular velocity $\omega_{\oplus}$ about the Z axis. The angular velocity vector describing the motion of $\mathscr{F}_{\mathrm{e}}$ with respect to $\mathscr{F}_{\mathrm{i}}$, expressed in $\mathscr{F}_{\mathrm{i}}$, is denoted as $\underset{\rightarrow}{\boldsymbol{\omega}}_{ei}$ and is defined as:

$$\underset{\rightarrow}{\boldsymbol{\omega}}_{ei} = \underset{\rightarrow}{\mathscr{F}}_{e}^{T} \begin{bmatrix} 0 & 0 & \omega_{\oplus} \end{bmatrix}^{T} \tag{3.6}$$

$$\omega_{\oplus} = 7.292115 \times 10^{-5} rad/s \text{ (all digits significant)} \tag{3.7}$$

3.5.3 Local Vertical Local Horizontal Frame $\mathscr{F}_{\mathbf{o}}$

In contrast to the ECEF $\mathscr{F}_{\mathrm{e}}$ and ECI $\mathscr{F}_{\mathrm{i}}$ frames, the LVLH frame $\mathscr{F}_{\mathrm{o}}$ is a satellite-based system. Its origin is centered at the satellite's center of mass, and its realization is easiest formulated by using the satellite's position and velocity vectors with respect to $\mathscr{F}_{\mathrm{i}}$ and expressed in $\mathscr{F}_{\mathrm{i}}$. The vectrix $\underset{\rightarrow}{\mathscr{F}}_{\mathbf{o}}$ corresponding to the LVLH frame $\mathscr{F}_{\mathrm{o}}$ is given by:

$$\underset{\rightarrow}{\mathscr{F}}_{\mathbf{o}} = \begin{bmatrix} \underset{\rightarrow}{\hat{\mathbf{o}}}_1 & \underset{\rightarrow}{\hat{\mathbf{o}}}_2 & \underset{\rightarrow}{\hat{\mathbf{o}}}_3 \end{bmatrix}^{T} \tag{3.8}$$

The basis unit vector $\underset{\rightarrow}{\hat{\mathbf{o}}}_3$, which is the frame's Z-axis, is aligned with the spacecraft geocentric position vector, but pointing towards the Earth (opposite direction). The basis vector $\underset{\rightarrow}{\hat{\mathbf{o}}}_2$ is aligned with the negative direction of the orbit normal. The orbit normal vector is the cross product of the position and velocity vectors expressed in the ECI frame. Finally, the basis vector $\underset{\rightarrow}{\hat{\mathbf{o}}}_1$, the frame's X-axis, completes the right hand rule with respect to the other two basis vectors.

3.5.4 Spacecraft Body Frame $\mathscr{F}_{\mathbf{b}}$

The satellite's body frame is denoted as $\mathscr{F}_{\mathrm{b}}$. The origin of the body frame is attached to its CoM. The basis vectors of $\mathscr{F}_{\mathrm{b}}$ are aligned with the orthogonal faces of the cuboid rectangular satellite bus structure. The vectrix $\underset{\rightarrow}{\mathscr{F}}_{\mathbf{b}}$ corresponding to the body frame $\mathscr{F}_{\mathrm{b}}$ is given by:

$$\underset{\rightarrow}{\mathscr{F}}_{\mathbf{b}} = \begin{bmatrix} \underset{\rightarrow}{\hat{\mathbf{b}}}_1 & \underset{\rightarrow}{\hat{\mathbf{b}}}_2 & \underset{\rightarrow}{\hat{\mathbf{b}}}_3 \end{bmatrix}^{T} \tag{3.9}$$

$\underset{\rightarrow}{\hat{\mathbf{b}}}_1$ is aligned with the spacecraft X axis, $\underset{\rightarrow}{\hat{\mathbf{b}}}_2$ is aligned with the spacecraft Y axis, and $\underset{\rightarrow}{\hat{\mathbf{b}}}_3$ is aligned with the satellite's Z axis. It is noted that the body frame axes do not generally align with the body's principal axes, the orthogonal triplet of basis vectors for which a diagonal moment of inertia matrix can be defined (i.e. the matrix composed by the eigen-vectors of the satellite's inertia tensor). Many times, payload and mission requirements as well as other mechanical constraints (e.g. the deployment mechanism) restrict the layout options for the placement of the satellite's components. This leads inevitably to misalignment between the body frame and the principal axes frame. As we will examine in Chapter 4, mechanical layout constraints, lack of space, and bulky payload instruments can drive design choices that may lead to death modes, requiring thorough evaluation and mitigation strategies.

3.6 Coordinate Transformations

Having defined the coordinate systems, this section describes the equations needed to manipulate and express quantities between those frames.

3.6.1 ECEF to ECI Frame, $\mathscr{F}_\mathrm{e}$ to $\mathscr{F}_\mathrm{i}$

In the context of simulating spacecraft attitude perturbations, the conversion of quantities from the ECEF frame $\mathscr{F}_\mathrm{e}$ to the ECI frame $\mathscr{F}_\mathrm{i}$ is important. For example, when computing the magnetic field flux density of the Earth during satellite propagation, this value is first calculated and expressed in the $\mathscr{F}_\mathrm{e}$ frame using models of the Earth's geomagnetic potential. However, the equations governing the satellite's orbital motion are expressed in the $\mathscr{F}_\mathrm{i}$ frame. The conversion from $\mathscr{F}_\mathrm{e}$ to $\mathscr{F}_\mathrm{i}$ (specifically the J2000 frame) and vice-versa is a delicate procedure that takes into account a series of phenomena that are described by a specific sequence of coordinate transformations. First of all, the sidereal rotation of the Earth is taken into account, expressed as the coordinate transformation matrix $\mathbf{C}_\Theta$. In addition, Earth axis nutation transformation is expressed as $\mathbf{C}_\Phi$, while the Earth's precession is expressed as $\mathbf{C}_\mathrm{N}$. Finally, the effect of the Earth's polar motion is expressed as $\mathbf{C}_\Omega$. The rotation sequence of those transformations is fundamentally important. For a satellite's position vector $\boldsymbol{r}_\mathrm{e}$ expressed in the ECEF frame $\mathscr{F}_\mathrm{e}$, the coordinate transformation $\mathbf{C}_\mathrm{ie}$ from $\mathscr{F}_\mathrm{e}$ to $\mathscr{F}_\mathrm{i}$ to get the position vector $\boldsymbol{r}_\mathrm{i}(\mathrm{t})$ in $\mathscr{F}_\mathrm{i}$ is expressed as:

$$\boldsymbol{r}_\mathrm{i}(t) = \mathbf{C}_\mathrm{ie}(t)\boldsymbol{r}_\mathrm{e} = \mathbf{C}_\Phi(t)\mathbf{C}_\mathrm{N}(t)\mathbf{C}_\Theta(t)\mathbf{C}_\Omega(t)\boldsymbol{r}_\mathrm{e} \tag{3.10}$$

The above process is referred in the literature as the *FK5 reduction*. It can also be encountered as *IAU-76/FK5* reduction, due to the use of the *IAU-76* model to describe the Earth's precession, nutation, and polar motion. For a detailed treatment of those transformations, as well as a discussion on the various models that the analyst can choose to realize those transformations, consult [37]. There is general consensus that, especially for the modelling of the polar motion, the models are not very precise for long propagation intervals into the future. The associated coefficients, models, and other constants that are needed for realizing these transformations need to be frequently updated. Typicaly, the latter data are provided at regular intervals from the International Earth Rotation and Reference systems Service, also abbreviated as *IERS* [38]. Depending on the required application and level of detail, analysts should be aware about the effect of data updates to the validity of results.

3.6.2 ECI to LVLH Frame, $\mathscr{F}_{\mathrm{i}}$ to $\mathscr{F}_{\mathrm{o}}$

As discussed in Section 3.5.3, the LVLH frame is easily realized from the ECI frame, therefore the transformation is straightforward. Consider the satellite's position and velocity vectors $\underrightarrow{\mathbf{r}}_i$ and $\underrightarrow{\mathbf{v}}_i$, expressed in the ECI frame $\mathscr{F}_{\mathrm{i}}$. The angular momentum $\underrightarrow{\mathbf{h}}_i$ of the satellite expressed in the ECI frame Hfi is defined as the cross product of the position and velocity vectors, according to $\underrightarrow{\mathbf{h}}_i = \underrightarrow{\mathbf{r}}_i \times \underrightarrow{\mathbf{v}}_i$. The representation of the LVLH basis vectors with respect to $\underrightarrow{\mathbf{r}}_i$, $\underrightarrow{\mathbf{v}}_i$ and $\underrightarrow{\mathbf{h}}_i$ is given by:

$$\boldsymbol{o}_{3,\mathrm{i}} = -\frac{\underrightarrow{\mathbf{r}}_i}{||\underrightarrow{\mathbf{r}}_i||}, \quad \boldsymbol{o}_{2,\mathrm{i}} = -\frac{\underrightarrow{\mathbf{h}}_i}{||\underrightarrow{\mathbf{h}}_i||}, \quad \boldsymbol{o}_{1,\mathrm{i}} = \boldsymbol{o}_{2,\mathrm{i}} \times \boldsymbol{o}_{3,\mathrm{i}} \tag{3.11}$$

The rotation matrix from the ECI frame $\mathscr{F}_{\mathrm{i}}$ to the LVLH frame $\mathscr{F}_{\mathrm{o}}$ then becomes:

$$\mathbf{C}_{\mathrm{oi}} = \begin{bmatrix} \boldsymbol{o}_{1,\mathrm{i}} & \boldsymbol{o}_{2,\mathrm{i}} & \boldsymbol{o}_{3,\mathrm{i}} \end{bmatrix}^T \tag{3.12}$$

3.6.3 ECI to Body Frame, $\mathscr{F}_{\mathrm{i}}$ to $\mathscr{F}_{\mathrm{b}}$

The rotation from the inertial frame $\mathscr{F}_{\mathrm{i}}$ to the spacecraft body frame $\mathscr{F}_{\mathrm{b}}$ is written as $\mathbf{C}_{\mathrm{bi}}$ and it is a fundamental expression of spacecraft attitude. Its computation forms part of the attitude estimation algorithms running on the spacecraft's on-board computer. More insight on the operation of OASYS and the on-board computation of the attitude determination algorithms is provided in [39].

3.7 Spacecraft Orbital Dynamics

This section provides a brief discussion of fundamental orbital mechanics concepts. For a detailed treatment, consult [37] and [40] . In addition, we provide a high-level description about SGP4, the closed-form orbital propagation algorithm that is used for all simulations of this thesis.

3.7.1 Orbital Elements and Equations of Motion

The starting point for orbital dynamics is the equation describing Keplerian two-body motion. Keplerian orbital motion is planar motion along an elliptical track, with the Earth placed at one of the ellipse's foci. It is derived from classical Newtonian mechanics, starting with the motion of a point mass about an orbiting body. The spacecraft's orbit can be expressed using six orbital elements, also known as Keplerian elements. The sixth element, called the *mean anomaly*, describes the time-varying component of the spacecraft's orbital position. Figure 3.3 visualizes three of the seven orbital elements, namely the inclination i, the right ascension of the ascending node Ω, and the argument of perigee ω. These three angles describe the orientation of the orbit plane with respect to the primary body. The inertial frame $\mathscr{F}_\mathrm{i}$ is also shown in the image for reference. Finally, the orbit's angular momentum vector $\underrightarrow{\mathbf{h}}_i$ can be shown, as it was defined in Section 3.6.2 above.

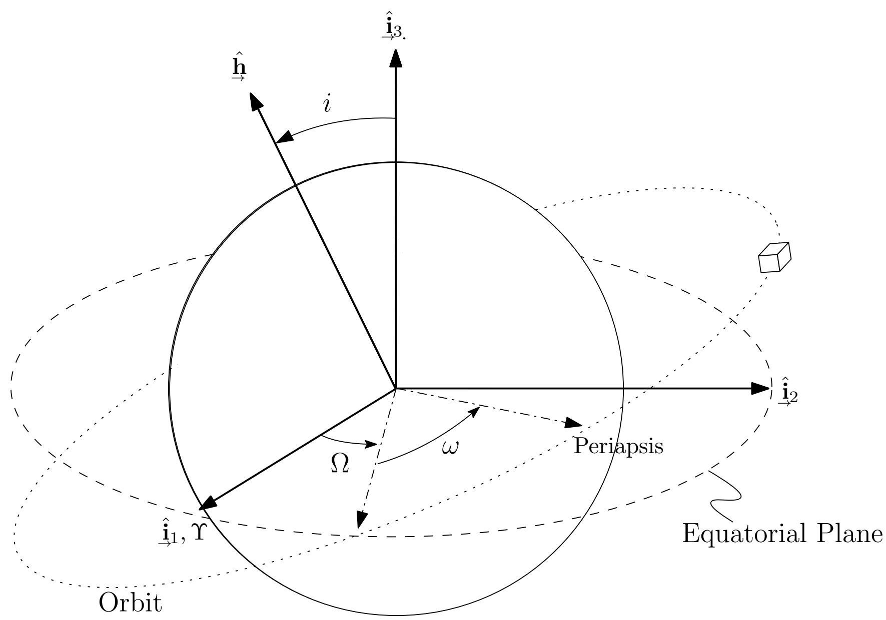

Figure 3.3: Depiction of orbital elements, i, Ω, ω [39]

A brief description of the orbital elements follows:

- a - The *semi-major axis* of the orbit's ellipse. The semi-major axis is the parameter that specifies the *size* of the orbit.
- e - *eccentricity*: a measure of a conic section's deviation from being circular, $0 < e < 1$ for an ellipse [40].
- i - *inclination*: the angle between the equatorial plane and the orbital plane, or equivalently the angle between the basis vector $\hat{\underrightarrow{\mathbf{i}}}_3$ of the ECI frame $\mathscr{F}_\mathrm{i}$, and $\underrightarrow{\mathbf{h}}$

- Ω - *Right Ascension of the Ascending Node (RAAN)*: the angle in between the vernal equinox, $\underset{\rightarrow}{\hat{\mathbf{i}}}_1$ and the vector $\underset{\rightarrow}{\hat{\mathbf{n}}}$, which points from the Earth's center of mass towards the ascending node (the point where the orbit crosses the equatorial plane, and the coordinate along $\underset{\rightarrow}{\hat{\mathbf{i}}}_3$ crosses from negative to positive)

- ω - *argument of perigee*: the angle between the ascending node and the perigee

- M_0 - *initial mean anomaly*: the initial mean anomaly $M_0 = M(t_0)$, which is typically used in place of the true anomaly. The *true anomaly* is the angle between perigee and the position of the orbiting body, at any given time.

3.7.2 Keplerian Two-body Equation of Motion

In order to propagate the satellite's orbital state as a Keplerian two-body motion, the following differential equation needs to be solved [40]:

$$\underset{\rightarrow}{\ddot{\mathbf{r}}} = -\frac{\mu}{||\underset{\rightarrow}{\mathbf{r}}||^3}\underset{\rightarrow}{\mathbf{r}} \tag{3.13}$$

Where μ is the gravitational constant of the primary body. [40]provides the analytical solution to the above equation.

The Keplerian two-body propagator is sufficiently accurate for short propagation time intervals up to several days, but it quickly becomes inaccurate due to the absence of adequate modelling for certain perturbations. In reality, a satellite's orbit is affected by other factors that make the orbit deviate from the ideal Keplerian motion. These factors include the non-spherical shape of the Earth, the presence of other celestial bodies and their gravitational fields, atmospheric drag, etc. Accurate modelling of orbit perturbations within the orbital propagator requires detailed numerical solutions to non-linear differential equations, whose computation time scale with complexity.

3.7.3 SGP4 Propagation Theory

As an alternative to the numerical propagation algorithms needed for highly accurate propagation, there exist closed-form analytical solutions that are sufficiently accurate for some applications. Since the 1970s, the US Air Force has been using the Simplified General Perturbation Theory to propagate a satellite's state using a set of initial orbital conditions. This set is packaged in a special, compact format and is known as Two Line Element (TLE) set. The official US SGP4 algorithm is open-source and is meant to be used as a standard for interoperability among satellite operators and analysts.

The SGP4 propagator is modelling the effect of the non-sphericity of the Earth up to a certain degree, and it also models atmospheric drag. For more information regarding the SGP4 Theory, consult [41]. The simulations performed in this thesis make use of the SGP4 algorithm as published and updated in the above reference. For more details about the definition and use of a Two Line Element, consult consult [41].

It is important to note that the accuracy of SGP4 drops over time, with an approximate error of 5 kilometers per day of propagation [41]. This is an acceptable level of accuracy for the purpose of our simulations, which span a maximum of 5 days. The accumulated error can be as high as 25 kilometers, which is acceptable since the key metrics that are evaluated in our study relate mostly to the spacecraft attitude dynamics.

3.8 Spacecraft Attitude Dynamics

This section describes briefly the equations of motion that describe the attitude of the spacecraft, treated as a rigid body.
First of all, the kinematic equation for the rotation from ECI frame $\mathscr{F}_\mathrm{i}$ to Body Frame $\mathscr{F}_\mathrm{b}$ is given by:

$$\dot{\mathbf{C}}_\mathrm{bi} + \boldsymbol{\omega}^\times \mathbf{C}_\mathrm{bi} = \mathbf{0} \tag{3.14}$$

Where $\boldsymbol{\omega}$ is the angular velocity of the spacecraft with respect to the ECI frame $\mathscr{F}_\mathrm{i}$ and expressed in the body frame $\mathscr{F}_\mathrm{b}$.
We define $\boldsymbol{\tau}_{\boldsymbol{\odot}}$ as the net external torque acting about the satellite's center of mass. We also denote $\mathbf{I}$ as the inertial matrix of the satellite expressed in the body frame $\mathscr{F}_\mathrm{b}$. The equation describing the attitude dynamics of the spacecraft in relation to the net torque applied by all disturbance torques is called the *Euler's rigid body equation* and is defined as:

$$\boldsymbol{\tau}_{\boldsymbol{\odot}} = \mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega}^\times \mathbf{I}\boldsymbol{\omega} \tag{3.15}$$

For the purpose of this analysis, no reaction wheel dynamics have been incorporated in the equations of motion. This is because we are interested in analyzing the spacecraft attitude without any means of active control mechanisms. For a formal treatment on the equations of motion using reaction wheel dynamics, consult [36], [39] and [42].

3.9 Environmental Disturbance Torques

The space environment imposes external torques that affect the attitude of a spacecraft. The key disturbance torques to consider are those caused by the Earth's magnetic field, aerodynamic effects, the gravity-gradient, and solar radiation pressure. For a complete discussion on attitude disturbance torques, please consult [39]. In this section, only two torques will be discussed. Simulations using SFL's MIRAGE simulation environment consider all disturbance torques to accurately model and control the spacecraft attitude.

3.9.1 Magnetic Field Torque

A magnetic disturbance torque can arise from the interaction of the Earth's magnetic field $\underrightarrow{\mathbf{b}}$ with the magnetic dipole moment of the spacecraft. Magnetic moments can occur from electrical current loops in the hardware wiring (e.g. solar panels), as well as the presence of ferromagnetic materials. The magnetic disturbance torque $\underrightarrow{\tau}_m$ is calculated from the cross product of the spacecraft magnetic dipole moment $\underrightarrow{\mathbf{m}}_{net}$ and the local magnetic field.

$$\underrightarrow{\tau}_m = \underrightarrow{\mathbf{m}}_{net} \times \underrightarrow{\mathbf{b}} \tag{3.16}$$

Permanent magnets (hard ferromagnetic materials) installed on the satellite contribute to $\underrightarrow{\mathbf{m}}_m$ and so do current loops. Some spacecraft equipment can produce magnetic moments at low levels during their operation, such as parasitic magnetic dipole moment stemming from the operation of reaction wheels or other components for radio communication with the ground station.

3.9.2 Hysteresis Damping Torque

This section describes the equations for the computation of the hysteresis torque. This torque is computed separately from the magnetic field torque, due to the nature of the dipole moment. While the magnetic field torque aggregates the effects of parasitic dipole moment and the presence of hard ferromagnetic materials (permanent magnets), the hysteresis damping torque aggregates the produced dipole moments only due to soft ferromagnetic materials.

Recall that we have assumed that the hysteresis rods are aligned with the spacecraft's body frame $\mathscr{F}_\mathrm{b}$. This is practical in terms of mechanical layout and mounting of the rods on the satellite bus, requiring simple design of mounting brackets. When a spacecraft equipped with ferromagnetic hysteresis rods is immersed in a magnetic field, the hysteresis torque produced is identified as $\underrightarrow{\tau}_{hyst}$. Additionally, we define the magnetic flux density produced within the ferromagnetic material as $\underrightarrow{\mathbf{b}}$. Then, the hysteresis torque can be computed as:

$$\underrightarrow{\tau}_{hyst} = \underrightarrow{\mathbf{m}}_{hyst} \times \underrightarrow{\mathbf{b}} \tag{3.17}$$

In matrix notation, Equation 3.17 becomes:

$$\boldsymbol{\tau}_\mathrm{hyst} = \boldsymbol{m}^\times_\mathrm{hyst}\boldsymbol{b} \tag{3.18}$$

$\boldsymbol{\tau}_\mathrm{hyst}$ is a 3×1 column matrix with each component defining the hysteresis torque that is produced about the corresponding spacecraft body axis and expressed in the body frame $\mathscr{F}_\mathrm{b}$. $\boldsymbol{m}^\times_\mathrm{hyst}$ is the skew symmetric matrix of $\boldsymbol{m}_\mathrm{hyst}$, a 3×1 column matrix with each of its components representing the dipole moment that is produced due to the presence of the hysteresis rods. In addition, $\boldsymbol{b}$ is a column matrix defining the vector of the Earth's magnetic field flux density in vacuum. All quantities in Equations 3.17 and 3.18 are expressed in the satellite's body frame $\mathscr{F}_{\text{b}}$. The SI units of $\boldsymbol{b}$ are in $N/(A \cdot m)$, also known as *Tesla* and abbreviated as T. The SI units of the magnetic moment $\boldsymbol{m}_{\text{hyst}}$ are in $A \cdot m^2$. The units of torque are in $N \cdot m$.

For reference, the typical symbol for magnetic flux density is B, in the field of electromagnetism. Here, abiding to Dr. Peter Hughes notation, we will be using $\boldsymbol{b}$ to make it clear that it is a column matrix. In vacuum, the magnetic flux density $\boldsymbol{b}$ is related to the magnetic field strength $\boldsymbol{h}$ through the magnetic permeability of vacuum constant μ_0, as:

$$\boldsymbol{b} = \mu_0 \boldsymbol{h} \tag{3.19}$$

$$\mu_0 = 4\pi \times 10^{-5} \frac{N}{A^2} \tag{3.20}$$

The SI units of the magnetic field strength $\boldsymbol{h}$ are in A/m. The SI units of the μ_0 constant are N/A^2. For reference, the typical symbol for magnetic field strength is H in the field of electromagnetism, but as discussed, we will be abiding to Dr. Peter Hughes notation and will be using the symbol $\boldsymbol{h}$ to make it clear that it is a column matrix.

It is reminded that the magnetic dipole moment due to the hysteresis rods $\boldsymbol{m}_{\text{hyst}}$ is a 3×1 column matrix that sums all contributions of dipole moment from all hysteresis rods mounted on the spacecraft structure, expressed in $\mathscr{F}_{\text{b}}$. At this point, we will explain how the dipole moment is computed for a single rod, before we move on to a formulation involving the installation of multiple rods.

When an external magnetic field $\boldsymbol{h}$ is applied to a hysteresis rod of volume v, the flux density produced within the rod is defined as $\boldsymbol{b}_{\text{rod}}$. Then, the magnetic dipole moment $\boldsymbol{m}_{\text{rod}}$ produced by the hysteresis rod is given by:

$$\boldsymbol{m}_{\text{rod}} = \frac{1}{\mu_0} v \, \boldsymbol{b}_{\text{rod}} \tag{3.21}$$

An important note needs to be made with respect to the properties of soft ferromagnetic materials. As discussed in the beginning of this chapter and by observing Figure 3.1, there are three important scalar values that characterize any soft ferromagnetic material; *saturation*, *coercivity*, and *remanence*. *Saturation* refers to the maximum amount of induced flux density that a soft ferromagnetic material can experience within any sufficiently large externally applied magnetic field. Saturation is denoted by b_s throughout the scope of this thesis. The points of saturation can be visualized as the points "a" and "d" in Figure 3.1 *Coercivity* is defined as the maximum amount of magnetic field strength for which the soft ferromagnetic material experiences zero magnetic flux density, when exposed to a periodically varying magnetic field. The points of coercivity can be visualized as the points "c" and "f" in Figure 3.1. Coercivity is denoted as h_c within this thesis. *Remanence*, or *Retentivity* is the amount of maximum residual magnetic flux in the soft ferromagnetic medium, even when the externally applied field is of zero magnetic field strength. Remanence is denoted as b_r within this thesis, and can be visualized as the points "b" and "b" in Figure 3.1. Using these three parameters, a simple empirical arctangent model of hysteresis magnetization can be created. This empirical model, as described in [23] and [25] forms the basis of modelling the induced flux density of soft ferromagnetic materials in this simulation.

Note that in equation 3.21, we assume that the orientation of the rod is aligned with the orientation of the externally applied field. Therefore, we need to proceed and generalize the computation of the dipole moment for any number of rods installed, as well as their dimensions and orientation within the satellite body frame $\mathscr{F}_{\mathrm{b}}$.

We define $\boldsymbol{v}$ as a $n \times 1$ column matrix with components $\boldsymbol{v} = \begin{bmatrix} v_1 & v_2 \ldots v_{n-1} & v_n \end{bmatrix}^T$, where n is the number of hysteresis rods that have been installed on the satellite. Each component of the column matrix $\boldsymbol{v}$ represents the volume of each cylindrical hysteresis rod. For the purpose of this thesis, all rods are cylindrical in shape. However, it is possible that each rod may be different in terms of shape or dimensions, for example, in cases where there would be an interest to provide stronger hysteresis damping along one of the spacecraft axes. The formulation below holds without loss of generality.

Each of the n rods installed on the satellite has a specific orientation of its cylindrical longitudinal axis with respect to the satellite body frame $\mathscr{F}_{\mathrm{b}}$. In some literature resources like [26], the longitudinal axis of an elongated soft ferromagnetic specimen is also encountered as the *easy magnetization axis*.

We define the matrix $\boldsymbol{W}$ as the *rod orientation matrix*, with dimensions $n \times 3$ and components:

$$\boldsymbol{W} = \begin{bmatrix} \underset{\to}{\hat{\mathbf{w}}}_1 & \underset{\to}{\hat{\mathbf{w}}}_2 & \ldots \underset{\to}{\hat{\mathbf{w}}}_{n-1} & \underset{\to}{\hat{\mathbf{w}}}_n \end{bmatrix}^T \tag{3.22}$$

For example, for the j$^{\text{th}}$ hysteresis rod, $\underset{\to}{\hat{\mathbf{w}}}_j$ is a column unit vector with components $\underset{\to}{\hat{\mathbf{w}}}_j = \begin{bmatrix} \underset{\to}{\hat{\mathbf{w}}}_{j,1} & \underset{\to}{\hat{\mathbf{w}}}_{j,2} & \underset{\to}{\hat{\mathbf{w}}}_{j,3} \end{bmatrix}^T$ describing the orientation of the j$^{\text{th}}$ rod's longitudinal axis with respect to the spacecraft body frame $\mathscr{F}_{\mathrm{b}}$. It is clear that all of the hysteresis rods will be exposed to the same magnetic field strength vector $\boldsymbol{h}$, but their response and resulting hysteresis moment will differ depending on the dimensions and their orientation, as hinted by Equation 3.21. They may also differ in terms of material properties, namely having different values of saturation b_s, coercivity h_c and remanence b_r.

At this point, an important simplification needs to be made. If we assume that all the rods are made out of the same material, then all of the three parameters of magnetic remanence b_r, coercivity h_c and saturation b_s can be the same for all rods. Under this assumption, we only need to compute a single vectorial value for the induced flux density, and then distribute the effect of that induced flux density according to the rod orientations and dimensions, as shown below.

We define $\boldsymbol{b}_{\mathrm{h}}$ as the induced flux density due to the Earth's magnetic field within the soft ferromagnetic material that is considered. It is a column vector with components $\boldsymbol{b}_{\mathrm{h}} = \begin{bmatrix} b_{h,1} & b_{h,2} & b_{h,3} \end{bmatrix}^T$, expressed in the spacecraft body frame $\mathscr{F}_{\mathrm{b}}$. Note that $\boldsymbol{b}_{\mathrm{h}}$ has the same direction as the Earth's magnetic field $\boldsymbol{h}$.

At this point, we define $\boldsymbol{B}$ as the *flux density distribution matrix*, with dimensions $n \times 3$ and components:

$$\boldsymbol{B} = \begin{bmatrix} \boldsymbol{b}_{\text{in,1}} & \boldsymbol{b}_{\text{in,2}} & \dots \boldsymbol{b}_{\text{in,n-1}} & \boldsymbol{b}_{\text{in,n}} \end{bmatrix}^T \tag{3.23}$$

For the j$^{\text{th}}$ hysteresis rod, $\boldsymbol{b}_{\text{in,j}}$ is a column vector with components $\boldsymbol{b}_{\text{in,j}} = \begin{bmatrix} \underset{\rightarrow}{\mathbf{b}}_{j,1} & \underset{\rightarrow}{\mathbf{b}}_{j,2} & \underset{\rightarrow}{\mathbf{b}}_{j,3} \end{bmatrix}^T$ describing the induced flux density of the j$^{\text{th}}$ rod's along its longitudinal axis with respect to the spacecraft body frame $\mathscr{F}_{\mathrm{b}}$ axes. According to the above, $\boldsymbol{B}$ can then be computed as:

$$\boldsymbol{B} = \boldsymbol{W}\,\mathrm{diag}(\boldsymbol{b}_{\mathrm{h}}) = \begin{bmatrix} \underset{\rightarrow}{\hat{\mathbf{w}}}_1 & \underset{\rightarrow}{\hat{\mathbf{w}}}_2 & \dots \underset{\rightarrow}{\hat{\mathbf{w}}}_{n-1} & \underset{\rightarrow}{\hat{\mathbf{w}}}_n \end{bmatrix}^T \begin{bmatrix} b_{h,1} & 0 & 0 \\ 0 & b_{h,2} & 0 \\ 0 & 0 & b_{h,3} \end{bmatrix} \tag{3.24}$$

With the above formulation, $\boldsymbol{B}$ encapsulates all vectors of induced magnetic flux contributed by each hysteresis rod, expressed in the body frame $\mathscr{F}_{\mathrm{b}}$. In the end, the magnetic moment $\boldsymbol{m}_{\text{hyst}}$ due to all hysteresis rods can be summarized as:

$$\boldsymbol{m}_{\text{hyst}} = \frac{1}{\mu_0}\boldsymbol{v}^T\boldsymbol{B} \tag{3.25}$$

Using Equations 3.19, 3.24 and 3.25 into Equation 3.18, we get:

$$\boldsymbol{\tau}_{\text{hyst}} = \left(\frac{1}{\mu_0}\boldsymbol{v}^T\boldsymbol{B}\right)^{\times}\mu_0\boldsymbol{h} = \left(\boldsymbol{v}^T\boldsymbol{W}\,\mathrm{diag}(\boldsymbol{b}_{\text{in}})\right)^{\times}\boldsymbol{h} \tag{3.26}$$

For details on the computation of the induced flux density vector $\boldsymbol{b}_{\text{in}}$ consult [23], [25], and [27]. The process involves the solution to a non-linear differential equation based on the arctangent model of the hysteresis major loop, using the values of saturation b_s, coercivity h_c, and remanence b_r.

3.10 Hysteresis Rod Sizing

By observation of the hysteresis torque Equation 3.26, we note that the magnitude of the hysteresis torque is linearly dependent on the volume of soft ferromagnetic material used. The equation, however, does not clearly capture the equally important concept of *dipole efficiency* and the effect to which the law of diminishing returns is in effect. With regards to cylindrical ferromagnetic rods (as well as thin ferromagnetic strips), experiments in [26] and [28] have demonstrated that *long and thin specimens produce higher dipole moment per volume*. This leads to a need to conduct experiments and assess the performance of an array of small size hysteresis specimens [28] versus the use of a single larger specimen. Arguments defending the use of single rods are based on the reduced complexity during spacecraft integration, as well as the negative and difficult to quantify effect of mutual magnetization between rods/strips mounted in parallel. Experiments conducted by Gerhardt [30] in a Helmholtz cage show that mutual magnetization can reduce dipole efficiency of each rod if the rods are installed at a distance of less than 30-40% of each specimen's length.

Specifically for NorSat-3, the initially considered rod length is based on hysteresis rods considered in trade studies of the EV9 mission, using the GNB bus (200 x 200 x 200 mm). The NorSat-3 bus nominal dimensions are 200 x 300 x 400 mm, therefore, if the rods are intended to be the same length, they all need to conform to the shortest bus dimension, leading back to the length chosen for EV9. The length of each rod planned to be used is approximately 152 mm, accounting for the thickness of the bus structure. If multiple rods were considered in order to double the dipole moment (and consequently the mass of the passive solution), then each rod would have to be spaced about 53mm apart. For using two rods, for example, each assembly would require a clear area of 152 x 53 mm for their installation on a panel with no other mechanical interferences. If three rods would be used, then this area would be 152 x 106 mm. Bracket size and mass would also increase with increasing number of rods. Available space and mass in the bus is always at a premium in microsatellite applications. Therefore, a single rod per satellite body axis is assumed to be used in the simulations, leaving the rod diameter as the last free variable in the trade space for optimal dipole moment efficiency. The simulation results prove that a single rod per axis is sufficient in achieving spin-up cancellation.

Under those assumptions, the rod diameter will determine both the rod dipole efficiency and the dipole moment, which are two somewhat conflicting figures of merit. Due to the availability of preexisting inventory of hysteresis rods at SFL, as well as the availability of a test setup with sense coils of the exact dimensions, specimens of 6.4 mm in diameter are used. The resulting length-to-diameter ratio l/D for a rod of length $l = 6.4mm$ and diameter $D = 152mm$ is therefore:

$$\frac{l}{D} = \frac{152\,mm}{6.4\,mm} = 23.75 \simeq 24 \tag{3.27}$$

According on [26], a length-to-diameter ratio of 24 is considered fairly low and will lead to some loss of dipole efficiency (dipole moment per unit of volume) in comparison to a rod that would be thinner. Of course, a thinner rod would produce less dipole moment, and considerations of multiple rods, mass budget discipline, volume space savings and mutual magnetization arise.

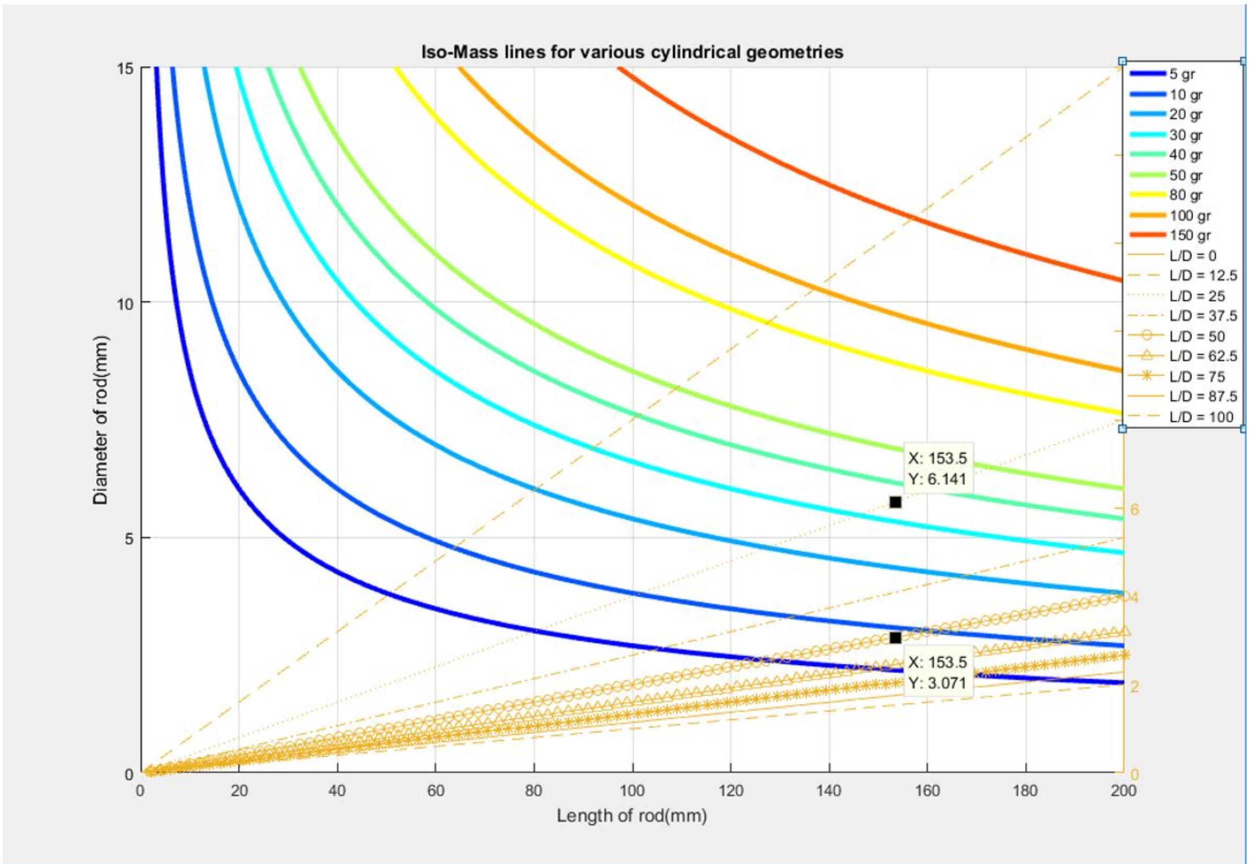

Figure 3.4: Iso-mass and Iso-efficiency lines versus the diameter and length of cylindrical specimens

In order to evaluate the added value of multiple rods versus a single rod, the plot in Figure 3.4 below could be of use and help in the trade study. The horizontal axis of this figure represents the possible length values of a hysteresis rod considered to be installed on the satellite, while the left vertical axis represents the possible diameter values of the same rod. The figure also illustrates two types of lines. Each of the colorful asymptotic lines represent the possible combinations of diameter and length for a rod of a specific weight (and by extension, volume). We call these lines as $iso-masslines$, and are produced by simply considering the computation of a rod's volume and mass, made our of a typical soft ferromagnetic material, in this case *High Perm 49*®. In the same figure, the straight lines starting from the origin represent what we define as $iso-efficiency$ lines. Each one of those lines represents potential combinations of a rod's diameter and length having the same length-to-diameter ratio (along each line). Now, given the unique constraints of a satellite with respect to mass budget discipline, available volume space, and requirement to detumble the spacecraft using hysteresis dissipative torque, the engineer can better visualize the trade space in this figure and identify potential combinations of length and diameter of rods that will accommodate all above constraints. Using this qualitative tool, the process is meant to be iterative, as the simulation results that accompany each proffered combination will prove or disprove the compliance to the detumbling requirement.

Based on [26], it is noted that the dipole efficiency (i.e. amount of dipole moment per volume) is approximately proportional to the rod elongation. If we are interested in reducing the mass of the system in order to achieve same hysteresis torque, it is possible to substitute one rod of about 6 mm diameter with two rods of 3.07 mm, and we will have effectively halved the mass. The dipole efficiency per rod will double, while the volume of each rod will shrink to a quarter of its value. Therefore, using two rods will halve the initial mass, for the same performance, while taking note of the additional space that will be required to avoid mutual magnetization. Given the described physical constraint of not being able to use more than three rods at most, coupled with the relatively low mass savings in absolute terms (20 grams instead of 40 grams per axis), the decision to use 1 rod per axis is implemented. In combination with the simulation results provided in Section3.12, it is proven that this rod schema selected is sufficient to cancel the potential spin-up of a satellite equipped with drag sails.

3.11 Material Selection

Selecting soft ferromagnetic materials is a combination of availability, cost, and performance. The performance within the context of material selection is defined as material's ability to have very high permeability, but at the same time have sufficient hysteresis losses expressed though high flux density. Figure 3.5 shows a material selection matrix based on those criteria, provided by Carpenter, a ferromagnetic material supplier [43].

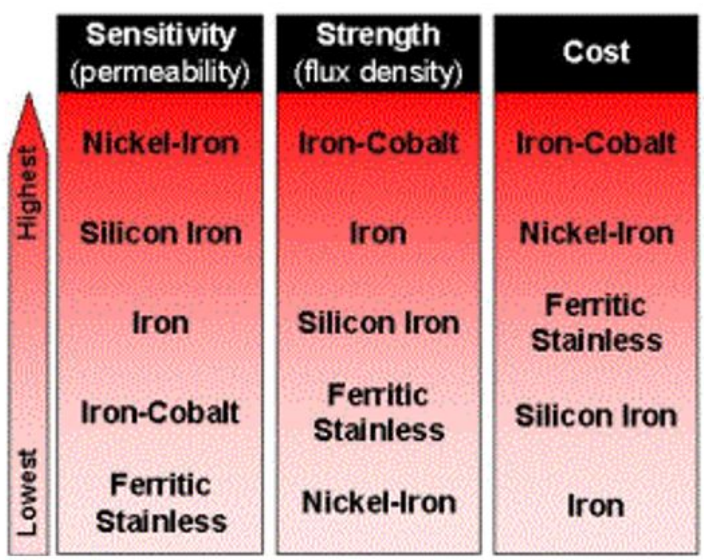

Figure 3.5: Magnetic Alloy Selection Matrix (Carpenter) [43]

Past missions have predominately been using Nickel iron alloys[21], with the main reason being the extremely high permeability in comparison to any other ferromagnetic alloy. Figure 3.6 below from Carpenter Material Selection Methodology illustrates this comparative advantage. Nickel-iron alloys like *HyMu 80®* and *High Perm 49®* dominate in terms of permeability and are well suited for magnetic hysteresis dissipation purposes.

In addition to the above, specific to the attitude stabilization strategy is the requirement for the material to be both extremely sensitive and ”spongy” (high permeability) but also “lossy” in nature and allow for hysteresis losses, contrary to other industries which require minimal losses, such as in power transformers. Some missions have selected *HyMu 80®* in the past. *HyMu 80®* a nickel-iron alloy with extremely high permeability but relatively low hysteresis losses. TRANSIT-1B and TRANSIT-2A used *AEM-4750*, also known as *High Perm 49®*. A nickel–iron alloy, the latter is more lossy than *HyMu 80®* with a larger hysteresis major loop, yet with lower maximum permeability. Generally speaking, the larger the area of a material’s H-B major loop, the larger the energy losses will be during the material excitation. The exact trade-off is difficult to make without simulating the exact conditions of excitation because the magnetic field strength of the Earth is far lower than the magnetic field used to characterize the outer major loop of the material. This becomes clear after the experimental characterization of the SFL rods, and the produced H-B plot in Section 3.13. Thus, choosing materials that maximize hysteresis losses (i.e the area enclosed by the major hysteresis loop) given a magnetic field excitation is very important. Recently, [28] demonstrates that Grain Oriented (GO) silicon-iron ($SiFe$) alloys show much promise in maximizing energy losses for attitude detumbling. However, the material has no flight heritage in any space mission yet.

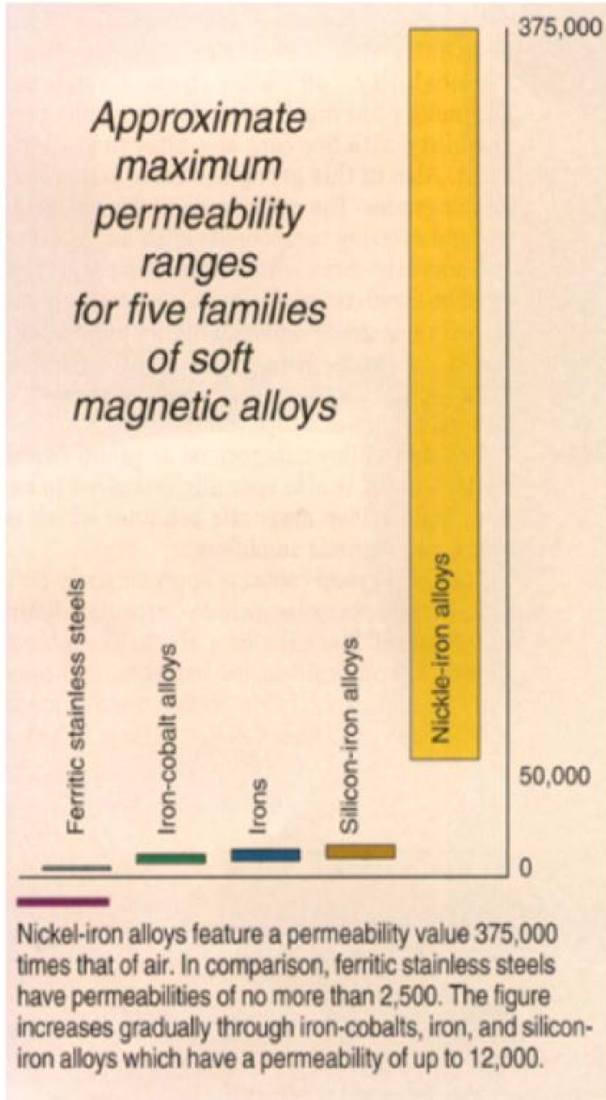

Figure 3.6: Comparison of permeability for different alloys. The vertical axis represents how many times the permeability of vacuum each material family demonstrates. [43]

HyMu 80® rods have been characterized for attitude detumbling of CubeSats by several sources, specifically by [26], [27] and [30]. The experimental setup of the latter reference shows more promise in validity of measurements due to the experiment being conducted in a Helmholtz cage that can isolate the measurements from the Earth's magnetic field. *High Perm 49*® rods will also be considered for the simulations, since it is the material of the rods in the laboratory's current inventory and due to its successful flight heritage.

3.12 Simulation Results

Table 3.1 below illustrates the important initial conditions of the simulations performed. Simulation #1 is the baseline simulation that was shown at the end of Chapter 2, yielding the largest spin-up observed by the spin-up permutation analysis (Section 2.6). All subsequent simulations use the same initial conditions for satellite body rate, attitude, and other modelling effects, with the only differences among them being the three parameters that characterize the hysteresis rods installed. The exact hysteresis parameters used for each simulation are shown in Table 3.1.

Table 3.1: Simulations performed and their initial conditions. Angular velocities are about the Sun-pointing axes

Sim Index	**Rods per satellite body axis**	**Material**	h_c	b_r	b_s	**Motivation**
1	0	N/A	N/A	N/A	N/A	Baseline sim, with spin-up conditions
2	1	*HyMu 80®*	0.3381	0.0006	0.300	Experimental properties [30]
3	1	*HyMu 80®*	12.000	0.004	0.025	Experimental properties [26]
4	1	*HyMu 80®*	1.590	0.350	0.730	Theoretical properties [30]
5	1	*High Perm 49®*	2.860	0.961	1.490	Theoretical properties
6	1	*HyMu 80®*	0.3381	0.004	0.300	Based on Sim #2, with larger b_r
7	1	*High Perm 49®*	0.609	0.0034	0.612	Scaled, based on Sim #2 and #6

As discussed, the simulations are performed using the NorSat-3 mechanical structure properties, with the setup as shown in the previous chapter in Figures 2.7 and 2.8. With the exception of Simulation #1, all simulations assume one hysteresis rod per spacecraft axis, of length 152.3 mm and diameter 6.4mm, which yield a length-to-diameter ratio of approximately 24, as shown in Equation 3.27. The mass of each rod is approximately 40 grams.

By observation of Table 3.1, the theoretical magnetic parameters for *HyMu 80®* alloy vary substantially with the measured results from [30] or [26]. Treating [30] as a more confident measurement due to the number of samples tested and the magnetic isolation of the setup in the Helmholtz coil, large discrepancies were still observed by the authors in magnetic properties even within the same lot of rods. However, it is important to note that such discrepancies should be expected in part because the theoretical magnetic parameters are a result of subjecting the specimen to maximum field amplitudes, in the order of 800 A/m. The averaged measured properties were collected from field cycles of 100 A/m, so it is expected for all measured values to be smaller because the specimens are cycled over a hysteresis loop that resides completely within the theoretical major loop. Discrepancies between specimens and the particularly high deviation of remanence b_r from the theoretical values have triggered the need for an experimental setup similar to [30] in our case, since rod performance needs to be validated at on-orbit magnetic field levels, in the order of 20-40 A/m. The test methodology followed in Section 3.13 relies largely on the same principles as that of [30].

Simulation #2 uses coercivity, remanence, and saturation values as those were determined through experimental fitted measurements from [30] for *HyMu 80*®. Simulation #3 uses another set of experimental values for *HyMu 80*®, as reported by [26]. Simulation #4 uses the theoretical values of *HyMu 80*® as reported by [30]. Simulation #5 uses the theoretical values of *High Perm 49*® as reported by the manufacturer of the SFL cylindrical rods in the inventory. Simulation #6 uses the same coercivity and saturation values as Simulation #2, but with a slightly higher remanence, to assess the effect of that parameter on the dissipative action in the simulation. This remanence value is selected at the upper range of experimentally observed values from the same source. It should be noted that *HyMu 80*® very high permeability, but very low coercivity and very low remanence, which minimizes losses so much as to not make it a good damping material, regardless of it having been used in some spacecraft missions. Parameters of Simulation #7 were determined using the following methodology. Based on the reported discrepancy of [30] between the theoretical and experimental values of b_r, b_s and h_c for *HyMu 80*®, a down-scale factor was determined for each of these parameters between their higher theoretical values and their lower experimentally determined value. Then, these down-scale factors were applied on the theoretical properties of *High Perm 49*®, in order to produce a set of pseudo-experimental magnetic properties, under the assumption that *HyMu 80*® and *High Perm 49*® would share the same discrepancy, if tested with the same methodology. This is done as an effort to introduce the same amount of margin to another material, for which no experimental data was found. After experimental testing of SFL's rods in Section 3.13, it is proven that the values match closely the results of the scaled-down remanence and coercivity, while saturation does not seem to scale linearly, but is well within acceptable margins. In the end, the experimental values of the rods prove to be adequate in dissipating the spin-up behavior.

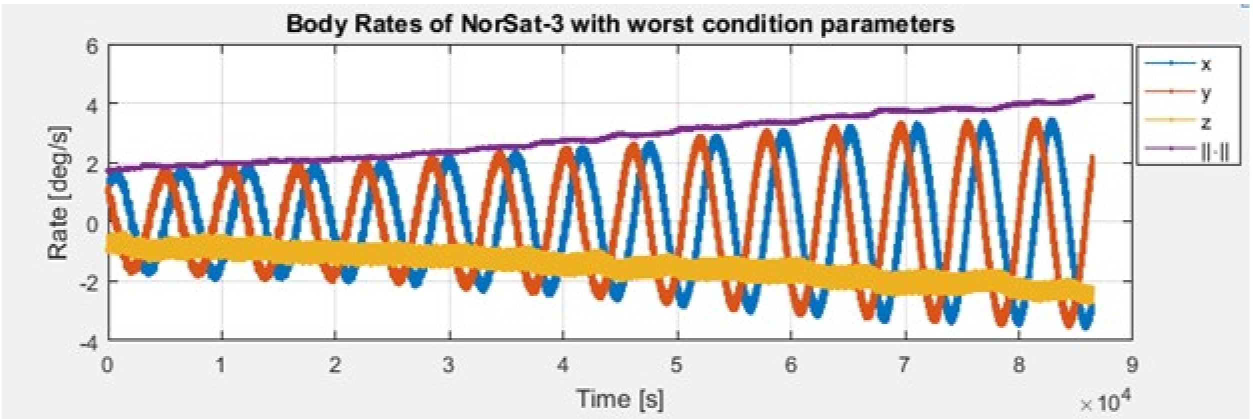

Figure 3.7: Simulation #1 Body Rates, no rods installed

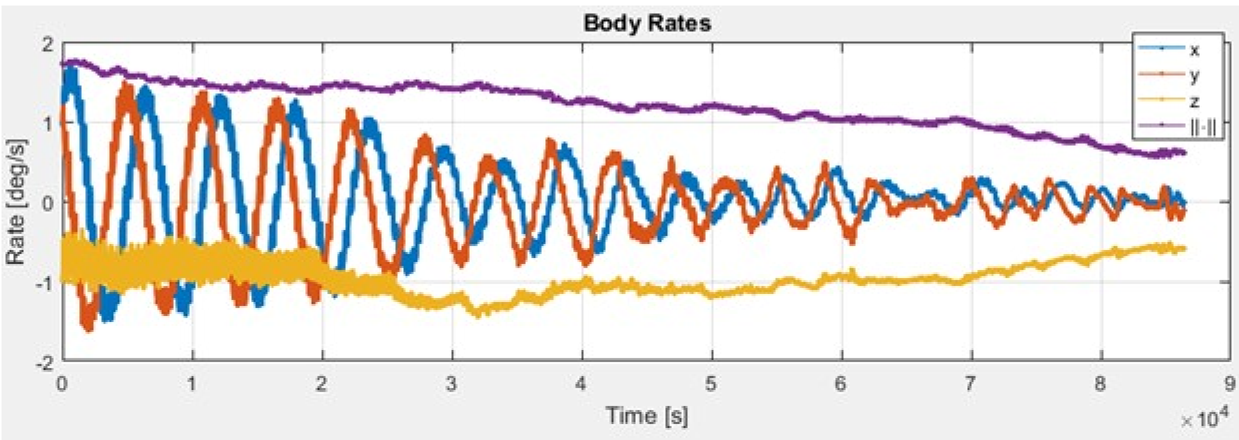

Figure 3.8: Simulation #6 body rates, using *HyMu 80®* rods

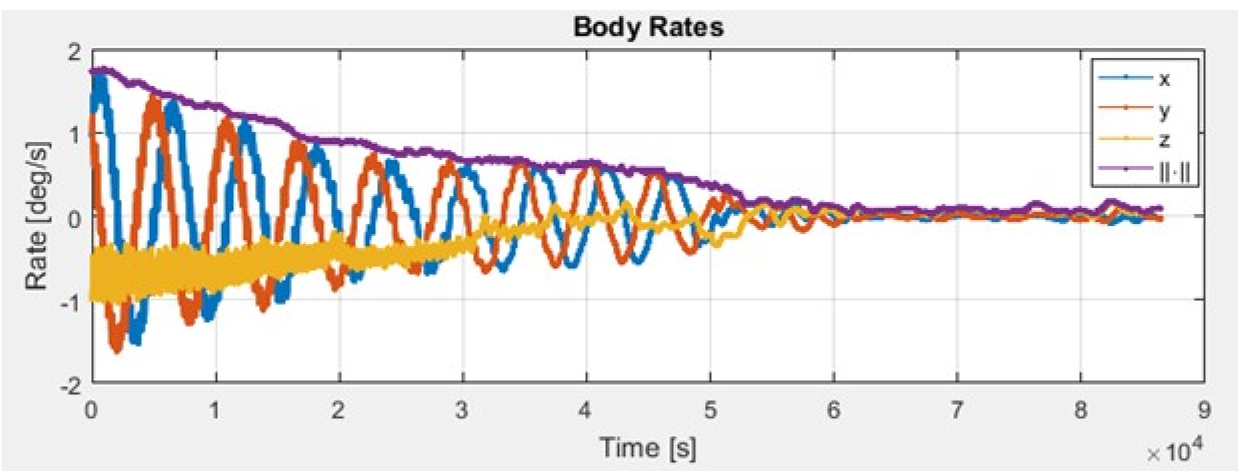

Figure 3.9: Simulation #7 body rates, using *High Perm 49®* rods

The results of Simulations #1, #6, and #7 are shown in Figures 3.7, 3.8, and 3.9 above.

Each figure shows the satellites body rates with respect to $\mathscr{F}_{\mathrm{i}}$, expressed in $\mathscr{F}_{\mathrm{b}}$, as well as the angular velocity norm, shown in purple color. In general, *High Perm 49®* seems to perform better than *HyMu 80®* mainly due to the overall higher coercivity and remanence, under the assumption that the same scale-down factors that apply to *HyMu 80®* will also apply to *High Perm 49®*. *High Perm 49®* is the same material used in SFL's NTS mission, designed around the GNB bus.

The simulation is performed using a simulator based on the core functions of the Mirage simulation environment, but with the added functions for computing the hysteresis damping torque, as described in 3.9.2. All environmental disturbances are being considered, and the worst-case initial conditions that trigger the spin-up are used, as explained in Chapter 2.11, Section 2.6. The next section expands on the experimental verification of the coercivity, remanence and saturation values used for *High Perm 49®* alloy.

3.13 Experimental Validation

This section illustrates the test results produced by in-house testing of the hysteresis rods. Six hysteresis rods of *High Perm 49®* alloy were tested. The material is also known as AER 4750, or MIL-N-14411C. Each rod has a diameter of 6.4 mm, length of 152 mm and weighs approximately 40 grams.

Few literature resources within the space community exist for the experimental setup required for hysteresis rod testing. A process using a single solenoid and a magnetometer attached at its one end is described in [26]. A similar setup is also used in [29]. ASTM standard ASTM A773/A773M provides a test method for direct-current magnetic property measurements of low-coercivity magnetic materials using hysteresis graphs [44]. However, the A773 standard suggests that the set-up is not optimal for hysteresis testing of nickel-iron alloys due to errors associated with integrator drift. ASTM suggests the use of ASTM A596/A596M to test soft ferromagnetic materials using the ballistic method, even though an oscilloscope with data capture and math function capabilities could avoid the integrator drift problem [45]. The procedures of the latter standard were used to perform the tests.

The procedure involves the electromagnetic induction of a secondary coil, by driving a sinusoidal current through a primary coil. The hysteresis rod fits tight within the secondary coil. The setup is placed within a Helmholtz cage that has been previously tuned with the help of a high-fidelity calibrated magnetometer to cancel out the local magnetic field within its bounding volume. The whole process is performed at SFL's magnetically clean chamber, in order to minimize the introduction of parasitic noise effects in the test. The magnetically clean chamber is part of the SFL ground facilities. It is a room shielded with in-wall steel beams, where magnetic measurements can be performed, isolated from the rest of the laboratory where spurious electromagnetic noise exists. The chamber also serves as a "dark room" where solar cells as well as other attitude sensors are tested. Within the chamber, a Helmholtz coil exists, driven by three independent power supplies. The Helmholtz coil is used to calibrate sensitive attitude determination sensors, such as magnetometers. The Helmholtz coil has been used as part of the test, to ensure a magnetically clean environment during the hysteresis rod characterization. As part of the experimental setup an operational amplifier is used at the output signal of the secondary coil to condition the signal. Both input and output signals are captured by an oscilloscope with integrator function capability and the results were stored in an external media for processing. Gerhardt and Palo [30] provide a detailed description of the experimental setup used, showing some resemblance to ASTM 596 practices. They used a Helmholtz cage as well as an instrumentation amplifier with the secondary coil to capture the B-H data through an oscilloscope.

A block diagram of the in-house experimental setup is shown in Figure 3.10 below. The block diagram features three types of connections, annotated with three different colors. Starting from the top-left, a computer drives three independent programmable power supplies, through the purple data connection *S_PC_1*. The power supplies are driven in such a way so that to maintain a stable sinusoidal magnetic field within the volume of the Helmholtz cage, by using the calibrated magnetometer in a closed-loop. The data connection with the magnetometer measurements to the PC is annotated as the purple line *S_PC_2*. Each of the programmable power supplies provides power to each coil in the 3-axis Helmholtz coil assembly, through the red power connection annotated as *P_HC_1*. The hysteresis rod with the sense coil is mounted inside the volume of the Helmholtz cage, and aligned with one of the coils axes. The applied voltage on this driving coil is input to the first channel of the oscilloscope, and this voltage signal connection is annotated as the green line *V_PS*. The secondary coil that surrounds the hysteresis rod is made out of AWG-40 copper-wire. Its senses the response of the hysteresis rod through the externally applied field from the Helmholtz coil, by producing a voltage across its ends. This low voltage of the sense coil is provided as input to the low noise amplifier through the signal connection *V_SC* to perform signal conditioning. In the end, the voltage signal from the output of the low noise amplifier is input to the second channel of the oscilloscope, through the connection labeled as *V_AMP*.

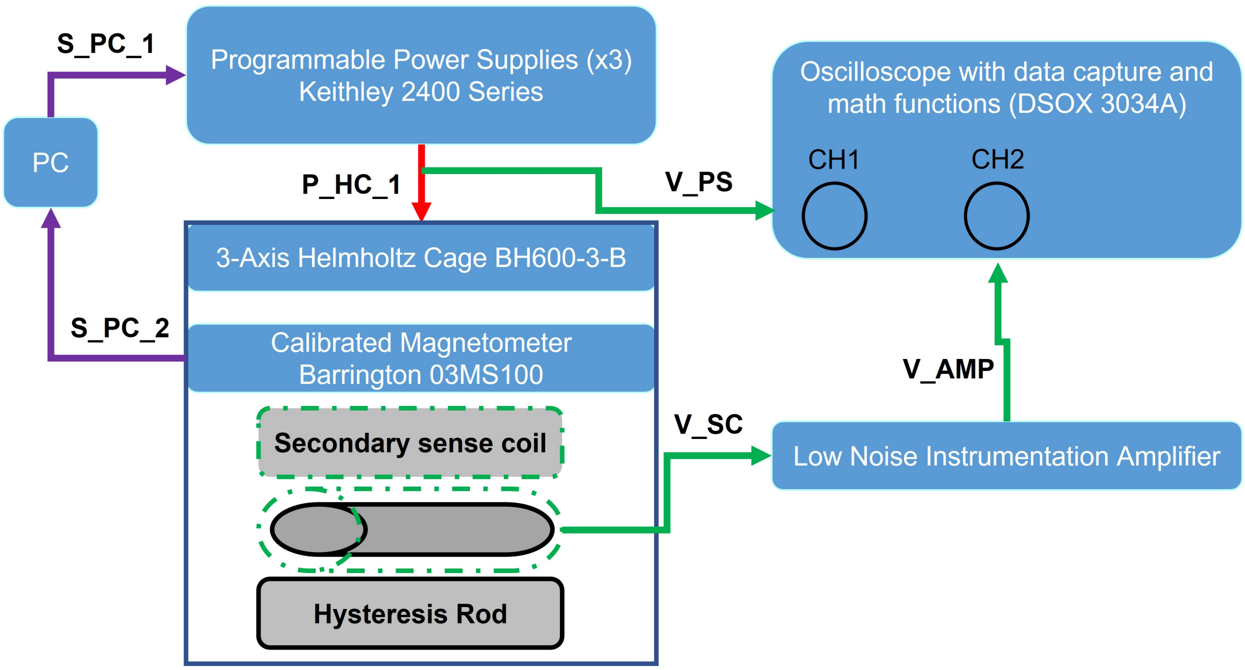

Figure 3.10: Block Diagram of the suggested experimental setup for hysteresis rod testing

It is noted that the experimental setup makes use of the Helmholtz cage for two reasons: First, it provides isolation from the external magnetic field environment by nullifying it, with the use of the 3-axis magnetometer as a feedback sensor. In addition, it provides an accurate source of a sinusoidal magnetic field within the cage's volume, by programmatically controlling the power supplies, operating in current source mode. In the middle of the cage, the secondary sense coil is installed such that its axial direction is parallel to one of the coil's axis as well as to that of the magnetometer's axis. The reason for this mounting solution is because it will require only one power supply to provide the required switching current/magnetic field, about a single axis. The hysteresis rod under test is placed within the sense coil, prior to its installation in the cage. With respect to the low noise amplifier, the AD8421 Op Amp Evaluation Board was used, due to its superior signal conditioning characteristics. The integration of the signal from the low noise amplifier (connection *V_AMP*) is performed using the oscilloscope. The signal needs to be integrated with respect to time before further being processed to a value representing magnetic flux density. Consider the secondary sense coil above, comprised of n number of turns, with a circular cross sectional area a. In addition, we define e as the induced voltage sensed by the secondary coil, amplified by the LNA (low noise amplifier), in units of Volts db. Then, the magnetic flux density difference Δb between current timestep t_1 and the previous timestep t_0 is given by the relation:

$$\Delta b = -\frac{1}{na}\int_{t_0}^{t_1} e\,dt \tag{3.28}$$

In order to ensure that the measurement will be characteristic of the hysteretic behavior of the material, the switching frequency of the magnetic field needs to be sufficiently low. Gerhardt and Palo, after discussing the test method used by Ivanov conclude that a frequency of 1Hz is sufficient, as it represents the golden mean between acquiring accurate measurements and the ability of the oscilloscope to sample both signals with adequate resolution. The latter is also known as ADC quantization, and represents the sensitivity of the oscilloscope in capturing faint signals with acceptable resolution. In addition, the switching frequency is related to the spacecraft's frequency of rotation, which should not be more than 0.1 Hz in magnitude. Therefore, switching frequencies in the order of 0.1-1 Hz should be implemented, with a preference to lower frequencies. The signal generator that was used was capable of frequencies as low as 20 mHz, which is considered as a quasi-static. The frequency of the applied sinusoidal signal was 28 mHz. Figure 3.11 below illustrates the signal generator that was used. The signal amplitude was 40 A/m resembling approximately the maximum magnitude of the magnetic field of the Earth in a Low Earth Orbit.

Figure 3.11: Tenma 72-7650 signal generator

Due to potential residual magnetization within the rod, the experimental procedure involved a degaussing process prior to the main test. Degaussing eliminates any direct current offset in the induced sensing voltage that would lead to the generation of asymmetric, not-centered hysteresis loops in the B-H plot. Hysteresis rods can maintain their magnetization to some extent depending on their short and long term storage environment, their distance from other ferromagnetic materials, handling, and any prior testing that has been performed.

Degaussing is a suggested technique that is also included in the ASTM A773 and ASTM ASTM A596 standards. For more information about the two standards, consult [44] and [45].

The process of degaussing involves the sourcing of a sinusoidal magnetic field with an exponentially decaying amplitude over time, starting from an amplitude that is large enough to ensure that little residual magnetization will remain in the end. Degaussing attempts to erase the hysteretic memory of the material as much as possible, so that the measurement that follows can start in the middle of the B-H plot and not outside of the expected hysteresis loop at the cycle amplitude of choice. Figure 3.12 demonstrates a degaussing profile. Figure 3.13 shows the sense coil assembly that was used during the experimental procedure. This assembly was installed within the Helmholtz cage and aligned with the coil that produced the sinusoidal magnetic field.

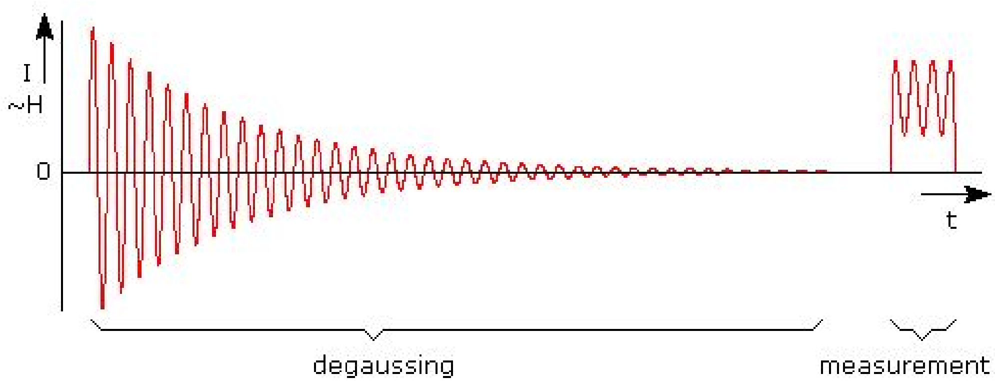

Figure 3.12: Degaussing profile. Prior to testing the rod, each sample needs to be degassed so that the any residual magnetism is completely removed from the specimen. [46]

The experimental procedure that has been followed is a robust method to characterize hysteresis rods in an isolated environment, at low frequencies which can be programmatically controlled to change sinusoidal or even match exact on-orbit changes in the magnetic field. Eddy current losses will be minimized due to low switching frequencies and the use of the instrumentation amplifier allows for testing at very low magnetic field strengths, provided that the common mode noise is sufficiently low.

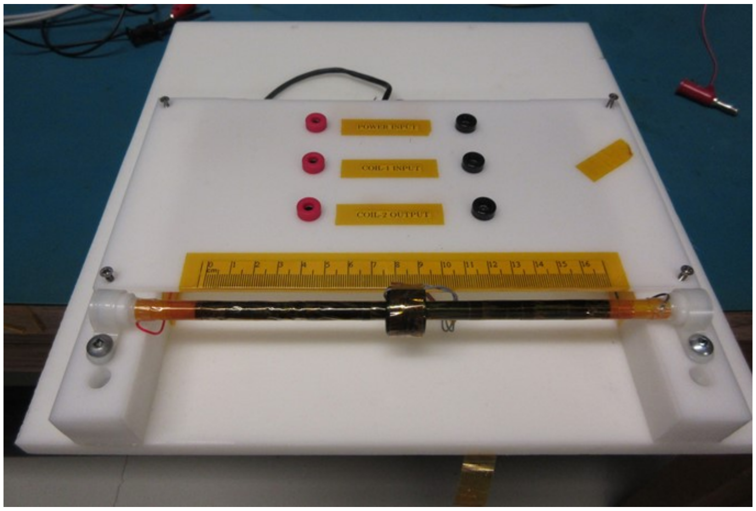

Figure 3.13: Primary and Secondary coil setup

The results of the hysteresis rod testing are processed to produce the H-B plots, presented in Figure 3.14 below. The H-B plot is also known as a *hysteresis graph*. The horizontal axis represents the value of the sinusoidal externally applied magnetic field, which is the independent variable. The vertical axis illustrates the value of the induced magnetic flux density within the hysteresis rod, as a response to the externally applied field.

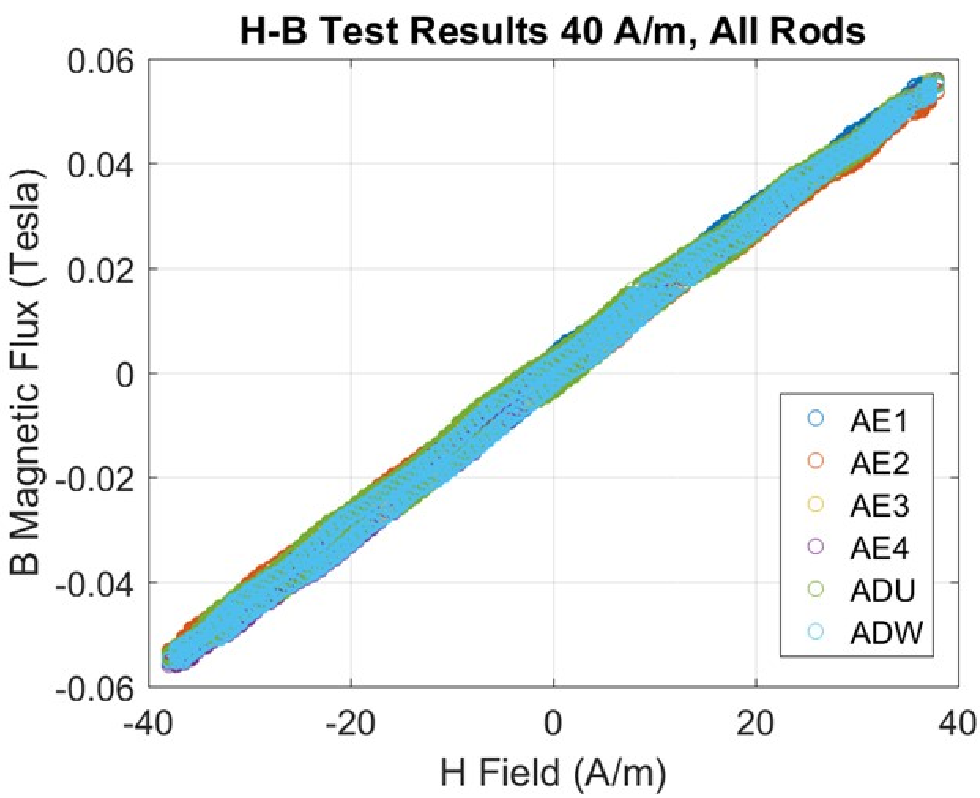

Figure 3.14: *High Perm 49*® hysteresis rod testing results

The plot illustrates the results of six cylindrical *High Perm 49®* hysteresis rods. The legend that accompanies the plot illustrates the part number that each hysteresis rod corresponds to. At SFL, all hardware follows a naming convention for parts, facilitating unit-testing and inventory purposes. Parts are typically named after a three-character alphanumeric code.

By observing the results, we see that all rods produce consistently similar curves. This proves that the cylindrical specimens have similar material properties. In addition, it is noted that the produced H-B curves are centered about the axis origin. This proves that the rods were effectively degaussed at the beginning of the test procedure. It also proves that the use of the Helmholtz cage was successful in isolating the hysteresis rod and sense coil from the Earth's magnetic field, while applying the sinusoidal field at the same.

By further examination of Figure 3.14, we notice that the curve of each rod does not exactly look like an arctangent function. This is expected, since according to [23], *the arctangent model approximates the characteristic major loop* of the material. When the material experiences magnetic fields that are smaller than the high magnetic fields that were used to characterize the rod, the material's response will reside within the major loop, forming leaf-shaped loops that slowly reassemble to an arctangent function as the magnetizing field increases sufficiently [23]. Furthermore, the low excitation magnetic field used produces curves that reside well within the characteristic sigmoid major loop shown in Figure 3.1. With regards to the magnetic parameters, we note that the remanence b_r for all rods is observed to be approximately 0.004 Tesla, the coercivity h_c is approximately 2 A/m, while the saturation b_s is approximately 0.056 Tesla. With the exception of saturation b_s, those values are all higher than the values used for *High Perm 49®* in Simulation #7, which represent a very conservative estimation of the rods dissipating capabilities. More specifically, we observe that, in relation to Simulation #7, the H-B curve that we have experimentally determined is wider (due to higher h_c) and also with slightly higher remanence, but with a lower saturation by a factor of 10. The discrepancy of the saturation means that the "pseudo-scaling" that was applied from the *HyMu 80®* material was not linear in nature. Regardless, it is noted that with regards to an H-B plot, the bigger the area that a hysteresis loop occupies, the larger the energy losses are. With a higher coercivity and remanence but smaller saturation, there is no quick conclusion that can be drawn at this point about whether the SFL rods are better or worse in dissipating rotational energy. Therefore, an additional simulation is needed with the aforementioned experimentally determined values of Figure 3.14.

The spin-up simulation was performed one last time with the averaged experimentally determined values of coercivity, remanence and saturation to observe the effect of the exact rod parameters on the detumbling performance. The results are presented in Figure 3.15 below, showing the satellites body rates with respect to $\mathscr{F}_{\mathrm{i}}$, expressed in $\mathscr{F}_{\mathrm{b}}$, in addition to the angular velocity norm, in purple color. The simulation was performed for a longer time span of ten days, and not just for one day, which was the time span of the previous simulations in Table 3.1.

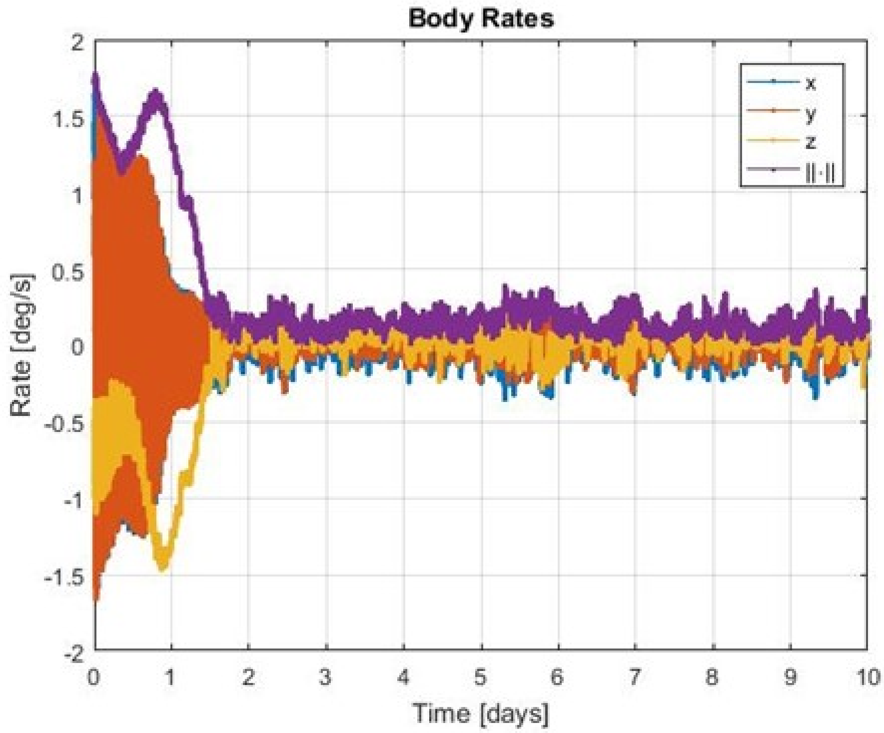

Figure 3.15: Final Simulation of spacecraft detumbling using averaged experimental hysteresis rod parameters

The addition of the rods with the experimentally determined values leads to complete stabilization of the body rates to within 0.5 deg/s per axis, with a steady state angular velocity magnitude no larger than 0.5 deg/s in total. This proves that the rod diameter, length, and material all contribute to a set of coercivity, saturation and remanence such that a spin-up initial condition of approximately 1 deg/sec per day can be avoided exclusively with the use of passive means. At this point, it is important to recall that this simulated spin-up and cancellation observed is based on careful setup of the initial conditions of initial body rates, attitude, and potential drag sail twist by a few degrees. Effectively, during the simulation, the forcing condition is still constantly present, as seen by the spin-up results of Simulation #1, mainly due to the assumption that the twist of the sails on the satellite structure is permanent, and unchanging. If other factors end up applying a constant torque on the satellite such that it spins up at a rate that is higher than simulated, then the stabilization of the satellite will take a longer time, and it is possible that more hysteresis rods will need to be added to ensure an adequate amount of dissipative torque. This will depend on the amount of drag sails installed, satellite geometry, and future improvements on the rigid body or integration of flexible dynamics within the simulation model, for better accuracy.

If we compare the results of Simulation #7 (Figure 3.9) with the results of this simulation on Figure 3.15, it is clear that in the latter case the stabilization occurs slower. In Simulation #7, stabilization was achieved within less than a day, while in the latest simulation, stabilization was achieved in approximately 1.5 days. Recall the discussion on the discrepancy between experimental magnetic parameters of our setup, and the parameters of Simulation #7, earlier in this section. Even though the coercivity and remanence of our rods were higher, the saturation of our setup was also determined to be a factor of 10 lower than Simulation #7. Also recall that the process of determining the parameters of Simulation #7 was through a judicious scale-down factor based on theoretical and experimental values of another material, due to lack of experimental data for *High Perm 49®* prior to testing our rods. This proves that our experimentally determined values are very conservative, with a carefully designed characterization procedure, and therefore more realistic. In fact, the literature review on hysteresis parameter characterization suffers from overestimating the dissipative capabilities of hysteresis rods. This has been briefly explained in the beginning of this chapter. Recall that [27] and [32] overestimated the stabilization time using hysteresis rods by as much as 60 days, which was confirmed from on-orbit telemetry afterwards. The experimentally determined magnetic parameters of the SFL rods allow for a more realistic approach on the dissipative capabilities of hysteresis rods, thus providing much clarity of the trade space of options. Corrective measures and improvements may include the addition of more rods, maintaining stricter mass discipline, changing drag sail configuration, or exploring other means of passive stabilization.

Finally, the hysteresis parameters and the rod geometry of the latter simulation were input to the Mirage simulations to assess the impact of the rods in the overall performance of the NorSat-3 active attitude control system. The simulations illustrated that all performance metrics were not significantly altered and all attitude control requirements were still met. This is expected, since the introduced torque due *High Perm 49®* is purely dissipative in nature, and would not contribute to a disturbance that would require feed-forward control or other means of active control to counter.

3.14 The Hyperbolic Model for Major Hysteresis Loops

As previously mentioned, the arctangent model used to perform the simulations of Section 3.12 is used as the basis for all simulations performed in [23], [25], [27] and [33]. In order to fit the experimental data to such a model, the user has the ability to vary the three magnetic parameters of coercivity, saturation and remanence. In the case where the experimental data may not be able to be fitted adequately, literature suggests that another simple model can be used instead, known as the hyperbolic model. It is noted that this model is provided for completeness of literature review, and also for future work on validation of the hysteresis rod testing results, as higher fidelity input to the simulations.

The hyperbolic model approach is based under the simple assumption that the magnetization process of a soft ferromagnetic material is the combined result of three distinct magnetization processes that occur in parallel and are each associated with a different magnetization mechanism of the domain walls; those processes are the domain wall movement, the domain rotation and the domain wall annihilation and nucleation. Each of these processes dominates the low, middle and near saturation region of magnetization respectively, and they can be separately modeled as hyperbolic functions. This model has been successfully used in a number of applications with well documented sources in the literature [47], [48] [22]. Among the space industry literature, the model has not yet drawn much attention.

The use of the hyperbolic tangent function to describe the hysteresis phenomena is not randomly selected. In fact, a formulation of the hyperbolic tangent function is an approximate solution to the integral form of the Preisach elementary operator, also known as Everett's integral. The reader is directed to [22] for more information on how the function is derived. In summary, the hyperbolic model is a simplified yet robust approximation to the stochastic Preisach model, considered as the most accurate model that exists to describe hysteresis behavior. The Preisach model takes into account the complete past magnetization history of the material to predict the magnetization, whereas the simpler models avoid this approach, to sacrifice accuracy for computational load, by considering only the immediately previous state of the material's induced flux density, and not its complete history.

3.15 Conclusion

In this chapter, we have demonstrated that the installation of three orthogonally mounted hysteresis rods within a NEMO class satellite equipped with two SFL drag sail modules can eliminate an undesirable spin-up condition that stems from an unfavourable initial condition of body rates, attitude, and potential drag sail twist of by a few degrees with respect to the satellite structure. After a thorough literature review on the phenomenology and fundamentals of magnetic hysteresis behavior of soft ferromagnetic materials, we have formulated an equation for computing the total dipole moment of a satellite due to the presence of hysteresis rods of arbitrary number, shape, and orientation within a satellite's structure. The equation has been used to model the effect of the hysteresis rods on providing dissipative rate-dependent torque, using theoretical as well as experimentally determined magnetic parameter values of remanence, coercivity and saturation. The rods considered in the experiment and simulation are 152mm length, 6.4mm diameter and made of *High Perm 49®*, which is a soft ferromagnetic material. Rod dimensions adequately fit within the mechanical structure of a NEMO class spacecraft bus. by comparing simulations using *HyMu 80®* and *High Perm 49®*, it is shown that *High Perm 49®* demonstrates better dissipative characteristics than *HyMu 80®*, mainly due to the higher coercivity and remanence. The latter characteristics assist in the increase of energy losses and the dissipation of rotational energy into heat. In addition, an experimental procedure for characterizing hysteresis rods in current SFL inventory has been performed. The verification process included two important features, namely the use of a Helmholtz case, and the use of a very low frequency sinusoidal signal with same amplitude as that experienced in Earth orbit. The results were fed back to the simulation to validate the stabilization performance and demonstrate compliance to the detumbling requirement.

Chapter 4

Passive Attitude Control for Death Mode Avoidance

A key enabler for highly successful microsatellite missions lies in the developer's focal attention to avoid death modes throughout the design life-cycle. A death mode is a condition when, in the absence of software or hardware failures, the spacecraft ceases to operate or to respond to commands. Death modes are the culmination of poor design choices (or lack thereof) during the trade study and can be related to one or many subsystems. In the Microspace doctrine, where cost savings are of paramount importance, diligent elimination-by-design of death modes is a way to maintain high mission reliability without investing unnecessary resources for overly redundant implementations.

One such example of a death mode relates to the spacecraft's inability to recharge its batteries, due to an undesirable attitude orientation. If, for example, the solar panels do not face the Sun for sufficiently long periods of time, the batteries may drain below a critical depth of discharge, such that recharging becomes impossible again. Similar examples from everyday life include electrical appliances equipped with batteries and poor circuitry design. Given enough time and negligence, a pair or rechargeable batteries equipping those devices will drain to such a point where recharging them will never be possible again.

Mitigating the aforementioned satellite death mode has been a key consideration for satellites developed at the Space Flight Laboratory. A design based on simple cuboid bus geometry with solar panels attached on every side can ensure power generation at any attitude, even at the loss of attitude control. The latter is a prime example of designing satellites with death mode avoidance considerations first. Modern attitude control hardware keeps breaking records pertaining to compact design and outstanding performance. As their miniaturization becomes affordable, adding more sensors and actuators for redundancy is now a feasible option in the developer's trade space, when mission requirements allow it. Highly capable attitude control has moderate power requirements that need to be accounted for in the satellite's power budget analysis.

A challenge emerges in the trade study when mission requirements call for the installation of large equipment within a satellite's limited volume. Such an example is the installation of a large optical aperture within a cuboid bus geometry. The Space Flight Laboratory has designed and flown a plethora of satellites with optical payloads, such as the successful GHGSat-D (Claire), HawkEye 360 Constellation, NEMO-HD, and many more. The installation of a large optical instrument on a cuboid satellite bus requires careful considerations for the thermal and electrical subsystems, among others. To ensure the fitting of the payload within the structure, the side on which the optical aperture is installed cannot be equipped with a solar panel (or solar strings, as commonly known by experts). This opens the possibility of an emerging death-mode. That death mode relates to the satellite pointing that face towards the Sun, for extended periods of time, without the ability to exit that condition (an inertial attitude lock). Without a solar panel attached on that face, the batteries could drain irreversibly. The inability to exit that condition can stem from the satellite being in a mode where no control authority exists (such as "Safehold mode"), while having a major axis spin with the no-cell face pointed at the sun. If the spin is about a minor axis, then it will eventually degenerate to a major axis spin, but care is required to ensure that this does not happen too late, or the major axis spin does not manifest to any other unforeseen death modes. The gyric stiffness introduced by the satellite's rotation can complicate the exit from an unfavourable attitude even more. Another set of conditions that could cause a solar stare death mode is the case where a 3-axis stabilized spacecraft with the no-cell face may point at the sun. At that moment, the satellite will loadshed and the reaction wheels will dump all their available momentum back to the spacecraft body. If the reaction wheels are orthogonally mounted to the satellite body frame, and if the reaction wheels perpendicular to the spin axis dump a significantly less amount of momentum compared to the aligned reaction wheel, then this inertial attitude lock can also take place. Although the aforementioned set of condition are quite remote and very unlikely to occur, they could be catastrophic for the mission. Abiding to the Microspace doctrine, this calls for a mitigation strategy.

This chapter investigates a solution such that the aforementioned "solar stare" does not occur. The solution has been applied with success to the next series of microsatellites designed by SFL with the mission of monitoring greenhouse gases, the GHGSat-C1 and GHGSat-C2 satellites.

4.1 GHGSat-C1 and GHGSat-C2 Mission

The increase of atmospheric concentration of greenhouse gases (GHG), such as carbon dioxide (CO2) and methane (CH4) is one of the factors contributing to climate change. Industrial operators are increasingly motivated by regulatory and operational imperatives to quantify their emissions, with the intent of ultimately reducing them [49] [50]. GHGSat Inc., based in Montreal, Canada, intents to become the global standard for greenhouse gas remote sensing. The Space Flight Laboratory designed, developed, and delivered the first satellite, GHGSat-D (Claire), launched successfully in June 2016. Since then, the satellite has successfully demonstrated greenhouse gas measurements around the world. GHGSat-C1 and GHGSat-C2 are the next two microsatellites that have been part of GHSSat satellite infrastructure [49]. GHGSat-C1 was successfully launched on September 3$^{\text{rd}}$ 2020, and GHGSat-C2 was successfully launched on January 24$^{\text{th}}$ 2021. With a mass of approximately 16 kg each, the design follows its predecessor Claire in leveraging SFL's Next Generation Earth Monitoring and Observation (NEMO) bus. Enhancements to the payload include reduced stray light, on-board calibration, and additional radiation mitigation.

Figure 4.1: Rendering of the GHGSat-C1/-C2 on orbit

4.2 GHGSat-C1 and GHGSat-C2 Satellite Platform

The following section outlines important features of the GHGSat-C1/C2 bus design, necessary for proceeding with the soar-stare death-mode investigation of this chapter.

4.2.1 Mechanical Design

The exterior design of GHGSat-C1/C2, along with the corresponding reference frame, is shown in Figure 4.2 below. The mechanical design of GHGSat-C1/C2 has been constrained to match the design of GHGSat-D as closely as possible. The –Y face is the direction on which the main and auxiliary payload optical instruments are mounted. It is the face that has no solar panels.

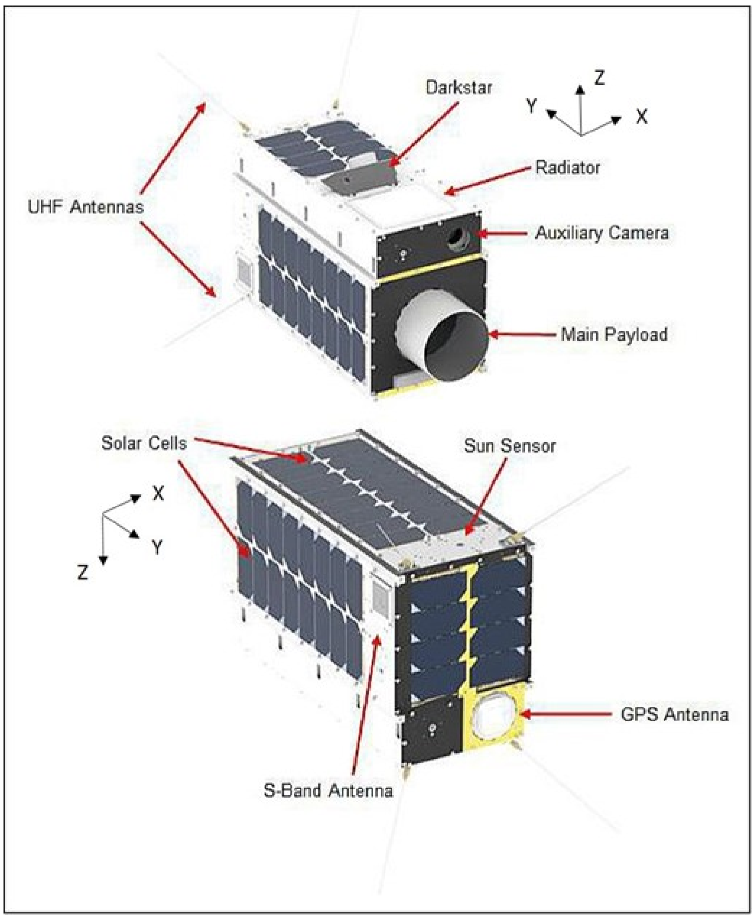

Figure 4.2: GHGSat-C1 Exterior solid model

Along the +Z axis of the satellite bus, a novel, dual-purpose instrument has been installed, known as Darkstar. Developed by Sinclair Interplanetary, Darkstar serves as a star tracker, as well as an optical downlink system. Darkstar's optical downlink system is designed to achieve rates up to 1Gbit/s, removing the system bottleneck of producing data faster than it can be downlinked to the ground [51]. As seen in Figure 4.2, Darkstar is installed on a bracket which is positioned at an angle with respect to the +Z face. The angle of this bracket has been optimally chosen to prevent impingement from the Sun or Earth in nominal operations.

4.2.2 Attitude Determination and Control hardware

With respect to attitude control sensors, the GHGSat-C1/-C2 suite is comprised of one three-axis magnetometer, one three-axis rate sensor, five sun sensors each mounted on one of the bus faces (except for the face where the optical aperture is mounted), as well as Darkstar operating as a star tracker for fine pointing attitude determination. With respect to attitude control actuators, the platform is equipped with three magnetorquers and four reaction wheels in skew configuration. The wheel configuration within the bus is illustrated in Figure 4.3 below. The fourth wheel provides redundancy with respect to arbitrary three axis stabilization, while the skew configuration allows for additional and equal control authority on all three satellite body axes.

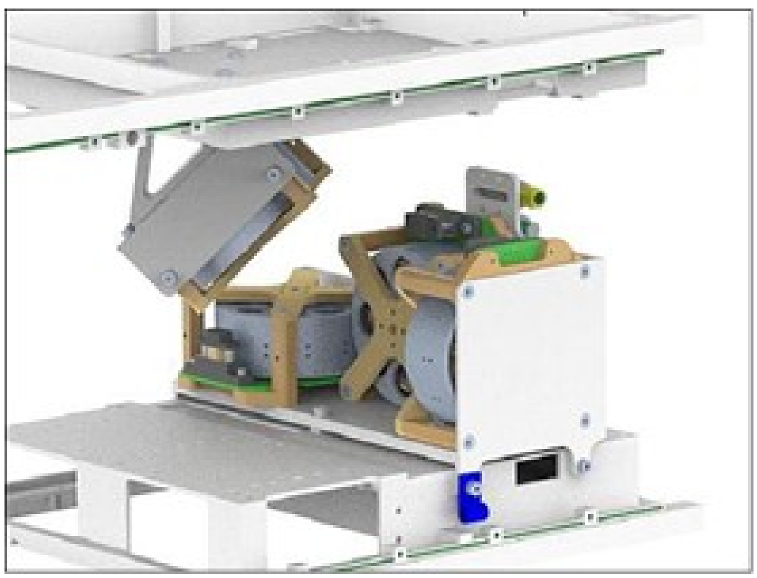

Figure 4.3: Reaction wheel configuration for GHGSat-C1/C2

4.3 GHGSat-C1/C2 Power Negative Solar Stare Analysis

This section will investigate the behavior of GHGSat-C1/-C2 bus staring at the Sun, in different initial attitudes. The attitudes considered here are based on the undesired orientation qualitatively described in Section 6.1; that is with the −Y panel, which has no solar cells, pointing at the Sun. A series of simulations will examine the effect of adding a permanent magnet on-board, such that the imparted magnetic dipole can provide the torque required to exit the undesirable attitude lock.

Five out of six faces of the spacecraft bus have solar panels mounted on the mechanical structure. Those solar panels are comprised of solar strings, which are a sub-assembly of solar cells. Each panel has a different number of solar strings, depending on the available surface area. The number of solar strings on the different panels is listed in Table 4.1 below.

Table 4.1: Number of solar strings on GHGSat-C1/C2 bus panels

Panel side	**No. of solar strings**
$+X$	2
$+Y$	1
$+Z$	1
$-X$	2
$-Y$	0
$-Z$	2

A power negative state can be realized if the –Y face is pointed at the Sun for extended periods of time. It is noted that this attitude is a possible death mode, because even if all satellite subsystems operate nominally, the satellite may inertially lock to an attitude that is unable to generate power. Given enough time, the state of charge could eventually drop below its critical threshold.

The inertia matrix **I** considered for the GHGSat-C bus is given as:

$$\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} = \begin{bmatrix} 0.346 & 0.002 & 0.004 \\ 0.002 & 0.144 & -0.005 \\ 0.004 & -0.005 & 0.304 \end{bmatrix} kg \cdot m^2 \tag{4.1}$$

The corresponding eigen-axes in the body frame are denoted as i_{xx}, i_{yy}, and i_{zz}. They are illustrated in Table 4.2 below:

Table 4.2: Principal moments of inertia and corresponding eigen-axes, expressed in the body frame of GHGSat-C1/C2

Eigen-Axis	**Principal Inertia** $kg \cdot m^2$	X	Y	Z
i_{xx}	0.344	0.996	0.016	0.094
i_{yy}	0.143	0.008	-0.999	0.034
i_{zz}	0.095	0.095	-0.033	-0.995

A number of simulations were performed to understand the conditions in which the satellite would end up in an inertially locked power negative attitude. Eleven (11) simulations were performed in total, with their initial conditions and rational outlined in the following section. The simulations were run using Mirage, SFL's high-fidelity attitude simulation platform. Mirage can interact with OASYS, SFL's On-orbit Attitude System Software, which is used as the foundation of the spacecraft's attitude control software stack. Its notable benefit to integrate with OASYS allows SFL to test the actual flight code before uploading to the spacecraft. Mirage contains models of the sensors and actuators developed over multiple missions based on ground testing and on-orbit results. It also has models of the spacecraft orbital dynamics, the astrodynamical ephemerides, and the attitude disturbance environment. When running simulations with Mirage, the developer can test the OASYS flight code, and examine and predict the performance of the AOCS subsystem architecture. For more information on the high-level architecture of Mirage and OASYS, [39] is a thorough reference. Examining the effect of a permanent magnet on-board of GHGSat-C satellites gives the developer the confidence of the solution's predictable behavior.

4.3.1 Initial Conditions

The two primary initial conditions under investigation are the initial attitude and the initial angular velocity, while simulations incrementally add the effects of environmental disturbances. With respect to the initial attitude, simulations are split between those that assume the -i_{yy} principal axis aligned with the Sun, while other's assume the body frame's -Y axis aligned with the Sun. The principal axis spin is more stable than the body frame -Y spin and comparing the behavior of those two scenarios in the absence of any environmental torques is a useful baseline. With respect to the initial angular velocity, some simulations are initialized with an angular velocity about the aforementioned axis with either 5 deg/s or 20 deg/s. By investigating Table 4.2 above, we observe that this spin condition is solely about a minor axis. According to [36] and [40], it is proven that minor-axis spins degenerate to major axis spins. In this case, the satellite would start off with a -i_{xx} spin, but eventually end up spinning about the principal i_{xx} axis. Both +X and -X panels are populated with solar strings, as noted in Table 4.1, so the initial bad attitude and resulting death mode will be eliminated. Yet, this study is performed to *evaluate the time interval* within which the satellite is in danger by the initial unfavourable attitude.

With respect to the satellite's orbit, variation of the exact orbital parameters of GHGSat-C1/C2 satellites do not need to form part of this analysis. The satellite is planned for injection in a Sun Synchronous orbit. It is known that the selection of the orbit's Local Time of the Descending Node (LTDN) can have an effect on the duration and periodicity of the sunlight – eclipse times. Although such differences could affect the power generation profile of the analysis, these power generation profiles are computed by combining the satellite inertial attitude and the inertial location of the Sun. Without any loss of generality, all simulations below were conducted for a 12:30 LTDN Sun Synchronous orbit, at an altitude of 720 km.

The initial angular velocities are about the axis pointed towards the Sun since this condition does not cause the satellite to be in a random tumble. An initial velocity of 20 deg/s and 5 deg/s about the –Y axis is comparable to dumping 568 rad/s and 142 rad/s from a 60 mNms reaction wheel, assuming a reaction wheel rotor inertia of 8.85 x 10-5 kg·m2, directly into that axis. The first three simulations are used to determine the difference in behavior between spinning about a principal and non-principal axis in the absence of environmental disturbances. The next two simulations are used to determine the effects of gravity gradient torques on the satellite motion. Simulations #6 to #11 are used to determine the effects of gravity gradient and magnetic disturbance torques, at various permanent dipole values and orientations. Aerodynamic and solar radiation pressure disturbances are not considered in the simulations because they are difficult to model and magnetic disturbance torques in practice are at least an order of magnitude larger, given the small area of the GHGSat-C bus. The last three simulations have spins about the -i_{yy} principal axis because these are the most conservative power limiting case. In addition, all simulations except #3 and #4 consider an initial angular velocity of 5 deg/s, as a more realistic scenario. In practice, it would be a remote scenario for a reaction wheel to unload momentum equivalent to the aforementioned 568 rad/s, because the attitude control algorithm performs reaction wheel momentum management to maintain reaction wheel speeds low (as well as the consumed power) and the control authority high. In any case, since this is a minor-axis spin, it would still reduce to major-axis spin without concern about the solar-stare death mode.

The power generated by the solar cells is considered in the analyses and is of focal importance. Each solar string is fixed to have a maximum instantaneous power generation of 8.5W, as derived from the power budget analysis for GHGSat-C1/C2 satellites. Consider the incidence angle θ as the angle between the solar panel normal vector and the Sun vector. The generated power of every solar string is a function of the incidence angle, specifically the cosine of the incidence angle. If we denote as p_inst the instantaneous power that a solar string generates, and as p_max as the maximum instantaneous power that the solar string can provide, then the power generated at any point in time is:

$$p_{inst}(t) = p_{max}cos(\theta(t)) \tag{4.2}$$

The incidence angle is a function of time and depends on the instantaneous orbital and attitude state of the satellite. In reality, it has been empirically determined that Equation 4.2 is not accurate for incidence angles below a certain threshold. More specifically, if the incidence angle θ is greater than 48 deg, then the following *Kelly cosine* relationship is used [52], replacing $cos(\theta(t))$ of Equation 4.2 as follows:

$$p_{inst}(t) = p_{max}(-0.018\theta(t) + 1.528) \tag{4.3}$$

Where $\theta(t)$ is expressed in degrees in Equation 4.3 above. An addition, if the incidence angle θ is greater than 84.8 deg, then no power is generated.

The GHGSat-C1/C2 satellites need to generate approximately 4 Watts of power to be power positive during Safehold mode, which is the satellite state with minimal power consumption. This value of 4 Watts is an average over the period of a complete orbit revolution. At an altitude of 720 km, the eclipse fraction f_e is 35.4%. We define $p_{gen,min}$ as the minimum amount of power generated during sunlight, and we also define as p_{safe} as the power generated during Safehold mode, averaged over an entire orbit. Since no power generation occurs during eclipse, the threshold $p_{gen,min}$ which allows the satellite to remain power positive during Safehold mode is given as:

$$p_{gen,min} = \frac{p_{safe}}{1 - f_e} = 6.2 \quad Watts \tag{4.4}$$

The end goal is to identify the direction and magnitude of permanent magnetic dipole that needs to be installed in GHGSat-C1/C2 such that power generation above 6.2 W can be achieved, while meeting the accuracy and stability requirements of the mission. This is achieved by proper sizing of a permanent magnet. The following simulations in Section 4.3.2 will contribute towards this goal.

Table 4.3: Simulations performed and their initial conditions. Angular velocities are about the Sun-pointing axes

Simulation Index	**Axis Pointed towards the Sun**	**Angular Velocity**	**Environmental Disturbances**	**Motivation**
1	-i_{yy}	5 deg/sec	None	Baseline simulation: Inertial -i_{yy} pointing, used to validate spacecraft dynamic stability
2	-Y	5 deg/sec	None	Baseline, body $-Y$ pointing, ensure zero power generation
3	-Y	20 deg/sec	None	Illustrate that Gravity Gradient is not enough to exit the inertial spin lock, body -Y sun pointing
4	-i_{yy}	20 deg/sec	Gravity Gradient	Same as #3, but with inertial i_{yy} Sun pointing, and high angular rate
5	-i_{yy}	5 deg/sec	Gravity Gradient	Same as #4, but with slow angular rate rate to illustrate looser gyric stiffness
6	-i_{yy}	5 deg/sec	Gravity Gradient and Magnetic	Addition of $+X$ dipole to illustrate faster exit from inertial lock (0.05 $A \cdot m^2$)
7	-i_{yy}	5 deg/sec	Gravity Gradient and Magnetic	Addition of $+Y$ dipole to illustrate faster exit from inertial lock. (0.05 $A \cdot m^2$)
8	-i_{yy}	5 deg/sec	Gravity Gradient and Magnetic	Addition of $+Z$ dipole to illustrate faster exit from inertial lock (0.05 $A \cdot m^2$)
9	-i_{yy}	5 deg/sec	Gravity Gradient and Magnetic	Same as #7, but stronger dipole (0.10 $A \cdot m^2$)
10	-i_{yy}	5 deg/sec	Gravity Gradient and Magnetic	Same as #7, but stronger dipole (0.15 $A \cdot m^2$)
11	-i_{yy}	5 deg/sec	Gravity Gradient and Magnetic	Same as #7, but stronger dipole (0.25 $A \cdot m^2$)

4.3.2 Simulation Results

This section summarizes the simulation results. The time span of all simulations is over a period of ten orbits. The reason as to why a duration of ten orbits is selected is explained in detail in Section 4.3.3 below. As a quick reference, the time span of ten orbits is equivalent to the amount of time that the satellite needs so that its depth of discharge is reduced by 40%, when operating in Safehold mode. Combining this with the discussion in Section 4.3.1, we are interested in identifying which simulations will demonstrate sustained average power generation of at least 6.2 W during sunlight intervals.

The results are visualized with the use of two figures per simulation. The first figure demonstrates the motion of the spin axis projected on a unit sphere with its origin attached on the satellite's Center of Mass, and with its unit vectors being aligned with the Earth Centered Inertial Frame $\mathscr{F}_{\mathrm{i}}$. For details on the realization of the ECI frame, consult Section 3.5.1. It is noted that the spin axis is equivalent to the satellite's angular momentum vector with respect to $\mathscr{F}_{\mathrm{i}}$, and expressed in $\mathscr{F}_{\mathrm{i}}$. The purpose of this figure is to visually assess how the spin axis will be "walking away" from its initial spinning attitude, in inertial space. Simulations that involve disturbance torques such as gravity gradient and magnetic torques are expected to produce strong motion of the spin axis, which is desirable because it is proof that the satellite will be quickly exiting any inertial attitude lock, a symptom of the solar-stare death mode. Simulations with less amount of disturbances, or even fully torque-free motion simulations, are expected to demonstrate the spin axis as more rigid in inertial space. This becomes more clear as the results are presented.

The second figure is a times-series plot demonstrating the amount of total power generated from all solar panels over time. For better readability of those plots, the time-series does not include the eclipse periods for which all power generation is zero.

It is noted that through all of those simulations of 10 orbit time span, minor to major axis spin degeneration was not observed. This mandates the use of a permanent magnet, as simulated in Simulations #6 to #11.

Simulation #1

Simulation #1 is the baseline simulation. The satellite initially has its -i_{yy} axis pointed at the Sun and is rotating about that axis at 5 deg/s, in the absence of environmental disturbances. Figure 4.4 shows the projection of the spin axis motion on a unit sphere aligned with $\mathscr{F}_{\mathrm{i}}$ and attached on the satellite's CoM, as explained previously. There are two small concentric circles that are visible on the surface of the unit sphere, a black circle, and a blue circle. The center of the blue circle represents the direction of the spin axis (i.e. angular momentum) expressed in $\mathscr{F}_{\mathrm{i}}$. It is denoted as a circle, instead of a point, so that it can be easily visualized on the larger surface of the unit sphere. The black circle surrounding the blue circle is in fact a series of black points forming a circle. Each of those points represents the direction of the body frame $\mathscr{F}_{\mathrm{b}}$ $-Y$ axis, expressed in the ECI frame $\mathscr{F}_{\mathrm{i}}$, and projected on the ECI unit sphere, attached on the satellite's CoM. The reason why the black points form a concentric circle around the spin axis is because there is a constant and fixed offset between the principal -i_{yy} axis and the mechanical -Y axis, and because the motion is torque free in Simulation #1.

Since the satellite is spinning about a principal axis, the inertial angular momentum (center of blue circle) aligns exactly with the proncipal axis and is unchanging throughout the simulation. Therefore in Figure 4.4, no "walk-away" behavior is observed.

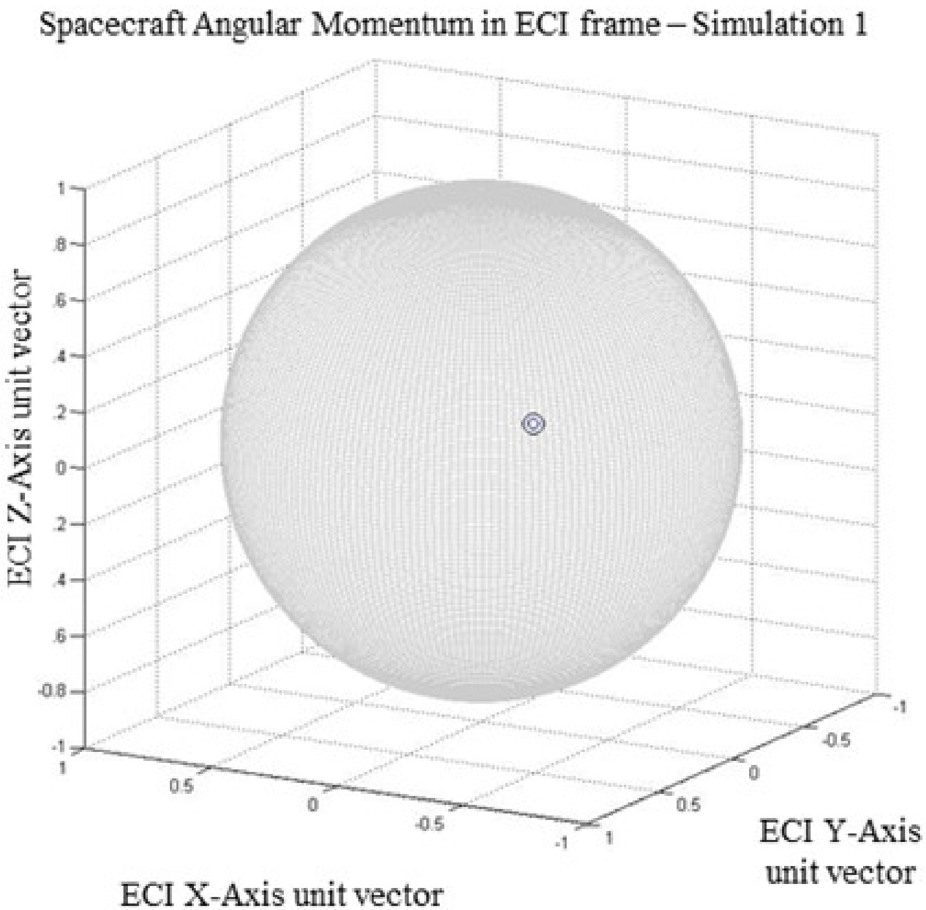

Figure 4.4: Simulation #1: Motion of satellite spin axis (angular momentum vector), projected in ECI frame $\mathscr{F}_{\mathrm{i}}$

Since the $+i_{yy}$ axis is approximately 2 degrees away from the body frame $-Y$ axis, a spin about $+i_{yy}$ is primarily exposing the $+X$ panel towards the Sun, for the duration of this simulation. As discussed in the previous section, the incidence angle of the Sun with that panel is so large during this period that no power will be generated (incidence angle greater than 84.8 deg). The average power generation in Simulation 1 is zero, as it is shown in

Figure 4.5 below.

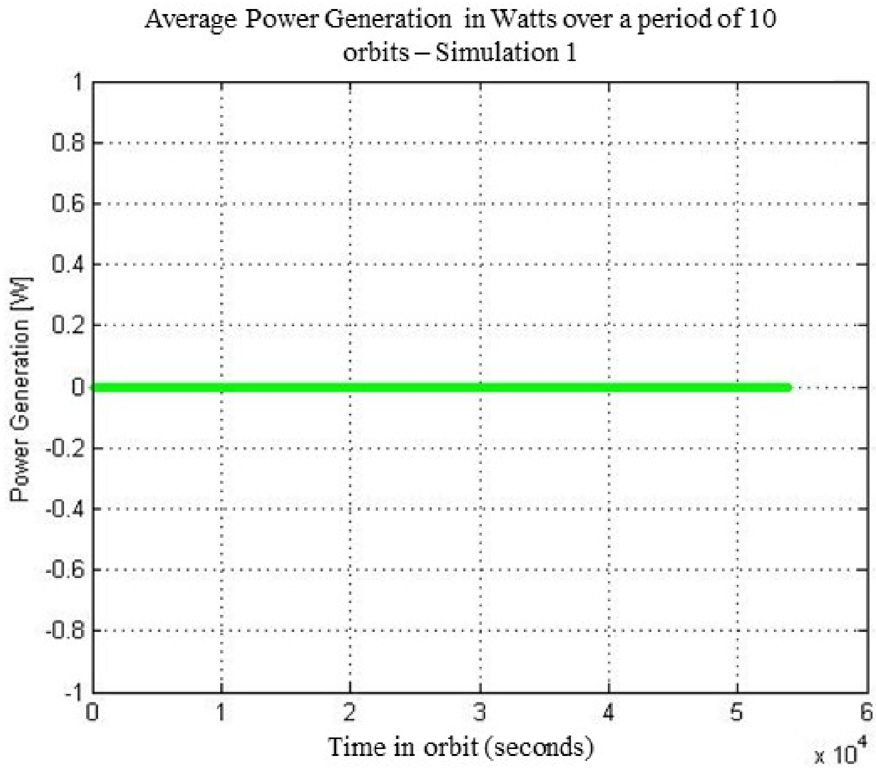

Figure 4.5: Simulation #1: Average power generated over a period of 10 orbits

It is reminded that this minor-axis spin will eventually degenerate to a major-axis spin. Instead of investigating when the minor-spin axis would degenerate into a major-spin axis that ends up being power positive, we investigate the power generated within the aforementioned fixed time interval of 10 orbits, which is a requirement from the power subsystem. As discussed, the motivation of those simulations is the addition of the permanent magnet so that the satellite generates the minimum Safehold power, before its batteries deplete below a certain depth of discharge, translated to orbit time of 10 orbits.

Simulation #2

Simulation #2 considers the satellite initially pointing its body frame $\mathscr{F}_\mathrm{b}$ $-Y$ axis at the Sun and rotating about that axis at 5 deg/s in the absence of environmental torques. Figure 4.6 illustrates the motion of the spin axis about the inertial angular momentum vector. The figure illustrates a unit sphere centered at the satellite's center of mass, and its unit vectors aligned with the ECI frame $\mathscr{F}_\mathrm{i}$. A blue circle and an intricate black shape are projected on the surface of the unit sphere. The center of the blue circle represents the direction of the satellite's angular momentum vector in $\mathscr{F}_\mathrm{i}$. Similarly to Simulation #1, this blue circle is fixed on the inertial sphere. This is because in both simulations, there are no external disturbance torque, thus the motion is torque free, and the direction of the inertial angular momentum is not changing.

The black shape is the pattern that the -$-Y$ axis is forming in inertial space. In contrast to Simulation #1, the spin axis is not inertially fixed, but it is rather dynamically stable as it perpetually processes about the inertial angular momentum vector. This happens because the inertia matrix is not diagonal; the -i_{yy} is not aligned with the -Y axis. Figure 4.6 only captures the attitude for the first 5 minutes of the simulation, otherwise the black pattern would not be recognizable and instead replaced by a black ring. Regardless, the motion of the spin axis is restricted within a narrow "ring" on the inertial sphere.

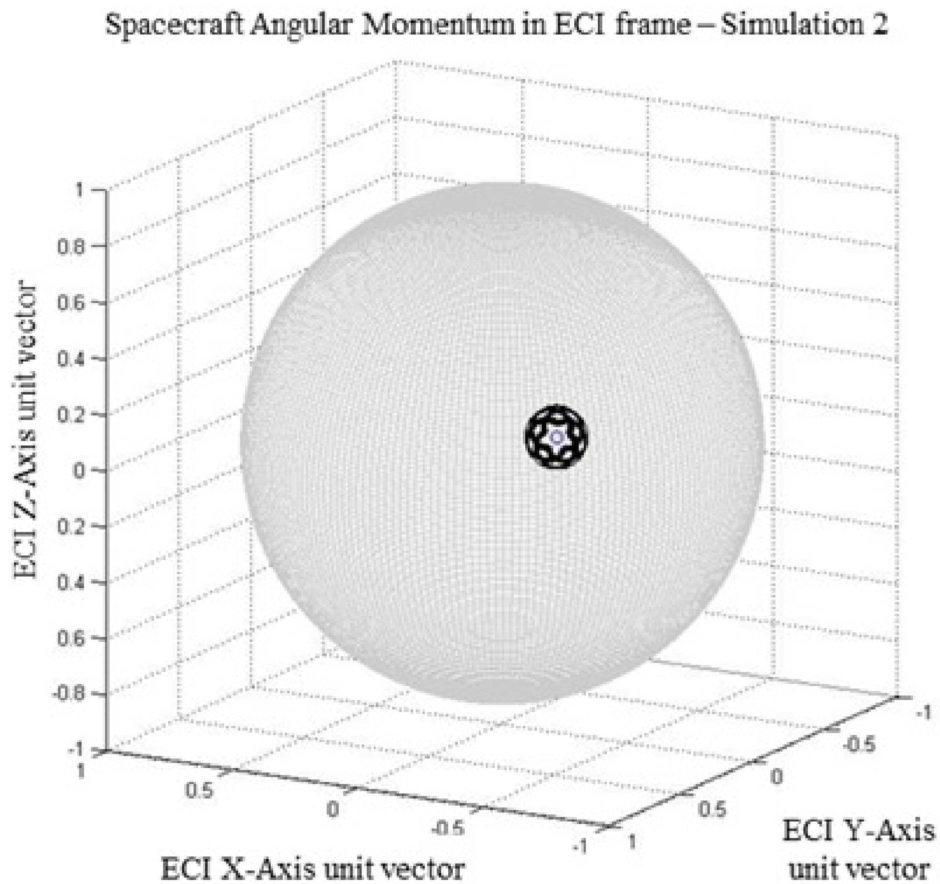

Figure 4.6: Simulation #2: Motion of satellite spin axis (angular momentum vector), projected in ECI frame $\mathscr{F}_{\mathrm{i}}$. the Angular momentum vector (center of blue circle) is inertially fixed due to absence of disturbance torques, while the spin axis motion (black shape) is dynamically stable within a narrow region on the satellite's inertial attitude sphere

In this simulation, the motion of the spin axis exposes slightly more of the body panels to the Sun, to the point where a small, yet insufficient amount of power is generated. Figure 4.7 shows the average power generated by the satellite's panels during sunlight. After a brief initial period until the satellite resorts to dynamic equilibrium, the satellite produces approximately 0.08 Watts of power during sunlight. It is reminded that Simulations #1 and #2 do not include environmental torques and no internal damping mechanisms. They are therefore a simplified depiction of reality, useful only as a baseline.

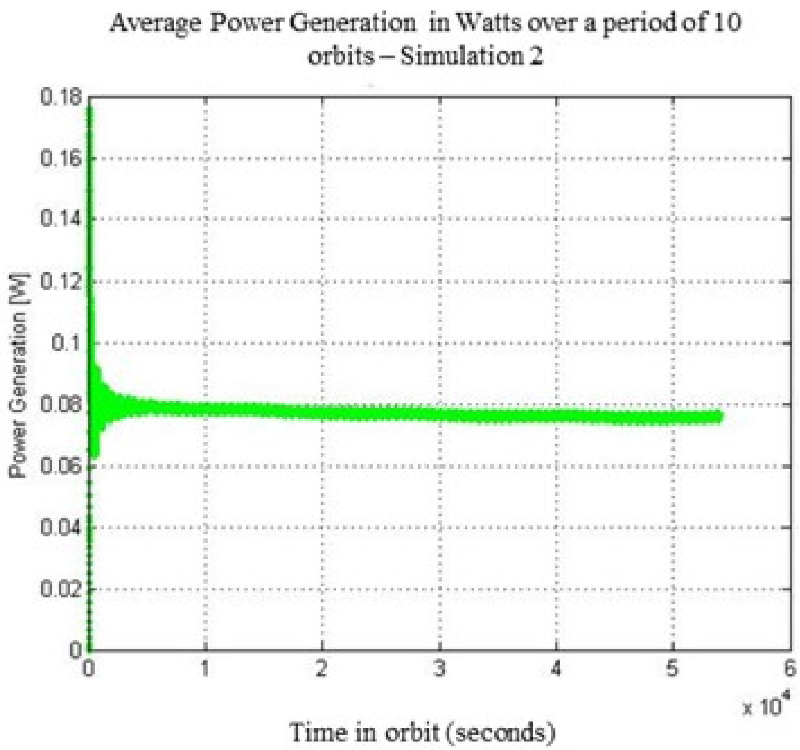

Figure 4.7: Simulation #2: Average power generated. After a brief stabilization period, the dynamically stable attitude generates a small but insufficient amount of power

Simulation #3

Simulation #3 considers the satellite initially pointing its –Y axis at the Sun and rotating about that axis at 20 deg/s in the presence of no environmental disturbances. It is similar to Simulation #2, differing only in the magnitude of angular velocity. Figure 4.8 illustrates the motion of the –Y axis (which is also the spin axis) with respect to the satellite's angular momentum vector, projected on a unit sphere attached at the satellite's center of mass, with its axes aligns with the ECI frame $\mathscr{F}_{\mathrm{i}}$. In comparison to the spin axis motion of Simulation #2, the spin axis in this simulation is also reaching a dynamic equilibrium, since its motion is also torque free. The black ring shown in Figure 4.8 is the path of the spin axis on the inertial attitude sphere of the satellite. It is clear that the angular momentum vector is inertially fixed, similarly to Simulation #2. The attitude is shown only for the first five minutes of the orbit, but it is clear that the faster angular rate clutters the visualization of the spin axis path, and no pattern can be easily recognized, as was the case for Simulation

#2.

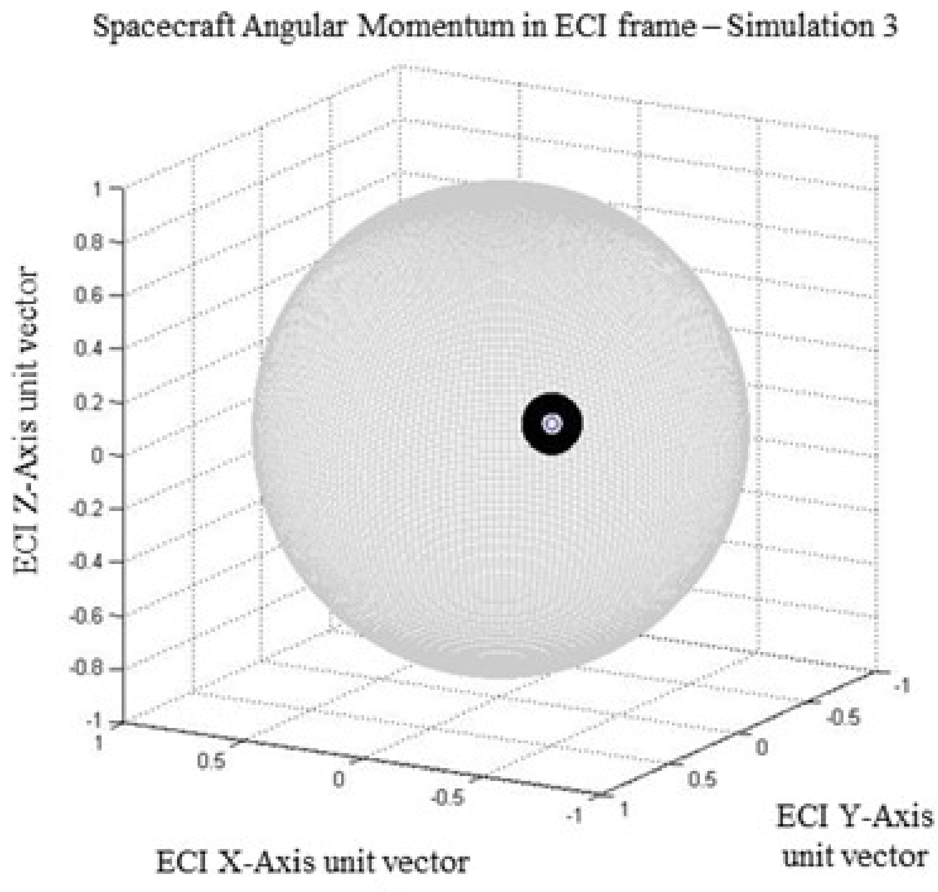

Figure 4.8: Simulation #3: Motion of satellite spin axis (angular momentum vector), projected in ECI frame $\mathscr{F}_{\mathrm{i}}$.

Figure 4.9 below shows the average satellite power generated during the simulation. Similarly to Simulation #2, there is a brief period of stabilization in the beginning , which manifests in the spike observed in this plot. The amount of generated power is similar to Simulation #2 at 0.08 Watts. This amount is still insufficient to keep the satellite power positive.

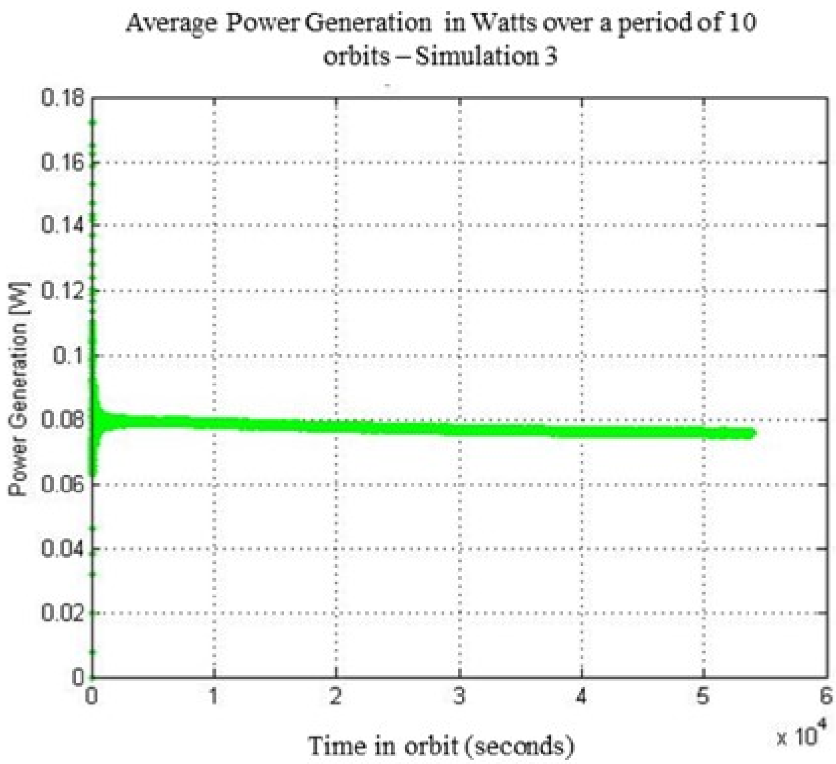

Figure 4.9: Simulation #3: Average power generated, very similar to Simulation #2. The increased angular velocity has no effect.

Simulation #4

Simulation #4 considers the satellite initially pointing its $-i_{yy}$ axis at the Sun and rotating about that axis at 20 deg/s in the presence of only gravity gradient disturbances. Similarly to the previous simulations, the motion of the -Y axis in the inertial frame $\mathscr{F}_{\rm i}$ is shown in Figure 4.10. In this case, we observe that the high angular rate of 20 deg/sec does not allow the spin axis (blue) or the -Y axis (black) to move around too much in inertial space, an artifact of the increased gyric stiffness. The motion of both axes in inertial space is restricted on a small footprint on the inertial sphere, yet there is minute amount of "walk-away" due to the presence of the weak gravity gradient torque.

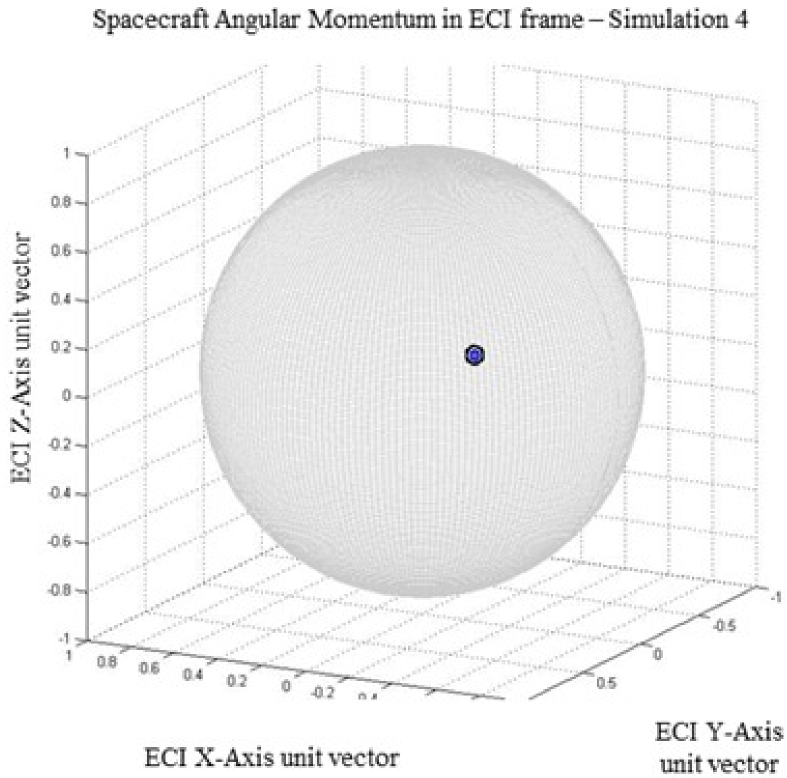

Figure 4.10: Simulation #4: Motion of satellite spin axis (angular momentum vector), projected in ECI frame $\mathscr{F}_{\rm i}$.

The total satellite power generation is shown in Figure 4.11. The addition of gravity gradient torque is not enough by itself to ensure exit from a power negative attitude. The attitude's stability leads to zero power generation.

If we compare Figure 4.4 to Figure 4.10, we observe that the inclusion of gravity gradient torque causes the inertial angular momentum, and consequently the $-Y$ face, to drift very slightly in inertial space, but still not being able to expose any of the other panels within a 10 orbit timeframe. As a result, Figure 4.11 demonstrates that zero power is generated.

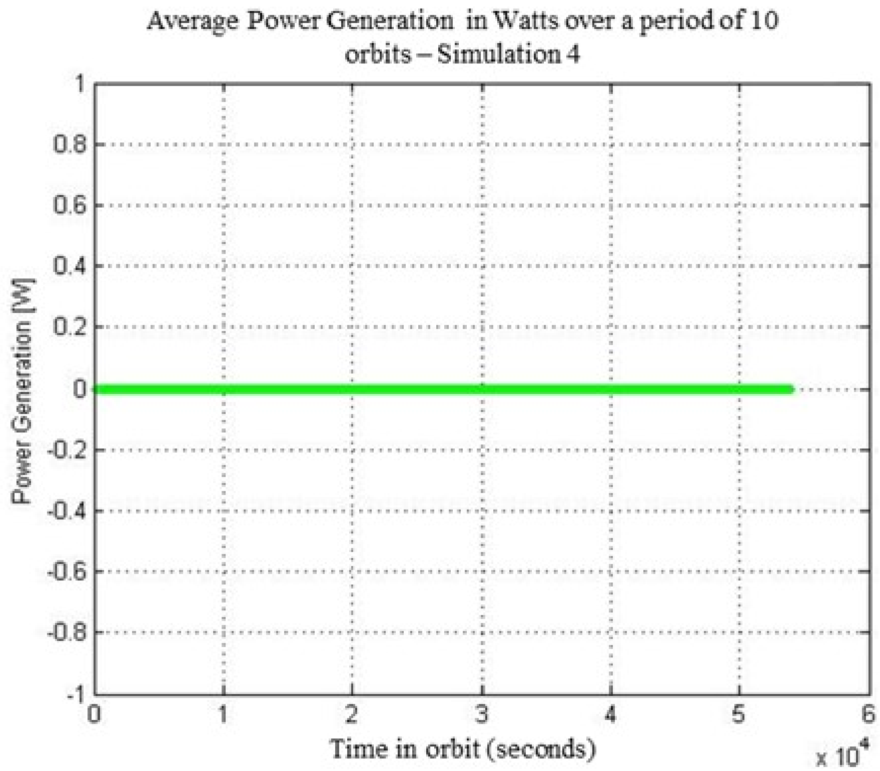

Figure 4.11: Simulation #4: Average power generated. The attitude's stability for this simulation leads to zero power generation

Simulation #5

Simulation #5 considers the satellite initially pointing its $-i_{yy}$ axis at the Sun and rotating about that axis at 5 deg/s in the presence of only gravity gradient disturbances. The inertial angular momentum and the $-Y$ axis, both expressed in the inertial frame $\mathscr{F}_\text{i}$ are shown in Figure 4.12. The blue circle represents the motion of the angular momentum vector expressed in $\mathscr{F}_\text{i}$ and projected on the unit sphere. The black circle represents the motion of the -Y axis, expressed $\mathscr{F}_\text{i}$ in as well. The two paths overlap significantly, but a very small amount of motion is visible. The addition of the gravity gradient torque, in combination with a small angular rate, leads to a tiny change in the direction of the satellite's angular momentum, but the fact that the spin occurs on a principal axis renders the motion fairly stable for the duration of the simulation. However, this is an indication that the addition of a disturbance torque can alter the inertial attitude. In subsequent simulations, the presence of magnetic disturbance torques will make that more clear. Comparing this simulation with Simulation #4, we note that the slow angular rate allows the spin axis to "wobble" a bit easier in the presence of gravity gradient torque. This is additional proof that the higher angular velocity in Simulation #4 introduced stronger gyric stiffness.

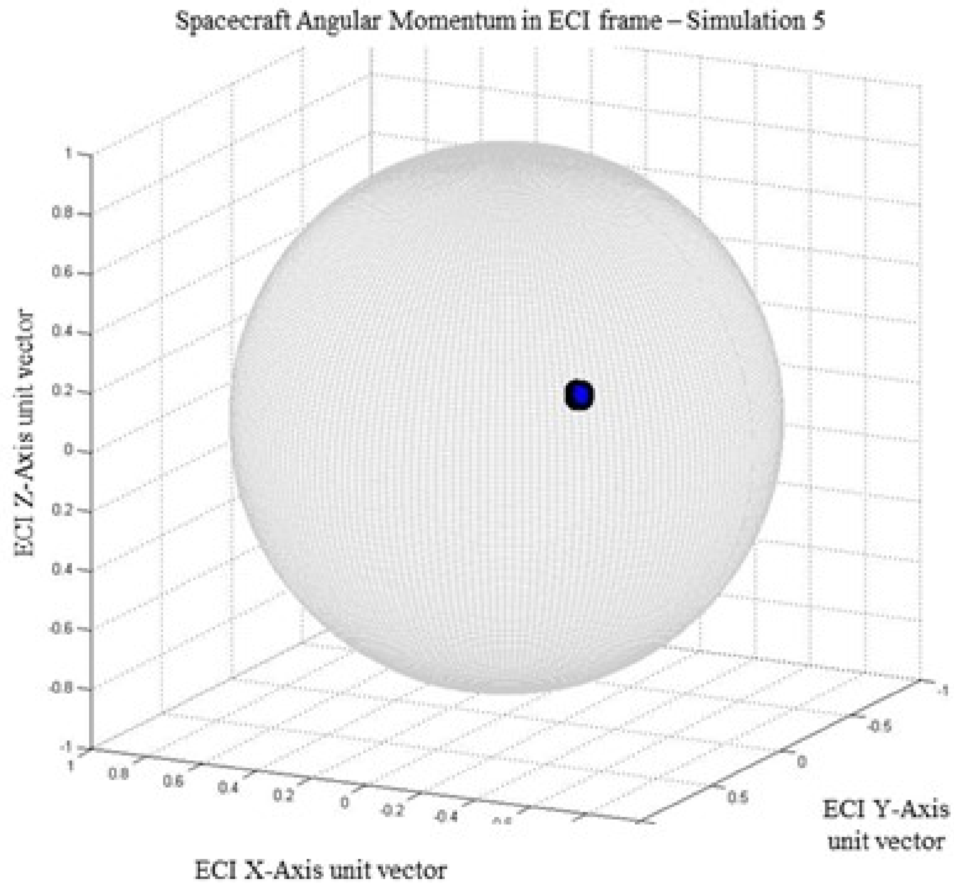

Figure 4.12: Simulation #5: Motion of satellite spin axis (black) and angular momentum vector (blue), projected in ECI frame $\mathscr{F}_{\mathrm{i}}$.

The total satellite power generation is shown in Figure 4.13. As expected, the average power generation is zero. This configuration is still insufficient to meet the needs of the satellite in Safehold mode.

The next simulations incorporate the inclusion of a permanent magnet to assist in exiting the observed attitude lock. It is reminded that the term ”lock” is only used with respect to the duration of this simulation, with a ten orbit timeframe, since minor-axis spins do degenerate to major-axis spins.

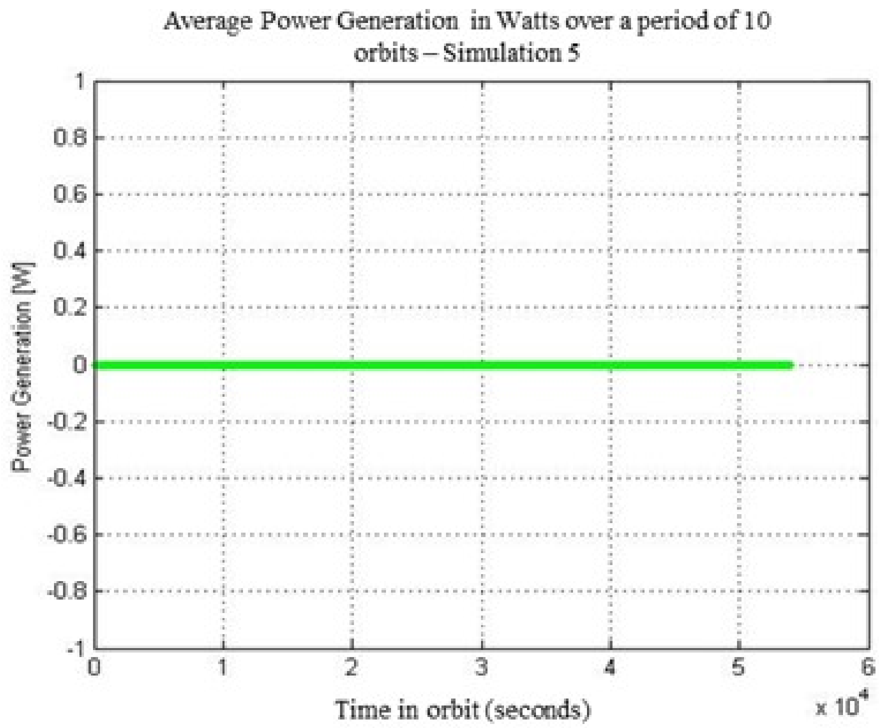

Figure 4.13: Simulation #5: Average power generated. Similarly to Simulation #4, the attitude's stability for this simulation leads to zero power generation

Simulations #6, #7, #8

Simulations #6, #7, and #8 consider the satellite initially pointing its $-i_{yy}$ axis at the Sun and rotating about that axis at 5 deg/s in the presence of gravity gradient and magnetic disturbance torques. Previous simulations have proven that relying on environmental disturbances alone does not guarantee that GHGSat-C1/C2 is safe from an undesired solar-stare death mode. Therefore, we need to magnetically bias the system such that it steers itself away from pointing the $-Y$ face towards the Sun for prolonged periods of time. A simple and cost effective method to magnetically bias the system is to install a permanent magnet. In this case, a key design choice is to identify the orientation at which the permanent magnet should be installed such that an undesirable attitude does not last at least as long as the 10 orbit requirement. The details of this requirement are discussed in the next section. The difference among these three simulations is the orientation that the magnetic dipole is installed with respect to the satellite body frame $\mathscr{F}_{\text{b}}$. Simulation #6 assumes that the permanent magnet is installed with its magnetic dipole vector aligned with the $+X$ axis, Simulation #7 assumes that the permanent magnet is installed with its magnetic dipole vector aligned with the $+Y$ axis, and Simulation #8 assumes that the permanent magnet is installed with its magnetic dipole vector aligned with the $+Z$ axis. For all three simulations, the same value of magnetic dipole moment is assumed, at 0.05 $A \cdot m^2$. This dipole value is fairly low and can be considered as parasitic dipole moment for the satellite. For reference, the operation of reaction wheels may produce low levels of dipole moment of the same order of magnitude, along the corresponding wheel spin axis. Subsequent simulations use stronger dipole values. The purpose is to comparatively identify which of the three body axes might be best suited for installation. The search of the magnet's optimal orientation might be of interest for future work. Regardless, installation along a body axis requires design of simple mounting brackets, and the added effort of the optimal orientation study might not contribute as much added value, if the requirement can be met at this level. Subsequent simulations investigate the sensitivity of the attitude to increasing values of dipole moment.

The motion of the -Y axis as well as the motion of the angular momentum vector expressed in $\mathscr{F}_{\rm i}$ and projected on the $\mathscr{F}_{\rm i}$ sphere is shown in Figures 4.14 for Simulation #6, 4.15 for Simulation #7, and 4.16 for Simulation #8.

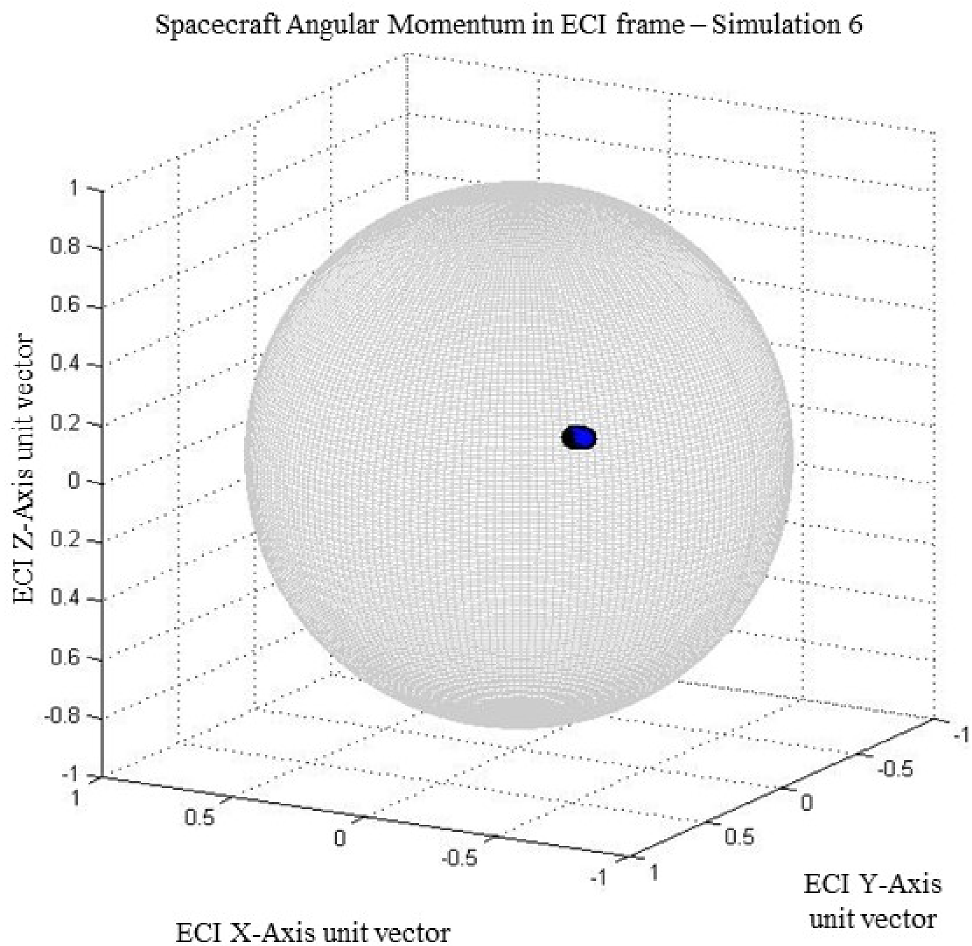

Figure 4.14: Simulation #6: Motion of satellite spin axis (black) and angular momentum vector (blue), projected in ECI frame $\mathscr{F}_{\rm i}$.

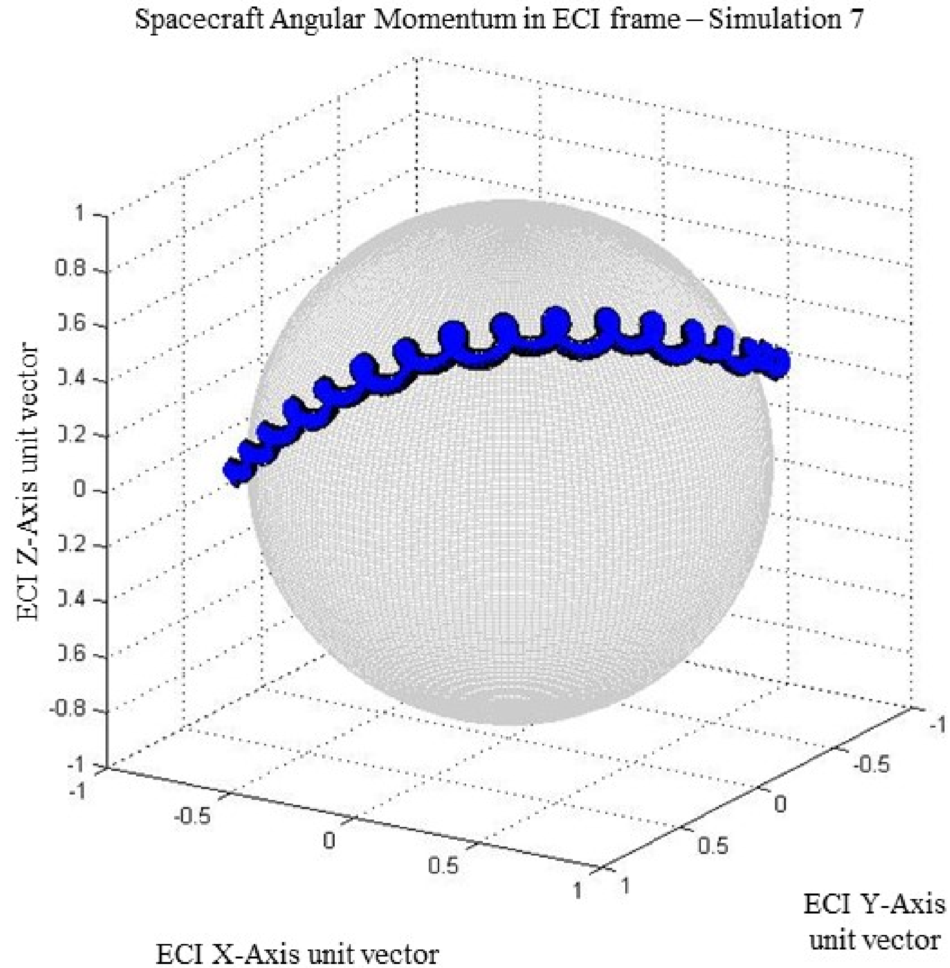

Figure 4.15: Simulation #7: Motion of satellite spin axis and angular momentum vectors, projected in ECI frame $\mathscr{F}_{\mathrm{i}}$.

By observing Figures 4.14, 4.15 and 4.16, we can see that when the magnet is installed along the $+Y$ axis of GHGSat-C1/C2, the satellite's attitude is strongly disturbed from the initial condition. In Figure 4.15, the blue path depicts the inertial direction of the angular momentum vector over time, while the black path depicts the inertial direction of the -Y axis over time. The black path closely follows the blue path, and therefore might be difficult to discern. It is reminded that the two paths do not match because of the misalignment between the principal axes frame and the body frame. This figure illustrates a desirable "walk-away" effect of the satellite's angular momentum vector, which means that even though the satellite started in an undesirable orientation, it has quickly diverged from its initial pointing direction, allowing exposure of all other solar-string mounted panels to the Sun. In order to visualize how quickly this "walk-away" occurs and whether it is sufficient for generating the minimum amount of power required for Safehold mode, we need to investigate the average power generated.

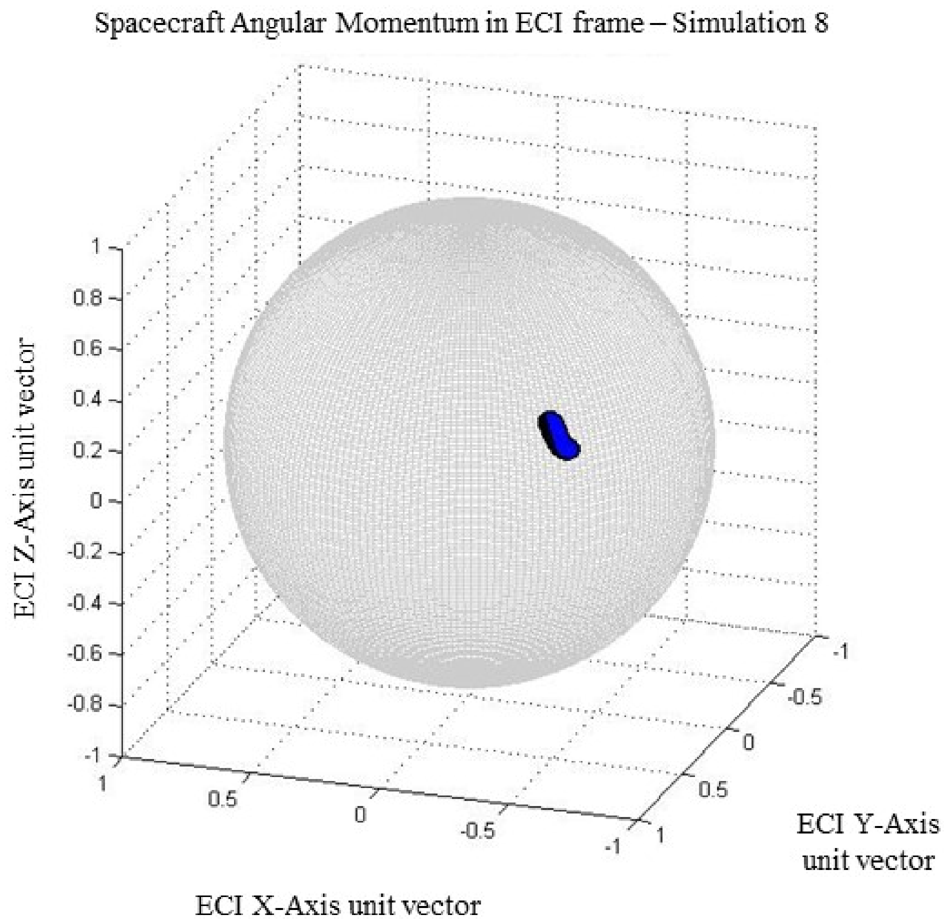

Figure 4.16: Simulation #8: Motion of satellite spin axis (black) and angular momentum vector (blue), projected in ECI frame $\mathscr{F}_{\mathrm{i}}$.

Figure 4.17 below illustrates the average power generated during sunlight. For better readability of the figure, eclipse periods where satellite produces zero power have been removed from the time-series plot. Regardless, the average power generation is computed over the complete course of the orbit. The figure shows that after approximately 3.3 orbits, the satellite is able to produce more than 6.2 Watts. After approximately 7 orbits, power generation is just shy of 12 Watts. This is a good indication that a dipole of 0.05 $A \cdot m^2$ meets the aforementioned requirement. It is concluded therefore that the $+Y$ axis is the best body axis along which the permanent magnet should be installed. The next simulations experiment with stronger dipole values along the $+Y$ axis to gauge the sensitivity of the attitude and resulting power generation.

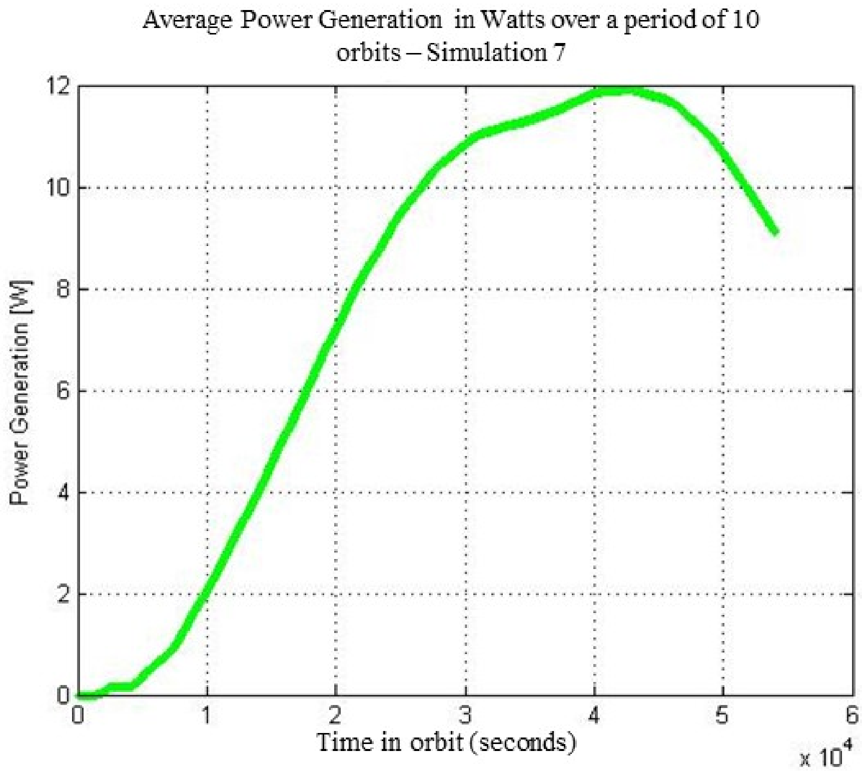

Figure 4.17: Simulation #7: Average power generated. Safehold power generation is achieved after 3.3 orbits.

Simulation #9

Simulation #9 considers the satellite initially pointing its $-i_{yy}$ axis at the Sun and is rotating about that axis at 5 deg/s in the presence of gravity gradient and magnetic disturbances, but with a magnetic dipole that is double of that of Simulations #6, #7, and #8. As discussed in the previous section, the previously chosen dipole value of 0.05 $A \cdot m^2$ falls within the noise regime of parasitic magnetic dipole that could be generated by solar strings loops, electronic equipment, or even reaction wheel motion. By doubling the magnetic dipole, the system is biased more strongly towards the indented direction. The reason why this is done is because, in practice, it is exceedingly difficult and costly to perform a complete magnetic cleanliness study at the satellite system level, particularly when abiding to Microspace design principles. Therefore, stronger magnetic bias will provide more confidence in the expected result, but care is required such that it does not interfere with pointing and control error requirements. With this simulation, the intention is to identify how faster the satellite will reach the minimum power generation threshold of 6.2 Watts.

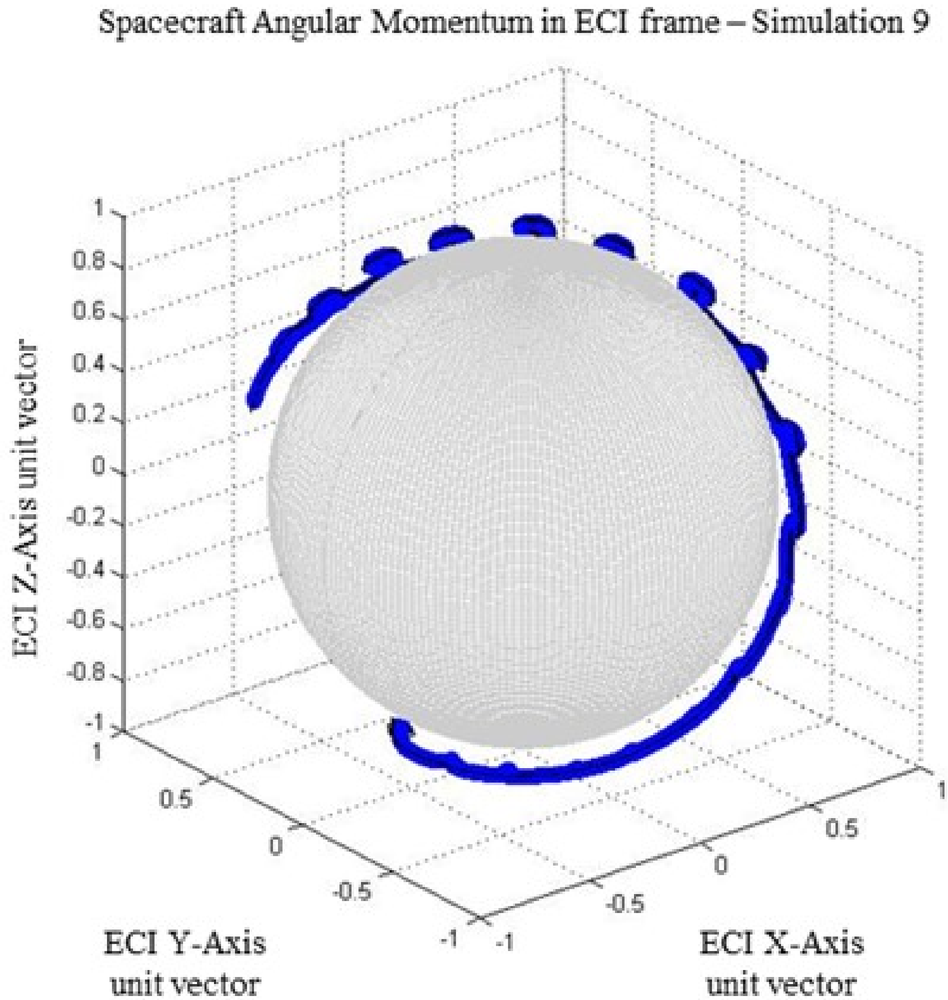

Figure 4.18: Simulation #9: Motion of satellite spin axis (black) and angular momentum vector (blue), projected in ECI frame $\mathscr{F}_\text{i}$.

The motion of angular momentum vector (blue) and the $-Y$ axis (black) over time both expressed in the ECI frame $\mathscr{F}_\text{i}$ and projected on the $\mathscr{F}_\text{i}$ unit sphere are shown in Figure 4.18. The figure is rotated to illustrate that the motion of the two vectors is quite vivid, and almost completes a full rotation about the inertial unit sphere. This means that this configuration is also going to expose enough of the panels with solar strings, leading to power generation. To assess that quantitatively, the satellite's generated power over time is shown in Figure 4.19.

The addition of the 0.10 $A \cdot m^2$ dipole installed along the $+Y$ axis shows that the satellite starts producing more than 6.2 Watts after approximately 2 orbits. Comparing that with the 3.3 orbits required when equipped with a 0.05 $A \cdot m^2$, this simulation shows that it is a better suited configuration.

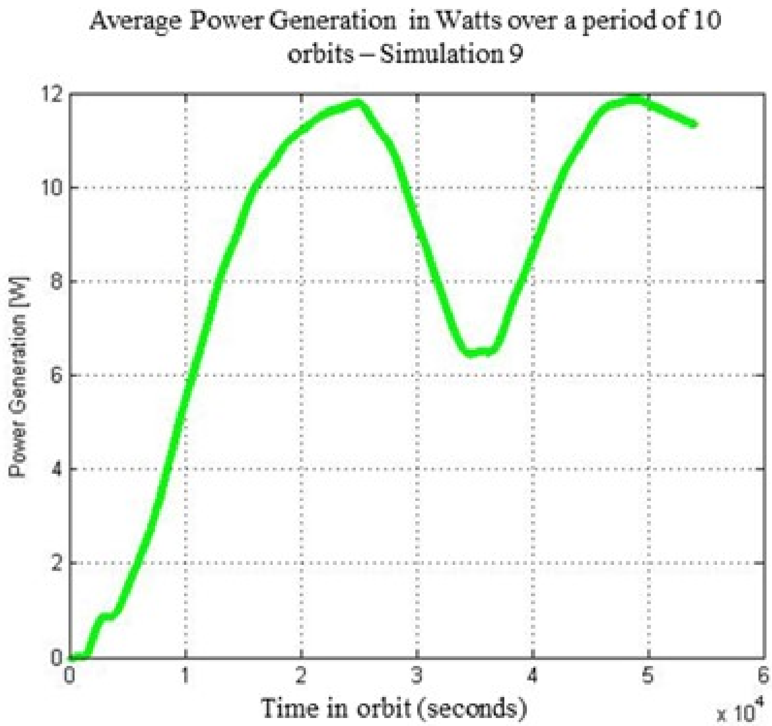

Figure 4.19: Simulation #9: Average power generated. Safehold power generation is achieved after 3.3 orbits.

Simulation #10

Simulation #10 considers the satellite initially pointing its $-i_{yy}$ axis at the Sun and rotating about that axis at 5 deg/s in the presence of gravity gradient and magnetic disturbances, with a magnetic dipole of 0.15 $A \cdot m^2$ installed along the $+Y$ axis. The motion of the inertial angular momentum and the $-Y$ axis, both expressed in the inertial frame $\mathscr{F}_{\mathrm{i}}$, are shown in Figure 4.20. The figure illustrates that the motion of the -Y axis is considerably disturbed from its initial condition, allowing the exposure of a larger portion of the other satellite panels towards the Sun vector.

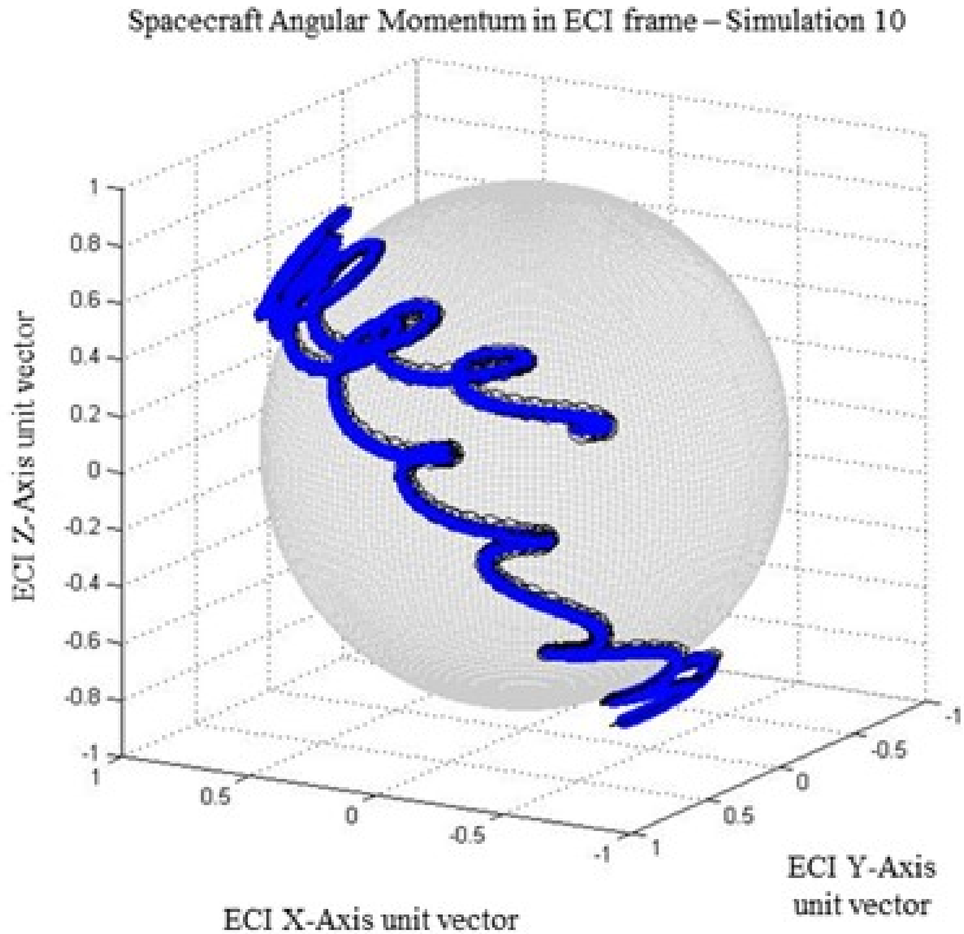

Figure 4.20: Simulation #10: Motion of satellite spin axis (black) and angular momentum vector (blue), projected in ECI frame $\mathscr{F}_{\text{i}}$.

The total satellite power generation is shown in Figure 4.21. The addition of the 0.15 $A \cdot m^2$ dipole installed along the $+Y$ axis shows that the satellite starts producing more than 6.2 Watts after approximately 1.5 orbits.

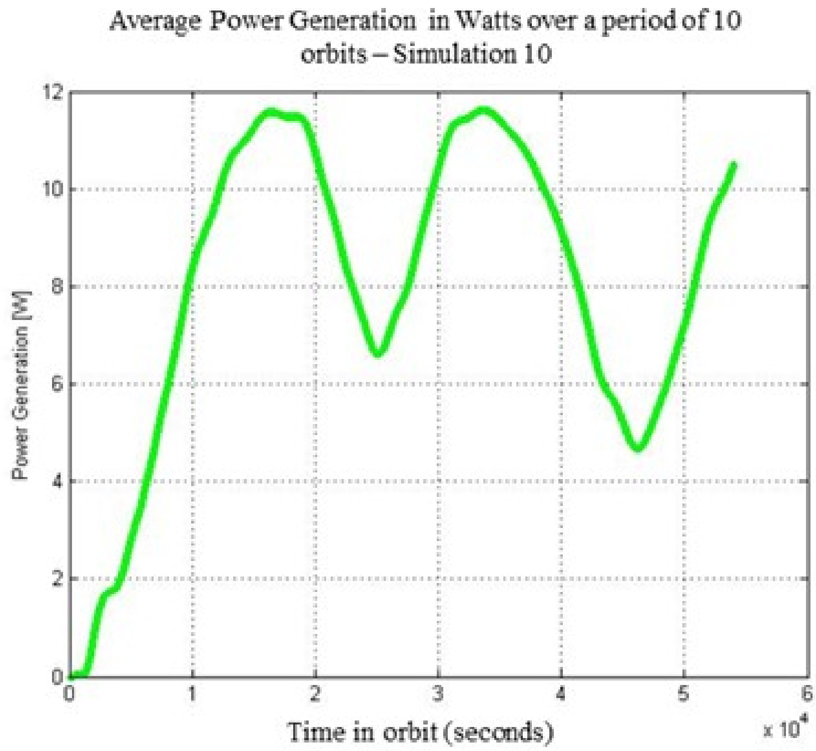

Figure 4.21: Simulation #10: Average power generated. Safehold power generation is achieved after approximately 1.5 orbits.

Simulation #11

Simulation #11 considers the satellite initially pointing its $-i_{yy}$ axis at the Sun and rotating about that axis at 5 deg/s in the presence of gravity gradient and magnetic disturbances. The magnetic dipole used in this simulation is 0.250 $A \cdot m^2$, installed along the $+Y$ axis. The inertial angular momentum and the $-Y$ axis, both expressed in the ECI frame $\mathscr{F}_{\mathrm{i}}$, are shown in Figure 4.22. The figure illustrates that the motion of the -Y axis is considerably disturbed from its initial condition, at time evolves, exposing an even larger surface area of the other satellite panels.

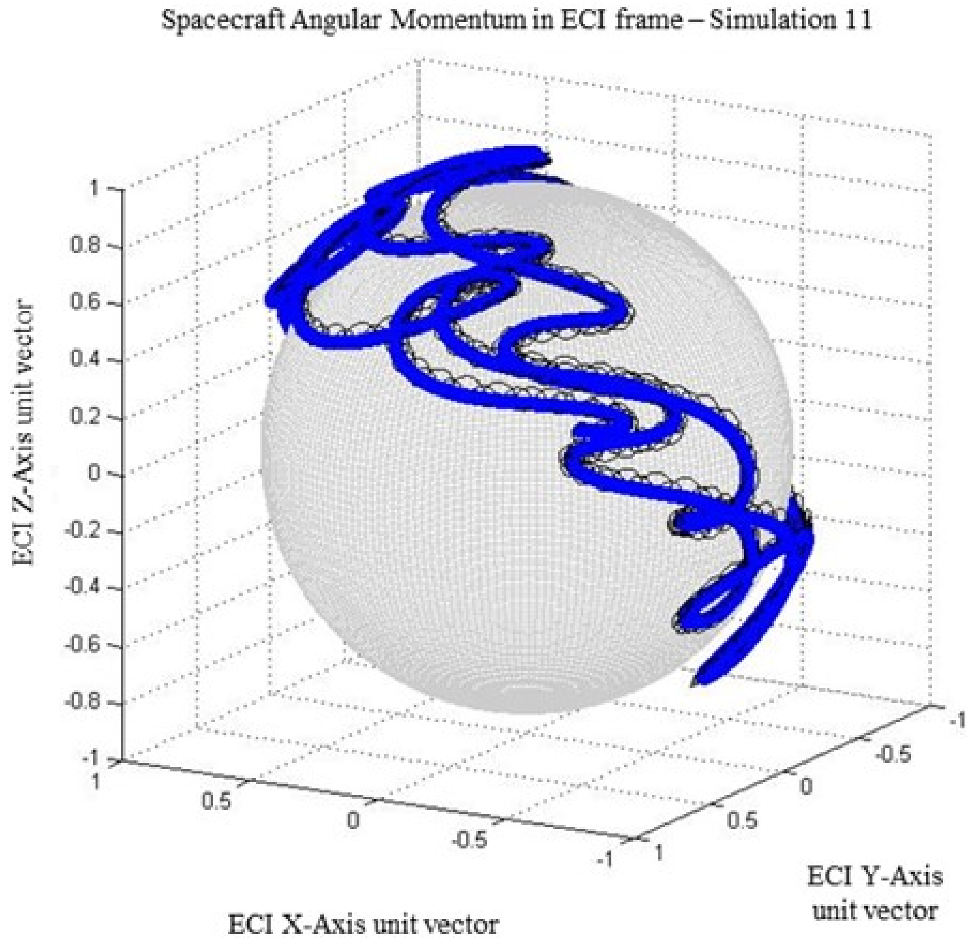

Figure 4.22: Simulation #11: Motion of satellite spin axis (black) and angular momentum vector (blue), projected in ECI frame $\mathscr{F}_{\mathrm{i}}$.

The total satellite power generation is shown in Figure 4.23. The addition of the 0.15 $A \cdot m^2$ dipole installed along the $+Y$ axis shows that the satellite starts producing more than 6.2 Watts after approximately one orbit. At this point, it is worth investigating what would be the effect of the added magnetic dipole on the pointing error and control error of the satellite under nominal terrestrial target tracking operations. This is discussed in the next section.

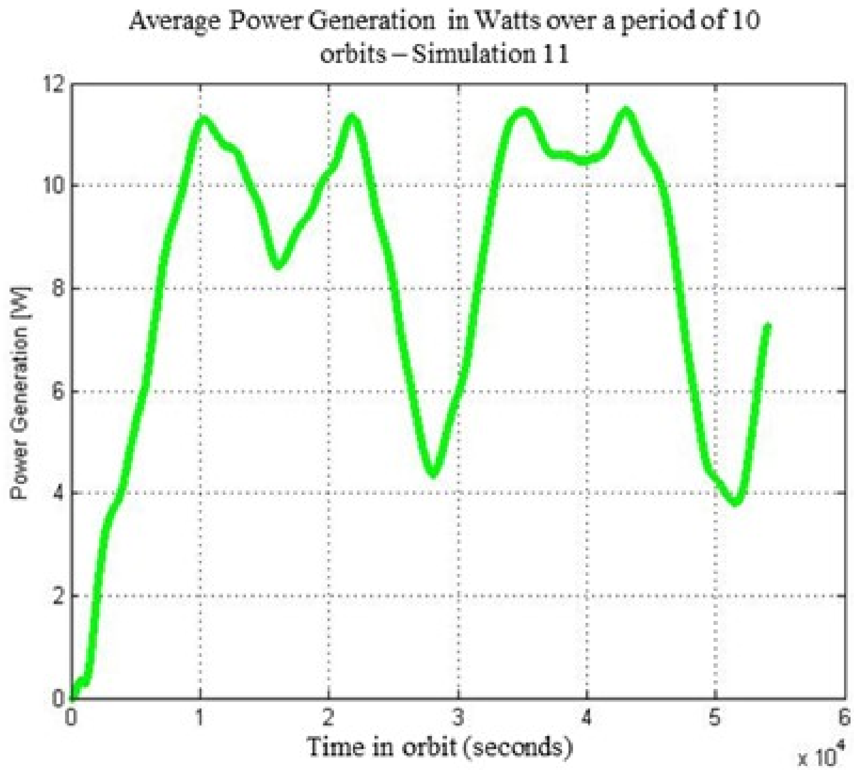

Figure 4.23: Simulation #11: Average power generated. Safehold power generation is achieved after approximately one orbit.

4.3.3 Dipole Value Selection

The simulations that were performed demonstrate that the inclusion of a permanent magnet along the $+Y$ axis of the spacecraft can effectively avoid a prolonged power negative, solar-stare attitude. Overall, the limiting condition is the spacecraft's rotation about the principal axis that is perpendicular to the power limiting panel. The larger the angle between that principal axis and the panel with no cells, the greater chance for power generation. Gravity gradient, aerodynamic and solar radiation pressure torques do not guarantee that the spacecraft can passively escape this undesirable attitude. Trying to actively escape an undesirable attitude using a combination of reaction wheels, magnetorquers and software is also not guaranteed, especially if the spacecraft is in a loadshed condition (Safehold mode). Magnetically biasing the satellite about the power limiting panel ensures, under all conditions, that the panel normal will be in motion in the presence of Earth's magnetic field.

By observing Table 4.2, it is reminded that the spin about the -Y axis pointing towards the Sun is a minor-axis spin. According to the major-axis rule explained in [36] and [40], all minor-axis spins eventually degenerate to major-axis spins. This means that given enough time, GHGSat would eventually exit the solar-stare death mode attitude. However, the time needed to wait until the minor-axis spin becomes major-axis spin would be longer than other operational constraints. More specifically, the satellite cannot be allowed to stay without charging its batteries for prolonged periods of time. In addition, it cannot be allowed to have its depth of discharge too low, even under the least power consuming conditions, which is the Safehold mode.

Power subsystem experts at the Space Flight Laboratory have imposed a requirement on the maximum allowable depth of discharge. This period is within the 10 orbit period before load shed starts from a fully charged battery. The period should also be smaller than the 5 orbit timeframe required for the depth of discharge of the battery to be 40%. In addition to this, the minimum power that needs to be generated in order to keep the spacecraft functioning nominally in Safehold mode is 6.2 Watts. In order to meet these strict time and power requirements, the addition of the dipole is necessary to timely exit the unfavourable attitude. According to the simulations, it has been determined that, for GHGSat-C1/-C2, a minimum dipole of 0.05 $A \cdot m^2$ ($1/10^{\text{th}}$ of the magnetorquer capacity) could be sufficient to steer the $-Y$ face away from an inertial attitude in a timeframe of 3.3 orbits. Higher dipole values yield better results, as was the case with Simulation #11 where a 0.25 $A \cdot m^2$ dipole allows spacecraft charging in about one orbit.

As discussed, the order of magnitude of this dipole is within the order of parasitic magnetic dipole that can stem from solar string loops and other equipment. Under these conditions, a slightly larger dipole also ensures that the system is magnetically biased towards the desired direction. This is important because in the case where the uncharacterized parasitic magnetic dipole is along the opposite axis from which we intend to install the magnet, then the net magnetic dipole moment would be zero. Therefore, there is a fine balance in choosing a large enough dipole to guarantee the correct net dipole moment orientation, and choosing a small enough dipole so that the strain and demands from the ACS system is minimal. Placement of the suggested magnet has been carefully thought to ensure that magnetometer measurements will not be affected. Aligning the dipole of the permanent magnet with the $+Y$ axis will ensure a quick exit from solar-stare. However, it is important to evaluate how the presence of a permanent magnet will affect the control and pointing error of the satellite, especially during operations. If the error is found to not comply with the requirements of the ACS system, then feed-forward control methods may be used. Feed forward control methods allow to estimate the magnetic torque and offset that with the use of magnetorquers of reaction wheel for improved stability. Regardless, the control authority of GHGSat is many times more capable to offset disturbances of this small magnitude.

Two important metrics are used to evaluate attitude control performance; the pointing error, as well as the control error. With respect to an Earth observation similar to GHGSat-C1/C2, the requirement that needs to be met for fine pointing accuracy is that the pointing error and the control error need to be 0.5 deg, 2-σ for nadir pointing. Additional Mirage simulations were performed to ensure this compliance. Three simulations were conducted, each varying the permanent magnetic dipole to 0, 0.125 and 0.250 $A \cdot m^2$. The simulations pertained to target tracking of twenty-one (21) targets over a course of one day. Each target was observed for twenty (20) seconds, during which the spacecraft is switched to its Fine Mode pointing, utilizing all actuators, as well as the rate sensor, magnetometer, and star tracker (Darkstar). The pointing and control error of those observation periods are compiled and their mean and standard deviation are computed.

Dipole Value (A m^2)	Pointing Error (deg)		Control Error θx (Pitch, deg)		Control Error θy (Roll, deg)		Control Error θz (Yaw, deg)		Control Error Angle φ (deg)	
	2σ	3σ	2σ	3σ	2σ	3σ	2σ	3σ	2σ	3σ
0	0.04	0.057	0.028	0.033	0.036	0.042	0.038	0.053	0.044	0.061
0.125	0.035	0.049	0.027	0.037	0.037	0.045	0.03	0.049	0.043	0.059
0.25	0.061	0.078	0.025	0.03	0.037	0.049	0.058	0.075	0.063	0.082
Relevant Requirement	0.5 deg 2-σ for Nadir pointing in Fine Mode		N/A						0.5 deg 2-σ for Nadir pointing in Fine Mode	

Table 4.4: Pointing and Control error in Fine Pointing mode for GHGSat-C, for three different sizes of permanent dipole

Table 4.4 above demonstrates the pointing and control error results from the Mirage-OASYS simulations. The sigma values are computed as the 95.45% and 99.7% percentiles of the Cumulative Distribution Function (CDF). The pointing error is the angle between the desired and actual vector specified as the -Y axis boresight, which is the direction of the payload's optical aperture.

The pointing error requirement is met with the addition of any size of permanent magnet of the ones considered. The control error requirement is also met with any of the specified dipole values. It is noted that these results are collected only during the fine pointing mode period, collected over the course of 20 seconds for 21 target tracking observations, using Darkstar as a star tracker and leading up to very low pointing and control error in the order of 0.05 degrees. Figure 4.24 illustrates the Cumulative Distribution Function results of the pointing error in the simulation where the 0.125 $A \cdot m^2$ dipole was installed. Figure 4.25 shows the corresponding CDF for the control error. In both figures, the horizontal green dashed line represents the 2-σ threshold at 95.45% while the red dashed line represents the 3-σ threshold at 99.97%. The control and pointing error is in the order of 0.05 degrees 2-σ. The 2-σ control error for the 0.125 $A \cdot m^2$ dipole is slightly better by 0.001 degrees in comparison to the case without the dipole. This is because these values of magnetic dipole are quite low, and the randomized measurement noise that is injected during the simulation is within the same order of magnitude, potentially causing low order differences in the control error. It is clear that the addition of the 0.125 $A \cdot m^2$ magnet has minimal impact on the attitude control performance.

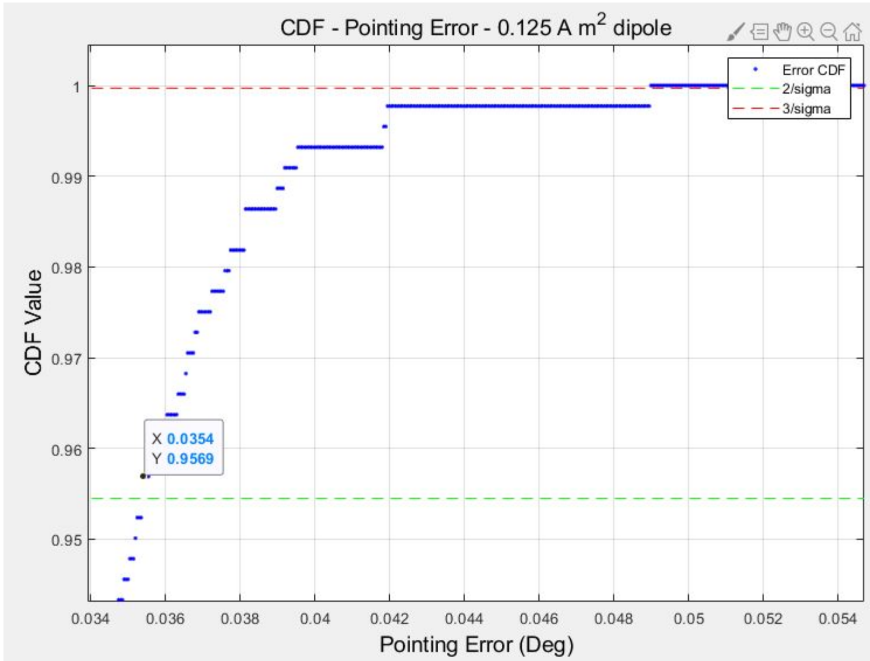

Figure 4.24: Cumulative Distribution Function of Pointing error of MIRAGE simulation with $0.125\ A \cdot m^2$ dipole installed

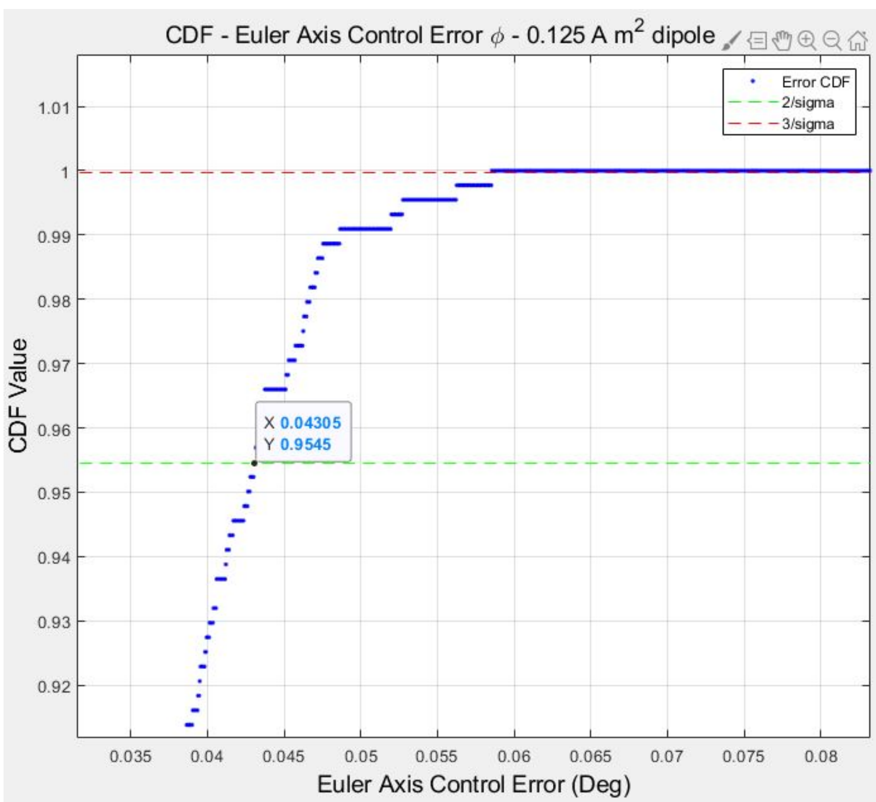

Figure 4.25: Cumulative Distribution Function of Control error of MIRAGE simulation with $0.125\ A \cdot m^2$ dipole installed

In the end, a commercial off-the-shelf neodymium magnet of type N52 was selected to be installed, with a nominal dipole of 0.133 $A \cdot m^2$. A total of 10 magnets of the same type were procured and tested to ensure that the dipole was within the suggested values. The magnets were successfully validated inside SFL's magnetically clean chamber, using the same high-fidelity equipment to test other critical components, such as SFL's magnetometers and magnetorquers.

4.4 Conclusion

This section investigated a number of analyses to understand the conditions in which the satellite is in a power negative attitude and how to subsequently escape one in if the satellite is in passive control or has load shed. A permanent magnet of 0.133 $A \cdot m^2$ was chosen to be installed in the GHGSat-C1 satellite. The value of the magnetic dipole is approximately double of the parasitic magnetic dipole levels, and fine point attitude control simulations of target tracking demonstrated that the solution complies to the mission attitude control requirements.

Chapter 5

Conclusion

In this chapter, we aggregate our observations, findings and conclusions with regards to the topics discussed in this thesis.

Overall, this thesis has investigated two topics of passive attitude stabilization and control. The first topic pertains to attitude stabilization and the use of soft ferromagnetic materials to detumble a spacecraft equipped with drag sails. The second topic pertains to passive attitude control and the use of a permanent magnet to avoid the occurrence of a solar-stare death mode. In the rest of this section, we revisit the key takeaways of each chapter.

In Chapter 1, we highlighted the fundamental role of the Microspace doctrine in setting a high level of expectations amidst the NewSpace revolution. Within the next decade, the small satellite market is expected to grow. However, the critical role of space in terrestrial applications has led to an increase in the number of satellites. Within the aforementioned environment, de-orbiting technology is of pivotal importance. The Space Flight Laboratory has developed and demonstrated a novel drag sail design that keeps breaking records daily, as it has allowed the CanX-7 satellite to reduce its orbital altitude by more than 100 kilometers in the last 4 years.

In Chapter 2, we started by taking a look at a unique behavior of CanX-7. The satellite experienced random tumbling periods of linearly increasing angular rates, along one of the transverse axis. The author's original contribution lies in the validation of this spin-up behavior through simulation. By varying the initial attitude and body rates of the satellite as well as the satellite geometry, the linear spin-up was replicated for CanX-7. As an extension, the author pursued to identify a set of conditions that would lead to a spin-up behavior for a NEMO class satellite, also equipped with drag sails. By varying the same parameters that led to the CanX-7 spin-up, it was possible to produce a linearly increasing tumble. A fundamental assumption of this thesis lied in the use of rigid body dynamics, and the fact that any sail twist on the satellite structure is unchanging and fixed. In the future, it would be of interest to integrate the equations of motion describing the flexible dynamics of CanX-7 and explore the possible reasons that would lead to the spin-up condition. In addition, it would be of interest to expand the search space of the permutation analysis performed with more parameters, being mindful about the computational load, or explore parallel/cloud computing solutions to speed up the permutation analysis.

In Chapter 3, the latter set of conditions was used as input in an investigation on potential mitigation solutions. Early on, it is identified that hysteresis damping provides a reasonable mitigation strategy. The author's original contribution in this chapter lied in the development and addition of the equations describing magnetic hysteresis torque within the simulation environment. Through this treatment, a compact formula was derived to compute the magnetic hysteresis torque of multiple rods regardless of their number, size or mounting orientation within the satellite. Using this framework, a series of simulations were performed with different hysteresis materials to assess their effectiveness in nullifying the linearly increasing spin-up due to the unfavourable initial conditions of Chapter 2. In parallel, the author performed an experimental procedure abiding to ASTM standards to characterize hysteresis rods available in SFL inventory. The results of the characterization were fed back to a simulation, and compared against simulations using theoretical magnetic values. It was determined that the experimentally characterized rods are adequate to nullify the spin-up to zero body rates. In general, it has been observed that simulated data from other literature sources overestimate the capabilities of the hysteresis rods. The in-house experimental process produced conservative magnetic parameters that led to a longer stabilization. In the future, it would be of interest to test more hysteresis materials, particularly silicon-iron grain oriented rods which provide higher energy losses. In addition, additional simulation and testing is needed to characterize the phenomenon of mutual magnetization between hysteresis rods that are mounted in close proximity. This becomes especially important if the designer chooses to increase dipole efficiency by installing thinner rods of a higher number, at the cost of satellite volume space.

In Chapter 4, we discussed about GHGSat-C1/C2 and the mitigation strategy for a potential solar-stare death mode. The satellite's bulky optical aperture leaves one of the satellite faces without solar panels, which is a risk to depleting the satellite's batteries, when that panel normal points at the Sun for a prolonged amount of time. However, it is highlighted that this axis is a minor axis, and therefore any minor spin will degenerate to a major axis spin. The challenge was to identify how long this will take, and if it takes too long, what solution can be explored to lead to power generation fast under safehold mode, where active attitude control is not available. Based on requirements for minimum safehold power generation, there is a need for the satellite to be power positive within less than 5 orbits. After a set of simulations, it was observed that the shift from minor to major axis spin would take too long to execute. In addition, disturbance torques themselves would still not lead to a power positive condition. The addition of a permanent magnet along the power limiting minor axis resulted in effective solar stare avoidance in less than the 5 orbit threshold. Various dipole values were used in simulation and tested successfully against compliance to active control requirements. In the end, a 0.133 $A \cdot m^2$ permanent magnet was chosen to be installed. In the future, it would be of interest to identify an optimal method for determining the best orientation of a magnet, rather than evaluating only three options limited to the satellite's body axes.

The findings of this thesis prove that passive attitude stabilization and control is still relevant today, amidst the abundance of miniaturized, highly performing and cost-efficient components-off-the-shelf for active attitude control. Passive solutions can be used as complimentary to active control solutions, as long as their addition does not interfere with pointing and control error requirements. Passive solutions require no moving parts, while energy requirement is limited at most to the additional input for feed-forward control as in the case of the added permanent magnet. No feed-forward control is necessary with the addition of hysteresis rods, since the nature of magnetic hysteresis torque is dissipative in nature.

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